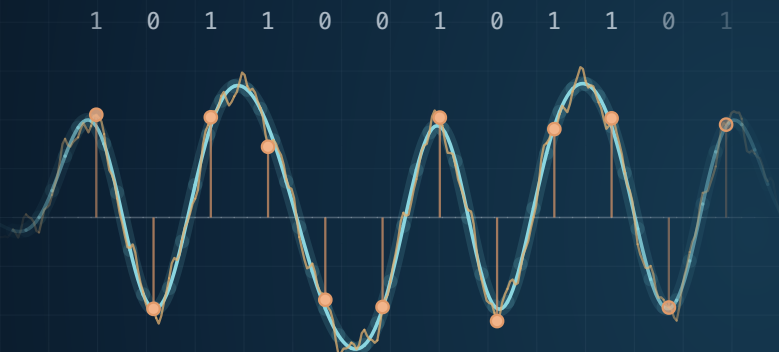




SAMPLING · DETECTION · MODULATION · CODING

Digital Communications

Student Workbook



Contents

Every question in the course, with no answer and no solution. Work each one on the page, then check it against the artifact or against the instructor edition.

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How to use it

The questions are in the order the course meets them, and each is labelled with what it asks for. Only the statement and its lettered parts are printed; the reasoning stays for you to supply. The question numbers are shared with every other edition, so D5-04 is the same question in the artifact, in this workbook and in the instructor solutions.

MODULE

1

Sampling, Quantization and PCM

30 questions on sampling, quantization and pcm. Write your reasoning in the space under each one.

D1-01 · PCM DESIGN FROM AN ACCURACY REQUIREMENT

Assume that a sinusoidal message signal is defined as $x(t) = V_{\max} \cos(12000\pi t)$, where $V_{\max}$ is the maximum amplitude of the message signal. This analog message signal is sampled at the Nyquist rate and quantized by using a uniform quantizer. The quantization noise is a uniform random variable between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$, where Δ is the step size. The quantization noise is required not to exceed $\pm 0.25\%$ of the peak-to-peak message signal. Quantized data are encoded by using a 4-level PAM system.

- [9 pts] What is the minimum number of bits per sample for this PAM system?
- [8 pts] Calculate the bit rate of this system.
- [8 pts] What is the symbol rate of this system?

D1-02 · PCM DESIGN FROM AN ACCURACY REQUIREMENT

Assume that a sinusoidal message signal is defined as $x(t) = V_{\max} \sin(7000\pi t)$, where $V_{\max}$ is the maximum amplitude of the message signal. It is sampled at a rate 25% greater than the Nyquist rate and quantized by using a uniform quantizer. The quantization noise is a uniform random variable between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$. It is required not to exceed $\pm 0.8\%$ of the peak-to-peak message signal. Quantized data are encoded by using an 8-level PAM system.

- [7 pts] What is the minimum number of bits per sample?
- [6 pts] Determine the sampling rate.
- [6 pts] Calculate the bit rate of this system.
- [6 pts] What is the symbol rate of this system?

D1-03 · PCM DESIGN FROM AN ACCURACY REQUIREMENT

Let $X(t)$ have a bandwidth of 3.2 MHz. This signal is sampled, quantized and binary encoded to obtain a PCM signal.

- [7 pts] Determine the sampling rate if $X(t)$ is sampled at a rate 25% greater than the Nyquist rate.
- [6 pts] If the samples of $X(t)$ are quantized by using a uniform quantizer with 2048 levels, determine the number of bits required per sample.
- [6 pts] Calculate the bit rate of this system in bits per second.
- [6 pts] Find the least channel bandwidth that can carry this PCM signal.

D1-04 · PCM DESIGN FROM AN ACCURACY REQUIREMENT

A PCM system carries the sinusoid $x(t) = V_{\max} \cos(10000\pi t)$. It samples at the Nyquist rate and quantizes uniformly over $[-V_{\max}, V_{\max}]$. The quantization noise is uniform between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$. The bits are sent by a 16-level PAM system, and the measured symbol rate is 17 500 symbols per second.

- a) [6 pts] Find the bit rate and the number of bits per sample.
- b) [6 pts] Find the number of quantization levels and the step size in terms of $V_{\max}$.
- c) [7 pts] Find the largest quantization error as a percentage of the peak-to-peak message signal.
- d) [6 pts] The design came from a requirement of $\pm p\%$ of the peak-to-peak signal. Find the range of p for which this number of bits is the minimum.

D1-05 · PCM DESIGN FROM AN ACCURACY REQUIREMENT

Assume that a sinusoidal message signal is defined as $x(t) = V_{\max} \cos(18000\pi t)$. It is sampled at the Nyquist rate and quantized by using a uniform quantizer. The quantization noise is uniform between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$. It is required not to exceed $\pm 0.1\%$ of the peak-to-peak message signal. The bits are sent by an M -level PAM system over a channel that accepts at most 60 000 symbols per second.

- a) [8 pts] What is the minimum number of bits per sample?
- b) [6 pts] Calculate the bit rate of this system.
- c) [5 pts] Find the smallest PAM order M , a power of two, that the channel accepts.
- d) [6 pts] Calculate the symbol rate with that M .

D1-06 · PCM DESIGN FROM AN ACCURACY REQUIREMENT

Let $X(t)$ have a bandwidth of 2.5 MHz. It is sampled at a rate 30% greater than the Nyquist rate, quantized by a uniform quantizer and binary encoded. The PCM signal must fit a link that carries 64 Mbit/s. Assume that 1 Mbit/s = 10^6 bit/s.

- a) [6 pts] Determine the sampling rate.
- b) [7 pts] Find the largest number of bits per sample, and the number of levels, that the link allows.
- c) [6 pts] Calculate the bit rate of the resulting system.
- d) [6 pts] For a full-scale sinusoidal test signal, calculate the SQNR in dB.

D1-07 · THE SPECTRUM OF A PRODUCT, A SQUARE OR A SINC

Let $x(t) = 10 \cos(3000\pi t) \cos(6000\pi t)$ be sampled and quantized by using a 512-level uniform quantizer. Assume that $1 \text{ kbit/s} = 1000 \text{ bit/s}$.

- a) [6 pts] Determine the minimum sampling rate if a guard band of 1 kHz is required.
- b) [6 pts] Calculate the bit rate of this system.
- c) [6 pts] If the data rate of this system is required as 108 kbit/s, what should the guard band be?
- d) [7 pts] Calculate the step size of the uniform quantizer.

D1-08 · THE SPECTRUM OF A PRODUCT, A SQUARE OR A SINC

Let $x(t) = 5 \sin(2000\pi t) \cos(8000\pi t)$ be sampled and quantized by using a 64-level uniform quantizer. Assume that $1 \text{ kbit/s} = 1000 \text{ bit/s}$.

- a) [6 pts] Determine the minimum sampling rate if a guard band of 1.5 kHz is required.
- b) [6 pts] Calculate the bit rate of this system.
- c) [6 pts] If the data rate of this system is required as 78 kbit/s, what should the guard band be?
- d) [7 pts] Calculate the step size of the uniform quantizer.

D1-09 · THE SPECTRUM OF A PRODUCT, A SQUARE OR A SINC

Let $x(t) = 8 \cos^2(3000\pi t)$ be sampled and quantized by using a 256-level uniform quantizer that spans the range of $x(t)$. Assume that $1 \text{ kbit/s} = 1000 \text{ bit/s}$.

- a) [6 pts] Determine the minimum sampling rate if a guard band of 2 kHz is required.
- b) [6 pts] Calculate the bit rate of this system.
- c) [6 pts] If the data rate of this system is required as 72 kbit/s, what should the guard band be?
- d) [7 pts] Calculate the step size of the uniform quantizer.

D1-10 · THE SPECTRUM OF A PRODUCT, A SQUARE OR A SINC

The signal

$$x(t) = \left(\frac{\sin(200\pi t)}{20\pi t} \right)^2$$

is sampled at the Nyquist rate. The samples are uniformly quantized with 256 levels over the range of $x(t)$. Use $\text{sinc}(u) = \sin(\pi u)/(\pi u)$. According to the information given above,

- a) [8 pts] Calculate the bit rate of this system.
- b) [8 pts] Calculate the step size of the uniform quantizer.
- c) [9 pts] Find the quantized value and the natural binary code word of the samples at $t = 0$, $t = 2.5$ ms and $t = 7.5$ ms.

D1-11 · THE SPECTRUM OF A PRODUCT, A SQUARE OR A SINC

The signal

$$x(t) = \frac{\sin(200\pi t)}{10\pi t} \cdot \frac{\sin(600\pi t)}{10\pi t}$$

is sampled at the Nyquist rate. The samples are uniformly quantized with 128 levels over the range of $x(t)$. Use $\text{sinc}(u) = \sin(\pi u)/(\pi u)$. According to the information given above,

- a) [8 pts] Calculate the bit rate of this system.
- b) [8 pts] Calculate the step size of the uniform quantizer. (Hint: $\min x(t) \cong -177$.)
- c) [9 pts] Find the smallest number of levels, a power of two, that makes the step size smaller than 1. Calculate the bit rate it needs.

D1-12 · THE SPECTRUM OF A PRODUCT, A SQUARE OR A SINC

The signal

$$x(t) = \frac{\sin(1000\pi t)}{\pi t} \cos(4000\pi t)$$

is sampled at the Nyquist rate. The samples are uniformly quantized with 1024 levels over the range of $x(t)$. According to the information given above,

- a) [8 pts] Find the Fourier transform $X(f)$ and the highest frequency in $x(t)$.
- b) [6 pts] Calculate the bit rate of this system.
- c) [5 pts] Calculate the step size of the uniform quantizer. (Hint: $\min x(t) \cong -902$.)
- d) [6 pts] Calculate the energy of $x(t)$ by Parseval's theorem.

D1-13 · A SUM OF SINUSOIDS THROUGH A UNIFORM QUANTIZER

The signal $x(t) = 2 \cos(3000\pi t) + 3 \cos(9000\pi t)$ is sampled at the Nyquist rate and samples are uniformly quantized with 256 levels. According to the information given above,

- a) [5 pts] Calculate the bit rate of this system.
- b) [10 pts] Calculate the step size of the uniform quantizer.
- c) [10 pts] Calculate the SQNR of the quantization scheme (in dB).

D1-14 · A SUM OF SINUSOIDS THROUGH A UNIFORM QUANTIZER

The signal $x(t) = 2 \cos(2000\pi t) + \cos(4000\pi t)$ is sampled at the Nyquist rate and samples are uniformly quantized with 128 levels. The quantizer spans the range of $x(t)$, from its minimum to its maximum. According to the information given above,

- a) [5 pts] Calculate the bit rate of this system.
- b) [7 pts] Find the maximum and the minimum of $x(t)$.
- c) [6 pts] Calculate the step size of the uniform quantizer.
- d) [7 pts] Calculate the SQNR of the quantization scheme (in dB).

D1-15 · A SUM OF SINUSOIDS THROUGH A UNIFORM QUANTIZER

The signal $x(t) = \cos(1000\pi t) + \cos(3000\pi t) + \cos(5000\pi t)$ is sampled at the Nyquist rate and samples are uniformly quantized with 512 levels. According to the information given above,

- a) [5 pts] Calculate the bit rate of this system.
- b) [10 pts] Calculate the step size of the uniform quantizer.
- c) [10 pts] Calculate the SQNR of the quantization scheme (in dB).

D1-16 · A SUM OF SINUSOIDS THROUGH A UNIFORM QUANTIZER

The signal $x(t) = 3 \sin(5000\pi t) + 4 \cos(5000\pi t)$ is sampled at the Nyquist rate and samples are uniformly quantized with 128 levels over the range of $x(t)$. According to the information given above,

- a) [5 pts] Calculate the bit rate of this system.
- b) [8 pts] Find the peak value of $x(t)$.
- c) [5 pts] Calculate the step size of the uniform quantizer.
- d) [7 pts] Calculate the SQNR of the quantization scheme (in dB).

D1-17 · A SUM OF SINUSOIDS THROUGH A UNIFORM QUANTIZER

The signal $x(t) = 4 \cos(2000\pi t) + 2 \cos(6000\pi t)$ is sampled with a guard band of at least 1 kHz and uniformly quantized over the range of $x(t)$. The bits must fit a link of 80 kbit/s. Assume that 1 kbit/s = 1000 bit/s.

- a) [5 pts] Determine the minimum sampling rate.
- b) [7 pts] Find the largest number of bits per sample that the link allows at that rate.
- c) [6 pts] Calculate the step size of the uniform quantizer.
- d) [7 pts] Calculate the SQNR of the quantization scheme (in dB).

D1-18 · A DENSITY WITH A CONSTANT AND A FINE UNIFORM QUANTIZER

A strict-sense stationary random process $X(t)$ is sampled. The sampled values X have the following PDF: $f_X(x) = k [1 + x^2]$ for $x \in [-1, 1]$. Sampled values are quantized by using a uniform quantizer with 128 levels. The quantization noise is a uniform random variable between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$, where Δ is the step size.

- a) [5 pts] Determine the value of k .
- b) [15 pts] Obtain the SQNR in dB.
- c) [5 pts] If the bandwidth of the signal is 4 kHz, what is the bit rate of the corresponding PCM system?

D1-19 · A DENSITY WITH A CONSTANT AND A FINE UNIFORM QUANTIZER

A strict-sense stationary random process $X(t)$ is sampled. The sampled values X have the following PDF: $f_X(x) = k [1 - |x|^{1/2}]$ for $x \in [-1, 1]$. Sampled values are quantized by using a uniform quantizer with 512 levels. The quantization noise is a uniform random variable between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$, where Δ is the step size.

- a) [5 pts] Determine the value of k .
- b) [15 pts] Obtain the SQNR in dB.
- c) [5 pts] If the bandwidth of the signal is 5 kHz, what is the bit rate of the corresponding PCM system?

D1-20 · A DENSITY WITH A CONSTANT AND A FINE UNIFORM QUANTIZER

The samples of a stationary source, $X(t)$, are distributed according to the PDF $f_X(x) = k(4 - x^2)$ for $|x| \leq 2$, and zero elsewhere. The samples are quantized by using a uniform quantizer with 64 levels over $[-2, 2]$. The quantization noise is a uniform random variable between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$.

- a) [5 pts] Determine the value of k .
- b) [15 pts] Obtain the SQNR in dB.
- c) [5 pts] If the bandwidth of the signal is 6 kHz, what is the bit rate of the corresponding PCM system?

D1-21 · A DENSITY WITH A CONSTANT AND A FINE UNIFORM QUANTIZER

The samples of a stationary source $X(t)$ have the PDF $f_X(x) = k e^{-|x|}$ for $|x| \leq 2$, and zero elsewhere. They are quantized by a uniform quantizer with 256 levels over $[-2, 2]$. The quantization noise is a uniform random variable between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$.

- a) [5 pts] Determine the value of k .
- b) [7 pts] Calculate the power of the samples.
- c) [7 pts] Obtain the SQNR in dB.
- d) [6 pts] Find the smallest number of levels, a power of two, that gives an SQNR of at least 50 dB.

D1-22 · A DENSITY WITH A CONSTANT AND A FINE UNIFORM QUANTIZER

The samples of a stationary source $X(t)$ have the PDF $f_X(x) = k(1+x)$ for $-1 \leq x \leq 1$, and zero elsewhere. The samples are quantized by a uniform quantizer with 32 levels over $[-1, 1]$. The quantization noise is a uniform random variable between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$.

- a) [5 pts] Determine the value of k .
- b) [8 pts] Calculate the mean and the power of the samples.
- c) [7 pts] Obtain the SQNR in dB.
- d) [5 pts] If the bandwidth of the signal is 3.5 kHz, what is the bit rate of the corresponding PCM system?

D1-23 · A DENSITY WITH A CONSTANT AND A FINE UNIFORM QUANTIZER

The samples of a stationary source $X(t)$ have the PDF $f_X(x) = k|x|$ for $|x| \leq a$, and zero elsewhere. The power of the samples is $E[X^2] = 2$. A uniform quantizer with L levels covers $[-a, a]$. The quantization noise is a uniform random variable between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$.

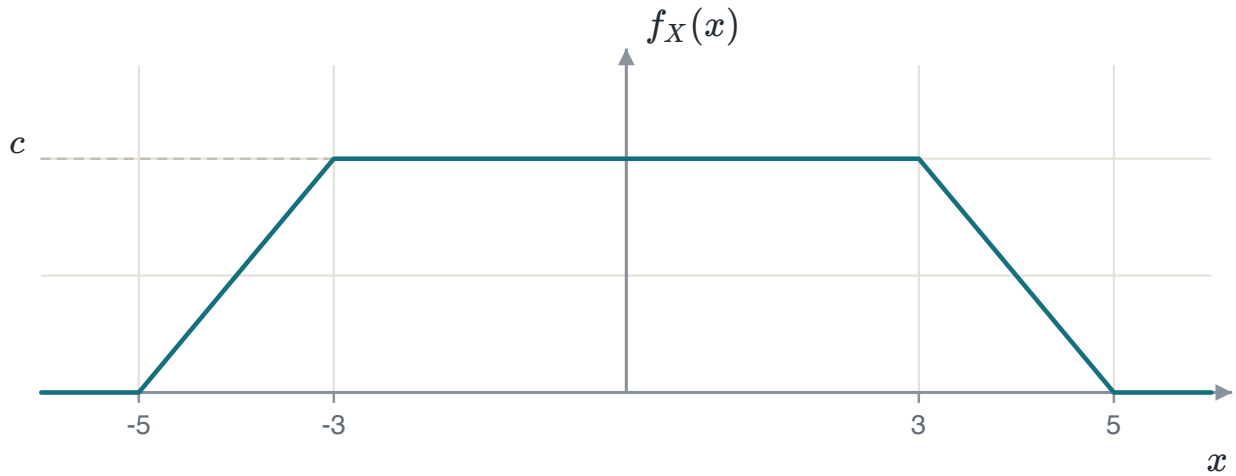
- a) [8 pts] Determine a and k .
- b) [7 pts] Find the smallest L , a power of two, that gives an SQNR of at least 40 dB.
- c) [5 pts] Calculate the step size and the SQNR obtained with that L .
- d) [5 pts] If the bandwidth of the signal is 6.5 kHz, what is the bit rate of the corresponding PCM system?

D1-24 · A DRAWN DENSITY AND A COARSE QUANTIZER

The samples of a stationary source, $X(t)$, are distributed according to the probability density function (PDF) drawn below. These samples are quantized using the following quantizer:

$$\hat{X} = \mathbb{Q}(X) = \begin{cases} -2, & -3 < X < 0 \\ 2, & 0 < X < 3 \\ 0, & \text{otherwise} \end{cases}$$

According to the information given above,



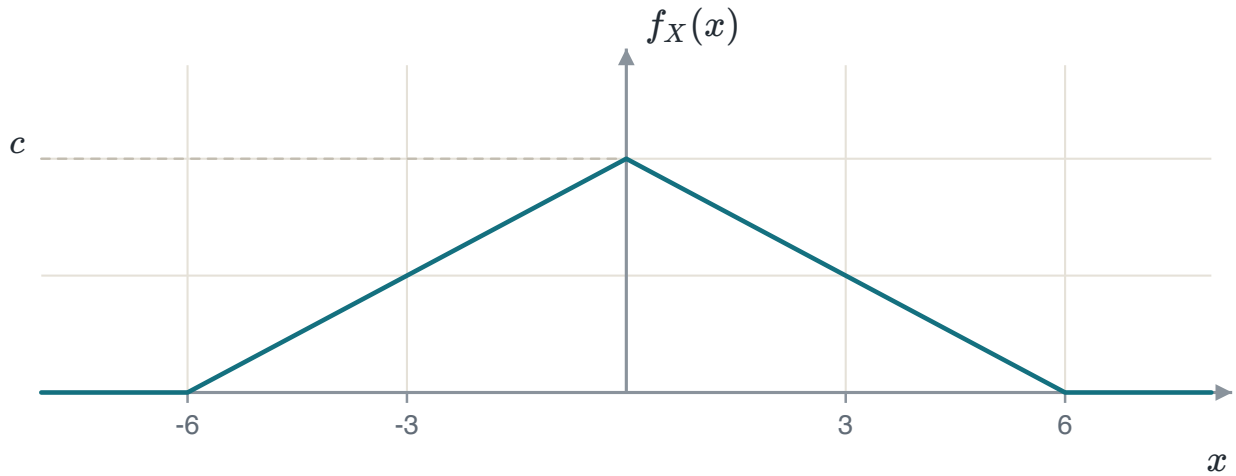
- [6 pts] Determine the value of c .
- [6 pts] Calculate the power of the samples of the stationary source.
- [6 pts] Calculate the power of the quantization noise.
- [7 pts] Obtain the SQNR in dB.

D1-25 · A DRAWN DENSITY AND A COARSE QUANTIZER

The samples of a stationary source, $X(t)$, are distributed according to the probability density function (PDF) drawn below. These samples are quantized using the following quantizer:

$$\hat{X} = \mathbb{Q}(X) = \begin{cases} -3, & -6 < X < 0 \\ 3, & 0 < X < 6 \\ 0, & \text{otherwise} \end{cases}$$

According to the information given above,



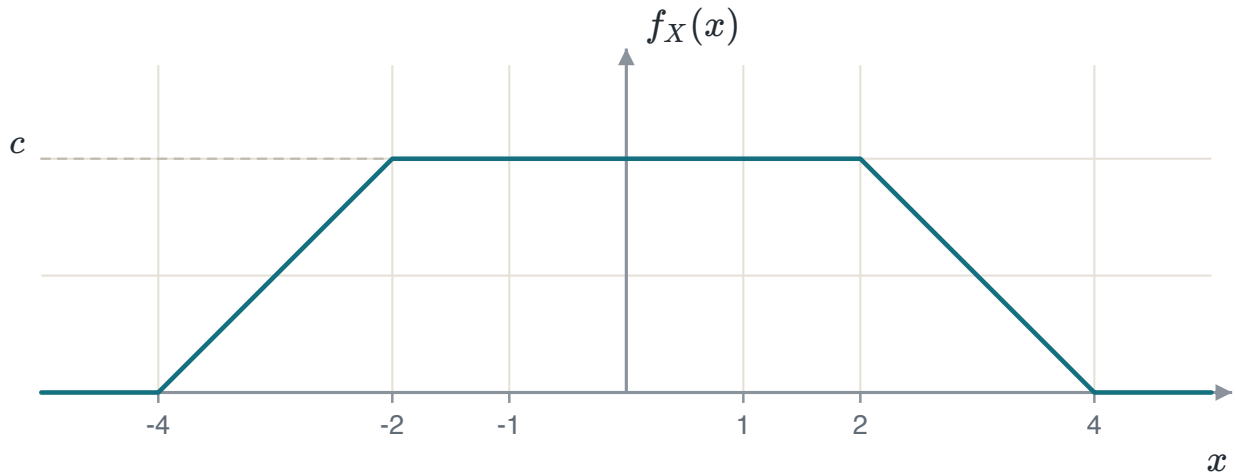
- [6 pts] Determine the value of c .
- [6 pts] Calculate the power of the samples of the stationary source.
- [6 pts] Calculate the power of the quantization noise.
- [7 pts] Obtain the SQNR in dB.

D1-26 · A DRAWN DENSITY AND A COARSE QUANTIZER

The samples of a stationary source, $X(t)$, are distributed according to the probability density function (PDF) drawn below. These samples are quantized using the following quantizer:

$$\hat{X} = \mathbb{Q}(X) = \begin{cases} -2, & -4 < X < -1 \\ 0, & -1 \leq X \leq 1 \\ 2, & 1 < X < 4 \end{cases}$$

According to the information given above,



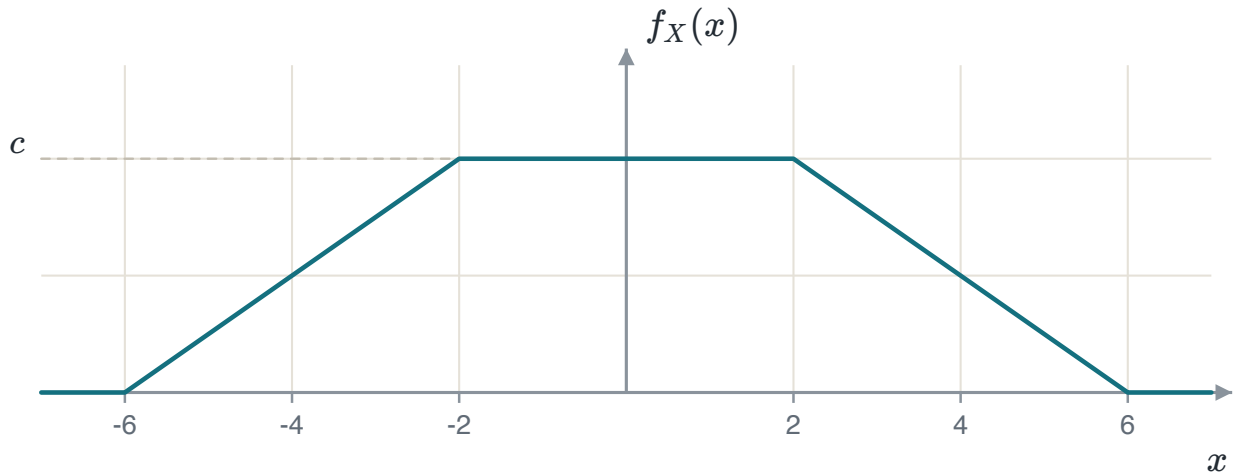
- [6 pts] Determine the value of c .
- [6 pts] Calculate the power of the samples of the stationary source.
- [6 pts] Calculate the power of the quantization noise.
- [7 pts] Obtain the SQNR in dB.

D1-27 · A DRAWN DENSITY AND A COARSE QUANTIZER

The samples of a stationary source, $X(t)$, are distributed according to the probability density function (PDF) drawn below. These samples are quantized using the following quantizer:

$$\hat{X} = \mathbb{Q}(X) = \begin{cases} -2, & -4 < X < 0 \\ 2, & 0 < X < 4 \\ 0, & \text{otherwise} \end{cases}$$

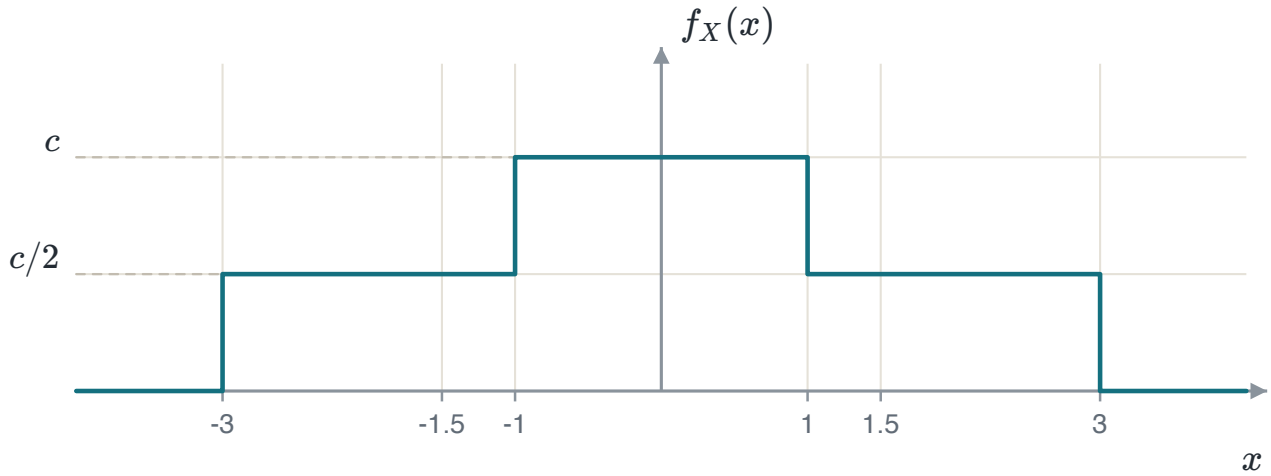
According to the information given above,



- [6 pts] Determine the value of c .
- [6 pts] Calculate the power of the samples of the stationary source.
- [6 pts] Calculate the power of the quantization noise.
- [7 pts] Obtain the SQNR in dB.

D1-28 · A DRAWN DENSITY AND A COARSE QUANTIZER

The samples of a stationary source $X(t)$ have the piecewise-constant PDF drawn below. They are quantized by a 4-level uniform quantizer over $[-3, 3]$, with outputs ± 0.75 and ± 2.25 and boundaries 0 and ± 1.5 .



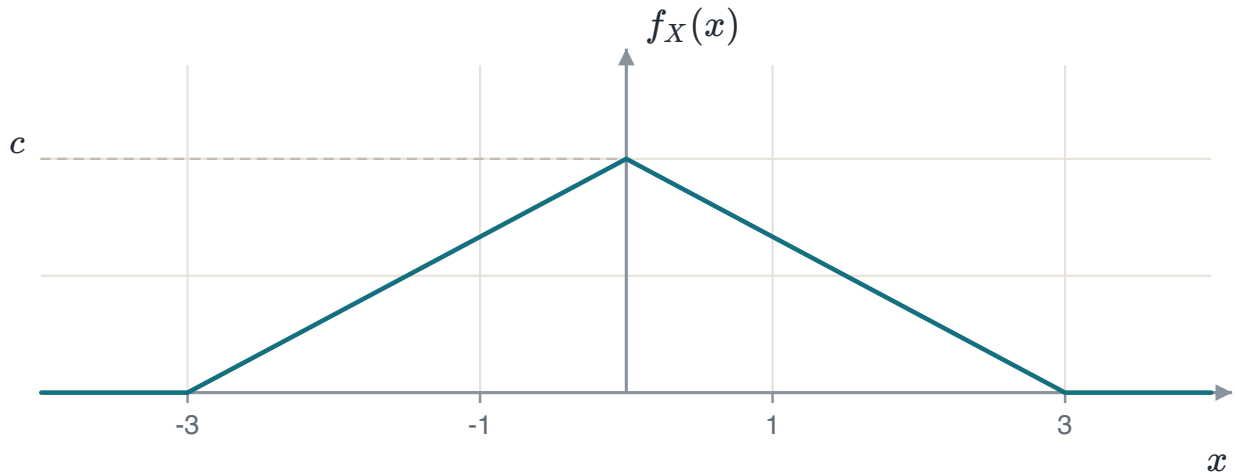
- [5 pts] Determine the value of c .
- [6 pts] Calculate the power of the samples.
- [8 pts] Calculate the power of the quantization noise, region by region.
- [6 pts] Obtain the SQNR in dB, and compare the noise power with $\Delta^2/12$.

D1-29 · A DRAWN DENSITY AND A COARSE QUANTIZER

The samples of a stationary source $X(t)$ have the triangular PDF drawn below. They are quantized by

$$\hat{X} = \mathbb{Q}(X) = \begin{cases} -b, & -3 < X < 0 \\ b, & 0 < X < 3 \\ 0, & \text{otherwise} \end{cases}$$

where $b > 0$ is a design constant.



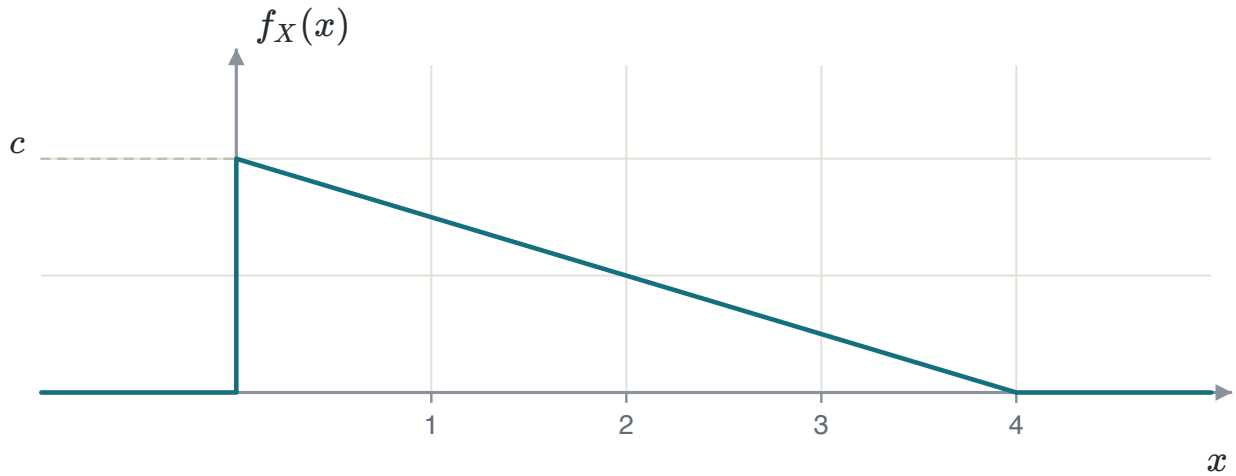
- [5 pts] Determine the value of c .
- [5 pts] Calculate the power of the samples.
- [7 pts] For $b = 2$, calculate the power of the quantization noise and the SQNR in dB.
- [8 pts] Find the value of b that makes the noise power smallest, and the SQNR it gives.

D1-30 · A DRAWN DENSITY AND A COARSE QUANTIZER

The samples of a non-negative stationary source $X(t)$ have the PDF drawn below. It falls linearly from c at $x = 0$ to zero at $x = 4$. They are quantized by

$$\hat{X} = \mathbb{Q}(X) = \begin{cases} 1, & 0 < X < 2 \\ 3, & 2 < X < 4 \\ 0, & \text{otherwise} \end{cases}$$

According to the information given above,



- [5 pts] Determine the value of c .
- [6 pts] Calculate the mean and the power of the samples.
- [8 pts] Calculate the power of the quantization noise.
- [6 pts] Obtain the SQNR in dB, and compare the noise power with $\Delta^2/12$.

MODULE

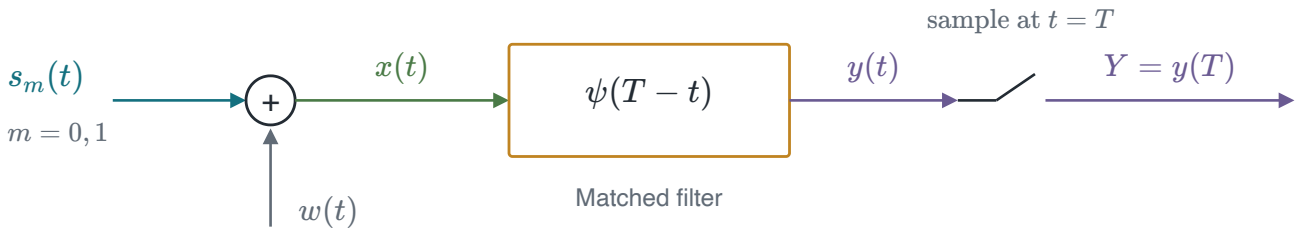
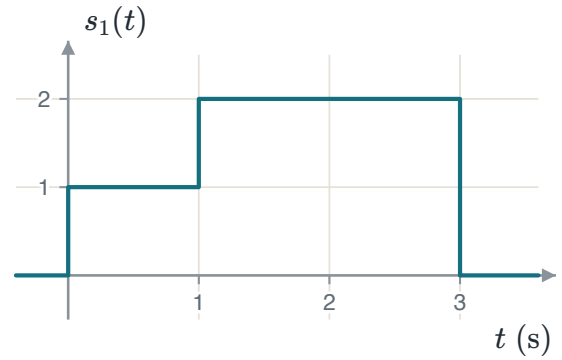
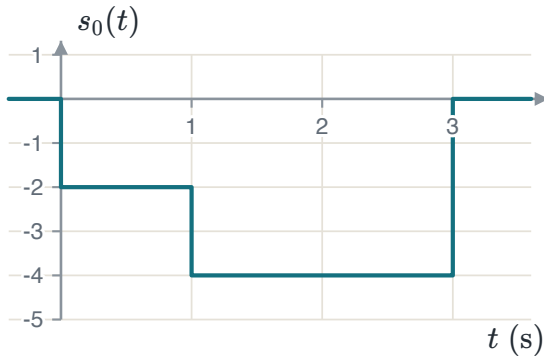
2

Baseband Transmission

30 questions on baseband transmission. Write your reasoning in the space under each one.

D2-01 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

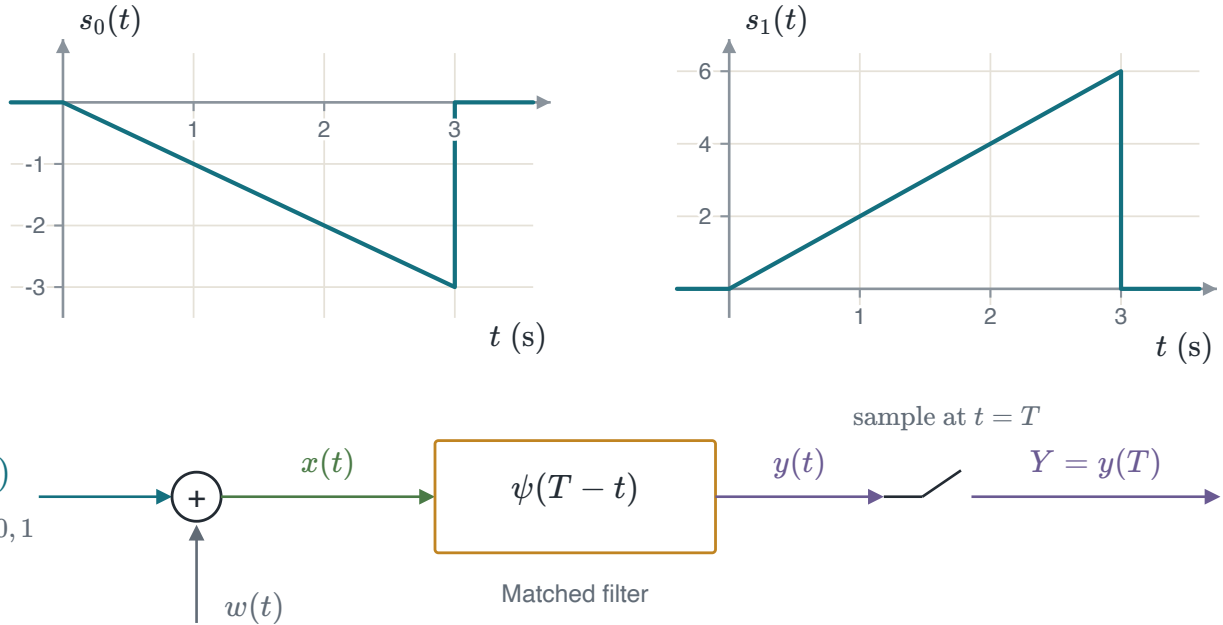
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 2.25$ W/Hz. Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below. These signals are passed through the following matched-filter type demodulator. Here $\psi(t)$ is the unit-energy basis signal (pulse shape) that represents these signals. It is also used in the impulse response of the matched filter. The noise $w(t)$ is zero-mean white Gaussian noise, and $T = 3$ s. According to the information given above,



- [5 pts] Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [10 pts] Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts] Find the optimal decision threshold (λ) and calculate the average probability of bit error (P_b). Use a table of the Q function for the numerical value.

D2-02 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

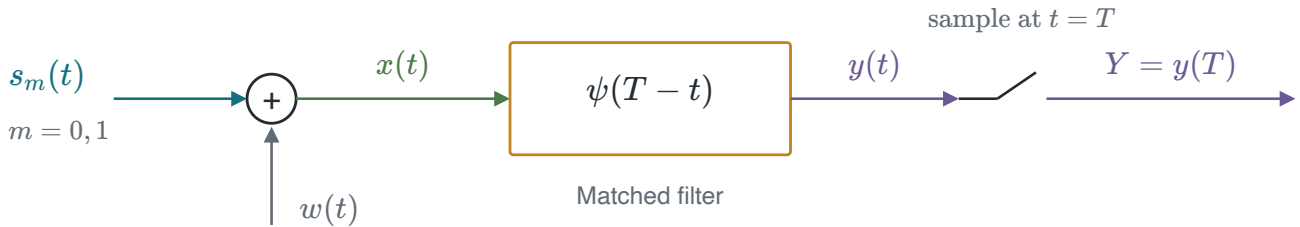
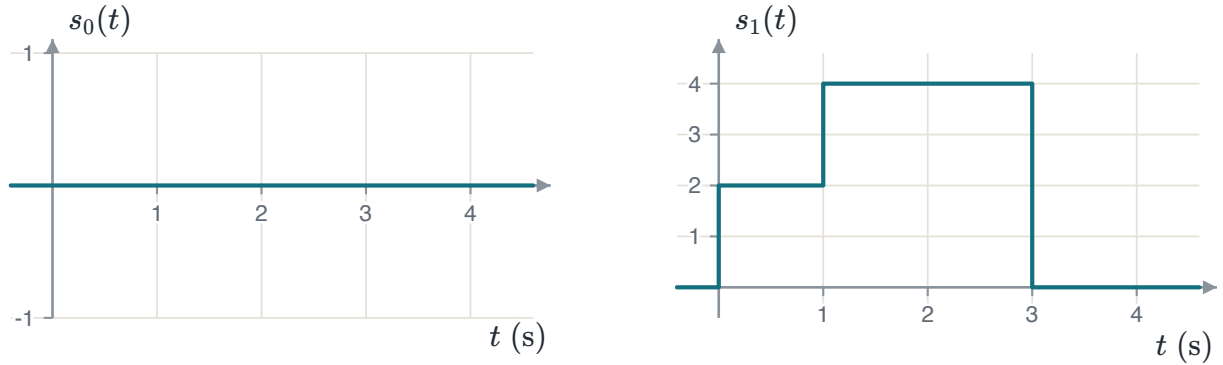
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 4$ W/Hz. Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below. These signals are passed through the following matched-filter type demodulator. Here $\psi(t)$ is the unit-energy basis signal (pulse shape) that represents these signals. It is also used in the impulse response of the matched filter. The noise $w(t)$ is zero-mean white Gaussian noise, and $T = 3$ s. According to the information given above,



- [5 pts]** Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [10 pts]** Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts]** Find the optimal decision threshold (λ) and calculate the average probability of bit error (P_b). Use a table of the Q function for the numerical value.

D2-03 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

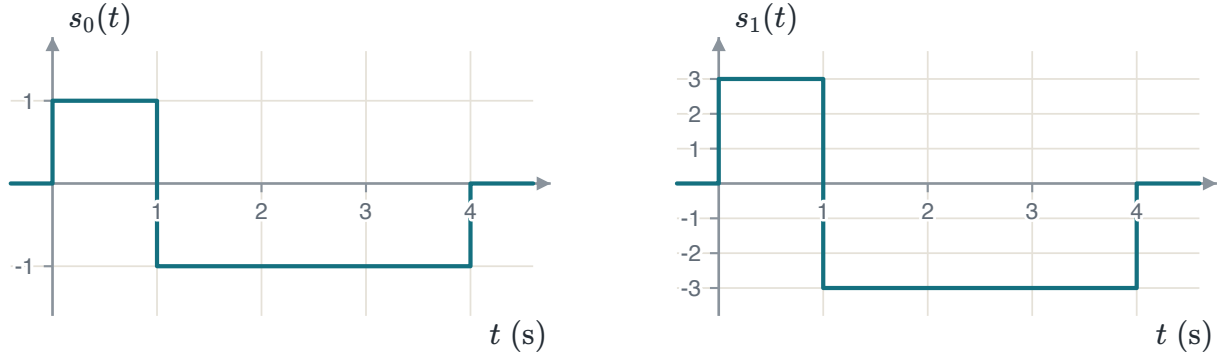
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 4$ W/Hz. Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below. These signals are passed through the following matched-filter type demodulator. Here $\psi(t)$ is the unit-energy basis signal (pulse shape) that represents these signals. It is also used in the impulse response of the matched filter. The noise $w(t)$ is zero-mean white Gaussian noise, and $T = 4$ s. According to the information given above,



- [5 pts]** Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [10 pts]** Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts]** Find the optimal decision threshold (λ) and calculate the average probability of bit error (P_b). Use a table of the Q function for the numerical value.

D2-04 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

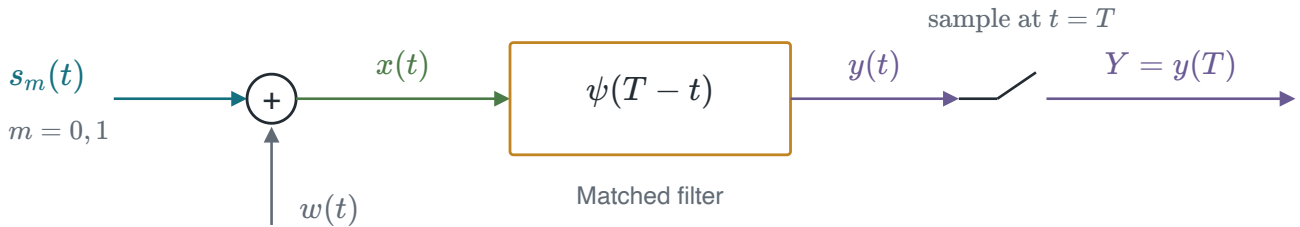
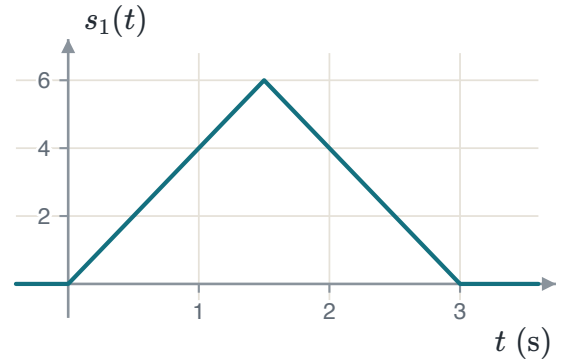
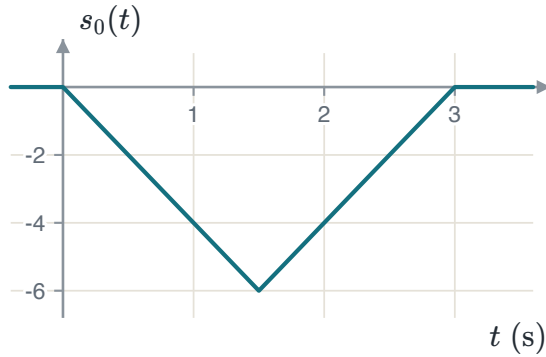
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 1$ W/Hz. Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below. These signals are passed through the following matched-filter type demodulator. Here $\psi(t)$ is the unit-energy basis signal (pulse shape) that represents these signals. It is also used in the impulse response of the matched filter. The noise $w(t)$ is zero-mean white Gaussian noise, and $T = 4$ s. According to the information given above,



- [5 pts]** Plot the impulse response of the matched filter, $h(t) = \psi(T-t)$.
- [10 pts]** Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts]** Find the optimal decision threshold (λ) and calculate the average probability of bit error (P_b). Use a table of the Q function for the numerical value.

D2-05 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

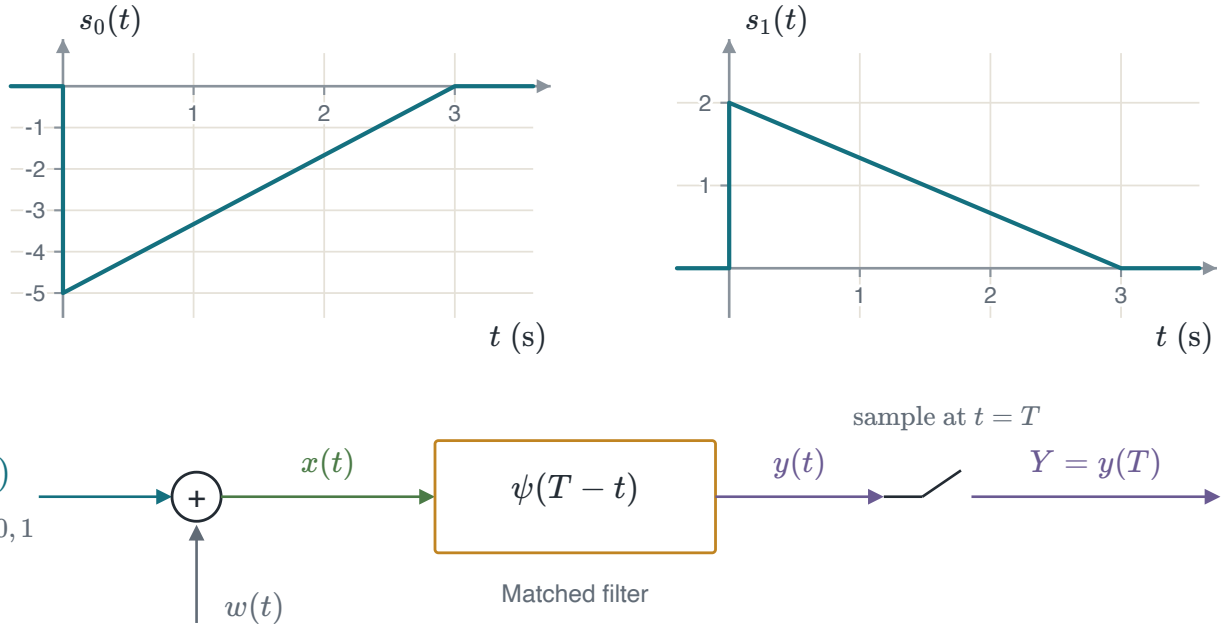
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 6.25$ W/Hz. Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below. These signals are passed through the following matched-filter type demodulator. Here $\psi(t)$ is the unit-energy basis signal (pulse shape) that represents these signals. It is also used in the impulse response of the matched filter. The noise $w(t)$ is zero-mean white Gaussian noise, and $T = 3$ s. According to the information given above,



- [5 pts]** Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [10 pts]** Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts]** Find the optimal decision threshold (λ) and calculate the average probability of bit error (P_b). Use a table of the Q function for the numerical value.

D2-06 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

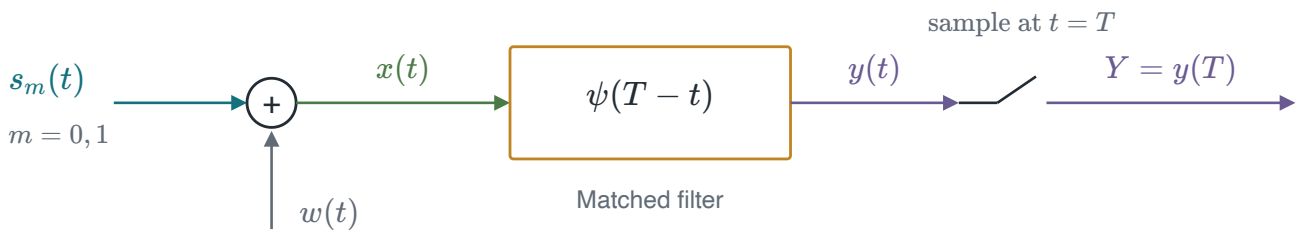
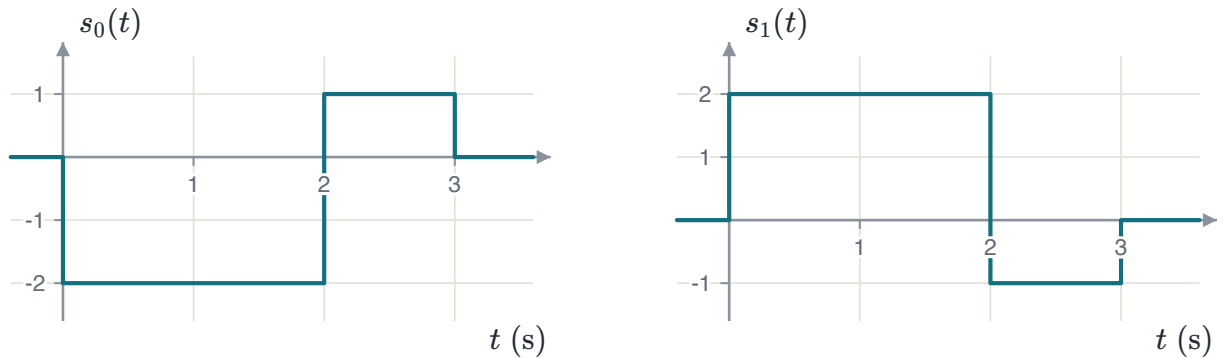
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 4$ W/Hz. Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below. These signals are passed through the following matched-filter type demodulator. Here $\psi(t)$ is the unit-energy basis signal (pulse shape) that represents these signals. It is also used in the impulse response of the matched filter. The noise $w(t)$ is zero-mean white Gaussian noise, and $T = 3$ s. According to the information given above,



- [5 pts]** Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [10 pts]** Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts]** Find the optimal decision threshold (λ) and calculate the average probability of bit error (P_b). Use a table of the Q function for the numerical value.

D2-07 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

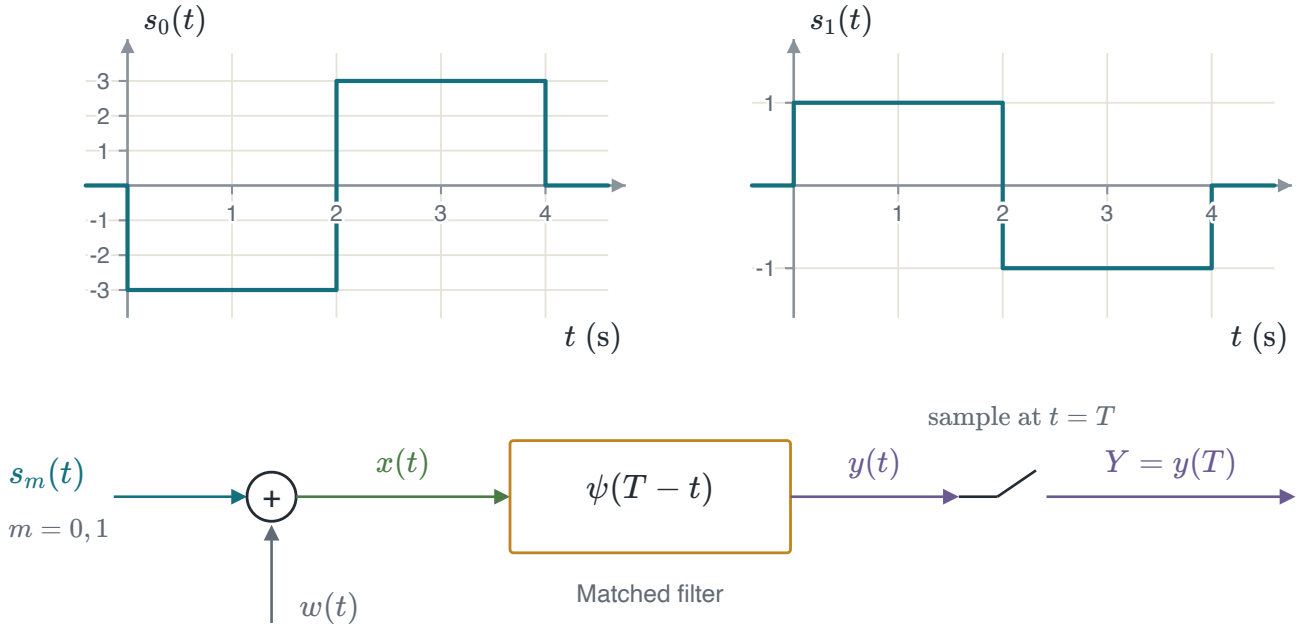
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 2.25$ W/Hz. Two messages are transmitted by the waveforms below: $s_0(t)$ for “0” and $s_1(t)$ for “1”. The message “0” is sent with probability 0.8 and “1” with probability 0.2. The signals pass through the matched-filter type demodulator shown, with the unit-energy basis signal $\psi(t)$ and $T = 3$ s. According to the information given above,



- [5 pts] Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [8 pts] Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [7 pts] Find the optimal decision threshold (λ).
- [5 pts] Calculate P_b at this threshold. Compare it with P_b for the threshold $\lambda = 0$.

D2-08 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

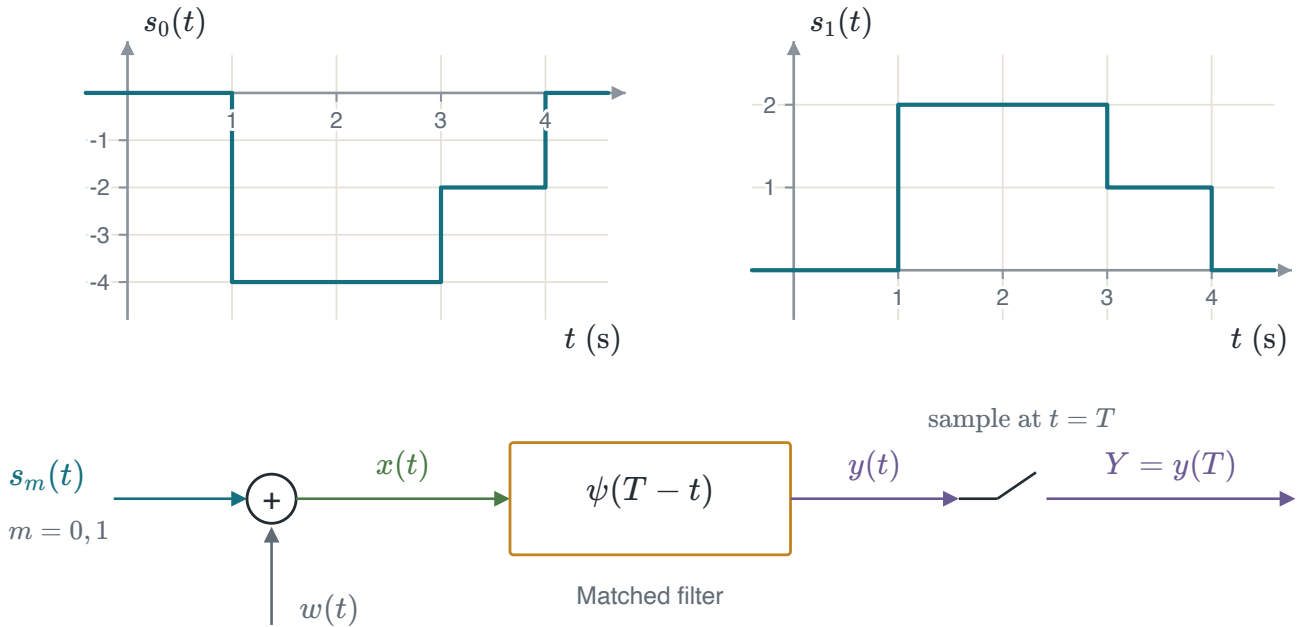
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 2.56$ W/Hz. Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below. These signals are passed through the following matched-filter type demodulator. Here $\psi(t)$ is the unit-energy basis signal (pulse shape) that represents these signals. It is also used in the impulse response of the matched filter. The noise $w(t)$ is zero-mean white Gaussian noise, and $T = 4$ s. According to the information given above,



- [6 pts] Show that one basis signal represents both waveforms. Give $\psi(t)$ and the coordinates of s_0 and s_1 .
- [5 pts] Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [7 pts] Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [7 pts] Find the optimal decision threshold (λ) and calculate P_b with a table of the Q function.

D2-09 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

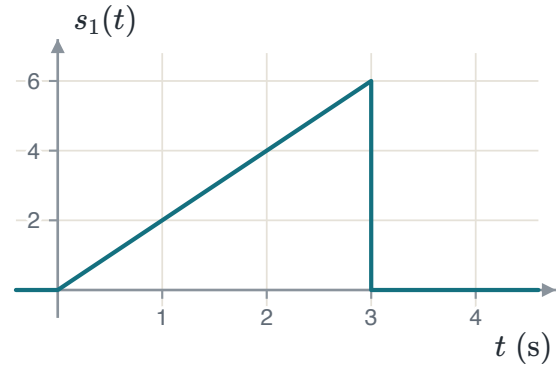
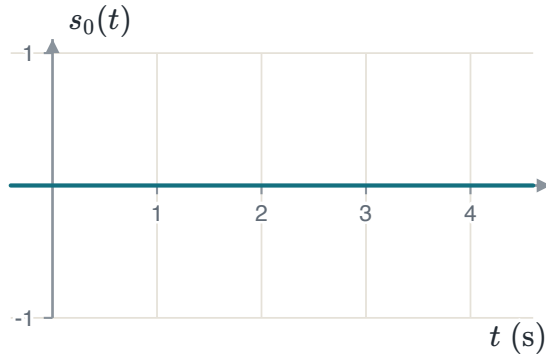
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 6.25$ W/Hz. Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below. These signals are passed through the following matched-filter type demodulator. Here $\psi(t)$ is the unit-energy basis signal (pulse shape) that represents these signals. It is also used in the impulse response of the matched filter. The noise $w(t)$ is zero-mean white Gaussian noise, and $T = 4$ s. According to the information given above,



- [5 pts]** Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [10 pts]** Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts]** Find the optimal decision threshold (λ) and calculate the average probability of bit error (P_b). Use a table of the Q function for the numerical value.

D2-10 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

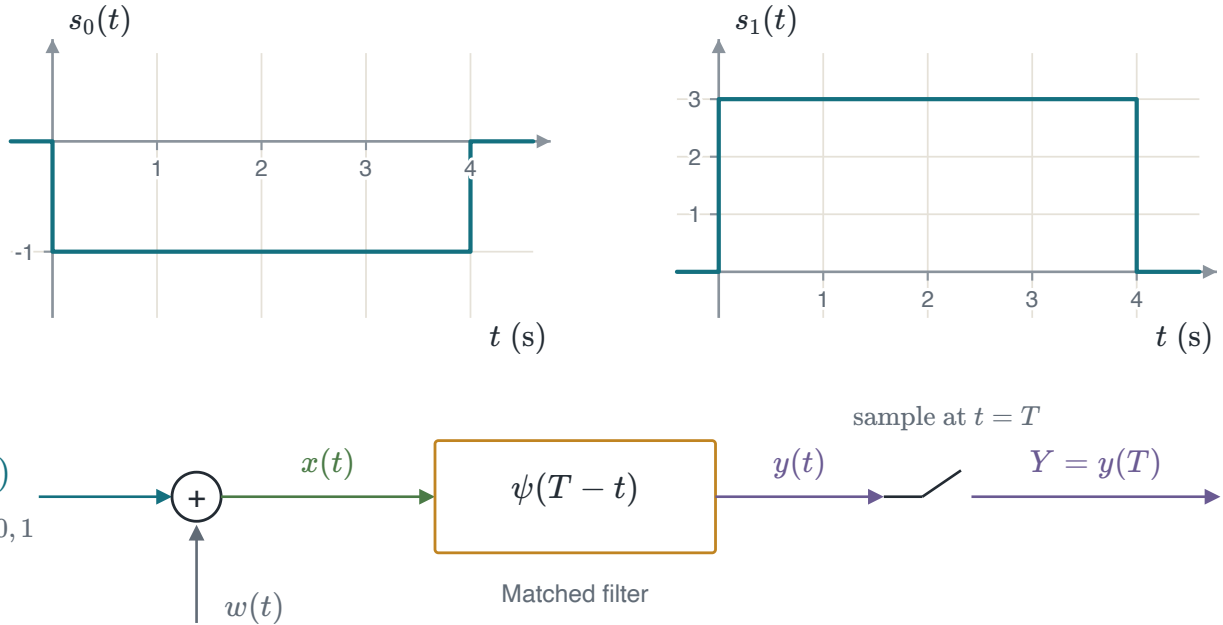
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 5.76$ W/Hz. Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below. These signals are passed through the following matched-filter type demodulator. Here $\psi(t)$ is the unit-energy basis signal (pulse shape) that represents these signals. It is also used in the impulse response of the matched filter. The noise $w(t)$ is zero-mean white Gaussian noise, and $T = 4$ s. According to the information given above,



- [5 pts]** Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [10 pts]** Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts]** Find the optimal decision threshold (λ) and calculate the average probability of bit error (P_b). Use a table of the Q function for the numerical value.

D2-11 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

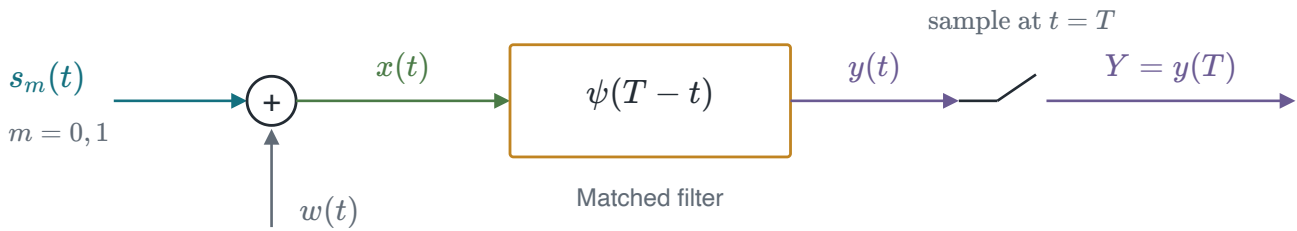
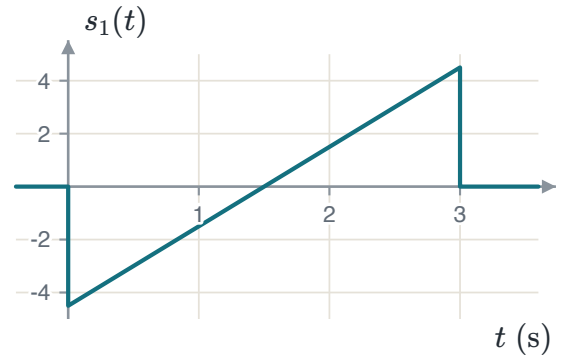
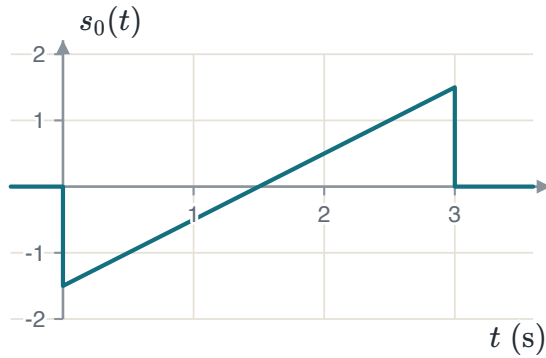
Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 6.25$ W/Hz. Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below. These signals are passed through the following matched-filter type demodulator. Here $\psi(t)$ is the unit-energy basis signal (pulse shape) that represents these signals. It is also used in the impulse response of the matched filter. The noise $w(t)$ is zero-mean white Gaussian noise, and $T = 4$ s. According to the information given above,



- [5 pts]** Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [6 pts]** Without noise, find and plot the filter output $y(t)$ when $s_1(t)$ is sent. Give its value at $t = T$.
- [7 pts]** Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [7 pts]** Find the optimal decision threshold (λ) and calculate P_b .

D2-12 · MATCHED-FILTER DEMODULATOR FOR TWO DRAWN WAVEFORMS

Consider an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 1$ W/Hz. The message “0” is sent by $s_0(t)$ with probability 0.6, and “1” by $s_1(t)$ with probability 0.4. These signals are passed through the matched-filter type demodulator shown, with the unit-energy basis signal $\psi(t)$ and $T = 3$ s. According to the information given above,



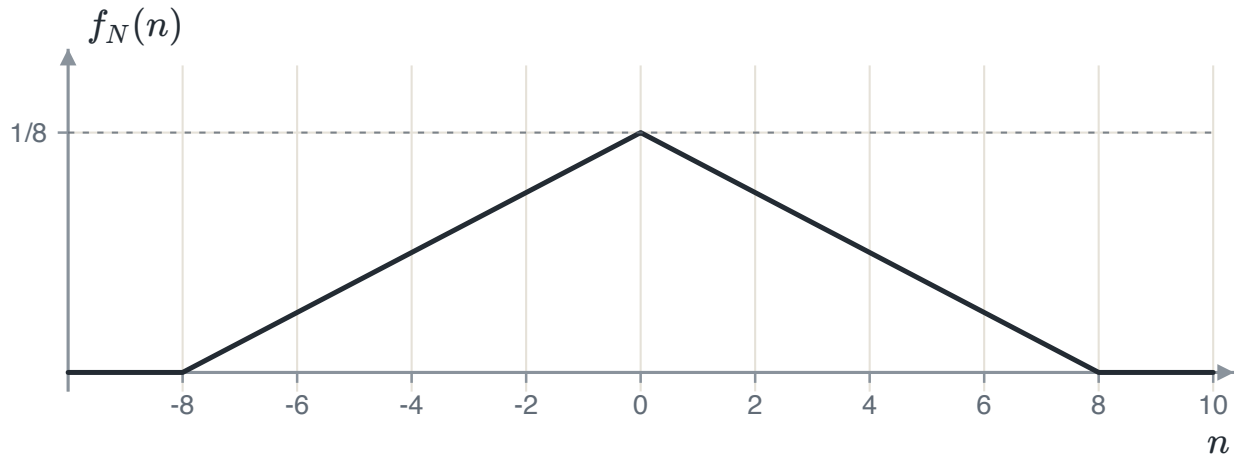
- [5 pts]** Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [10 pts]** Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts]** Find the optimal decision threshold (λ) and calculate the average probability of bit error (P_b). Use a table of the Q function for the numerical value.

D2-13 · BINARY DECISION IN NOISE THAT IS NOT GAUSSIAN

In a binary digital communication system, the received sample is $Y = N$ if “0” is sent and $Y = 5 + N$ if “1” is sent. The probability density function of the noise sample N is shown below:

$$f_N(n) = \begin{cases} \frac{1}{8} \left(1 - \frac{|n|}{8}\right), & -8 \leq n \leq 8 \\ 0, & \text{otherwise.} \end{cases}$$

Assume that “0” and “1” are transmitted with equal probabilities.



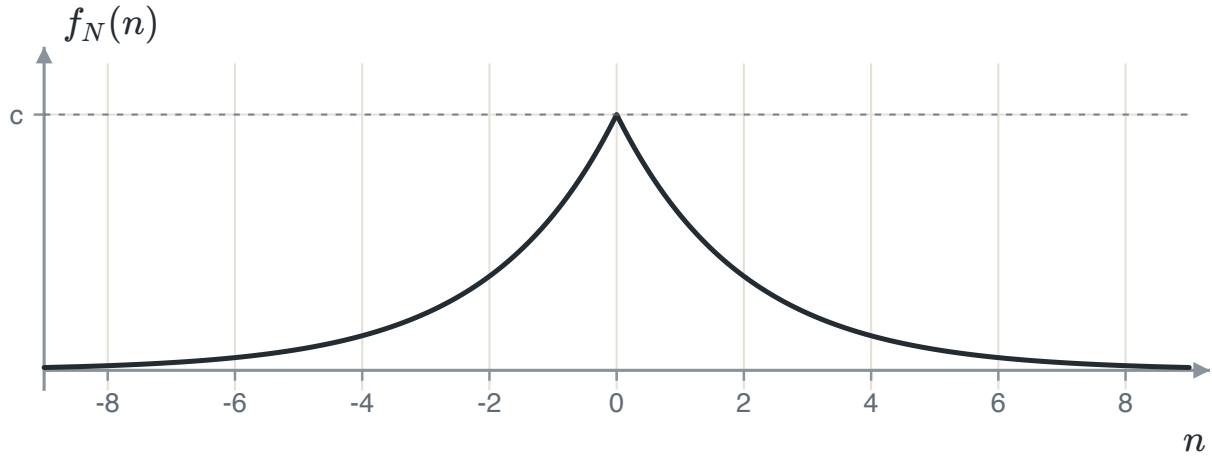
- a) [9 pts] Calculate the average probability of bit error (P_b) for the decision rule $Y \underset{0}{\overset{1}{\gtrless}} 3$, that is, decide “1” if $Y > 3$ and “0” if $Y < 3$.
- b) [8 pts] Determine the optimal decision threshold (λ).
- c) [8 pts] Calculate P_b for the λ value obtained in part (b).

D2-14 · BINARY DECISION IN NOISE THAT IS NOT GAUSSIAN

In a binary digital communication system, the received sample is $Y = -2 + N$ if “0” is sent and $Y = 2 + N$ if “1” is sent. The PDF of the noise sample N is

$$f_N(n) = c e^{-|n|/2}, \quad -\infty < n < \infty,$$

where c is a constant. Assume that “0” and “1” are transmitted with equal probabilities.



- a) [9 pts] Find c . Then calculate the average probability of bit error for the decision rule $Y \underset{0}{\overset{1}{\geq}} 1$, that is, decide “1” if $Y > 1$ and “0” if $Y < 1$.
- b) [8 pts] Determine the optimal decision threshold (λ).
- c) [8 pts] Calculate the average probability of bit error (P_b) for the λ value obtained in part (b).

D2-15 · BINARY DECISION IN NOISE THAT IS NOT GAUSSIAN

In a binary digital communication system, the received sample is denoted by Y . If “0” is transmitted, Y is exponentially distributed with mean 2. If “1” is transmitted, Y is exponentially distributed with mean 6. The input bits are equiprobable. An exponential PDF with mean μ is $f_Y(y) = \frac{1}{\mu} e^{-y/\mu}$ for $y \geq 0$, and 0 otherwise.

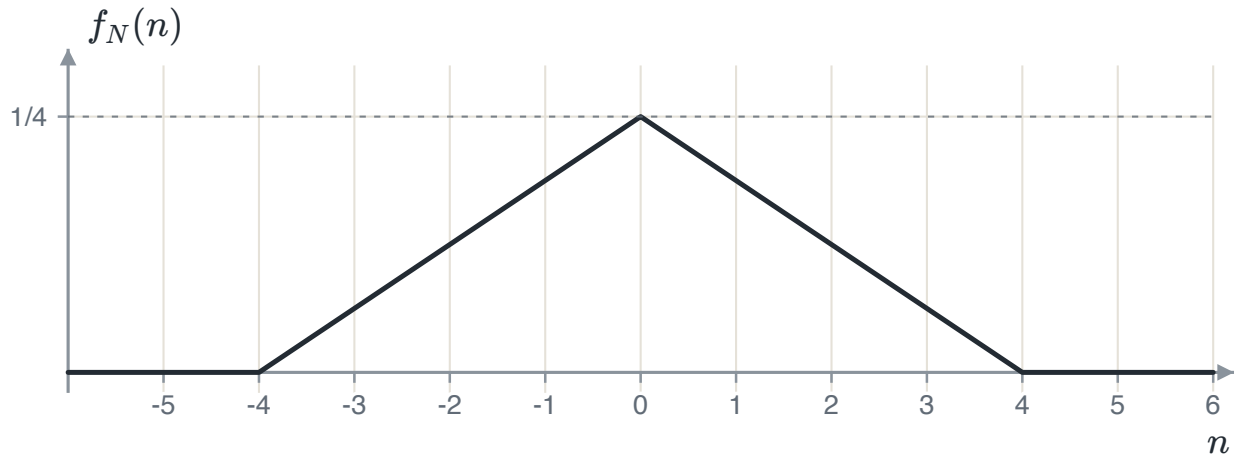
- a) [9 pts] Calculate the average probability of bit error for the decision rule $Y \underset{0}{\overset{1}{\geq}} 4$, that is, decide “1” if $Y > 4$ and “0” if $Y < 4$.
- b) [8 pts] Determine the optimal decision threshold (λ).
- c) [8 pts] Calculate the average probability of bit error (P_b) for the λ value obtained in part (b).

D2-16 · BINARY DECISION IN NOISE THAT IS NOT GAUSSIAN

In a binary digital communication system, the received sample is $Y = N$ if “0” is sent and $Y = 6 + N$ if “1” is sent. The noise sample N has the triangular PDF shown below:

$$f_N(n) = \begin{cases} \frac{1}{4} \left(1 - \frac{|n|}{4}\right), & -4 \leq n \leq 4 \\ 0, & \text{otherwise.} \end{cases}$$

Assume that “0” and “1” are transmitted with equal probabilities.



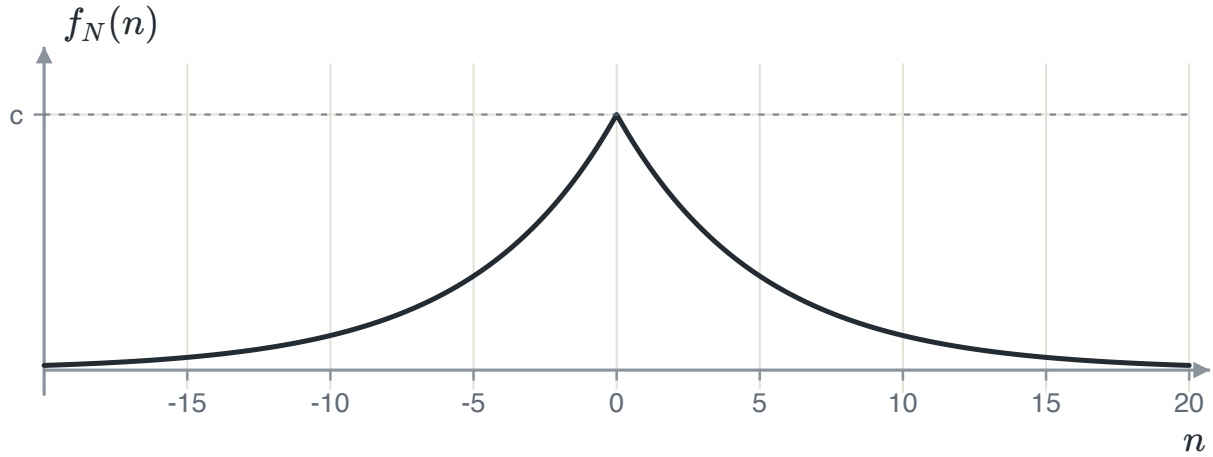
- a) [9 pts] Calculate the average probability of bit error for the decision rule $Y \underset{0}{\overset{1}{\geq}} 5$, that is, decide “1” if $Y > 5$ and “0” if $Y < 5$.
- b) [8 pts] Determine the optimal decision threshold (λ).
- c) [8 pts] Calculate the average probability of bit error (P_b) for the λ value obtained in part (b).

D2-17 · BINARY DECISION IN NOISE THAT IS NOT GAUSSIAN

In a binary digital communication system, the received sample is $Y = N$ if “0” is sent and $Y = 8 + N$ if “1” is sent. The PDF of the noise sample N is

$$f_N(n) = c e^{-|n|/5}, \quad -\infty < n < \infty,$$

where c is a constant. Assume that “0” and “1” are transmitted with equal probabilities.



- a) [9 pts] Find c . Then calculate the average probability of bit error for the decision rule $Y \underset{0}{\overset{1}{\geq}} 3$, that is, decide “1” if $Y > 3$ and “0” if $Y < 3$.
- b) [8 pts] Determine the optimal decision threshold (λ).
- c) [8 pts] Calculate the average probability of bit error (P_b) for the λ value obtained in part (b).

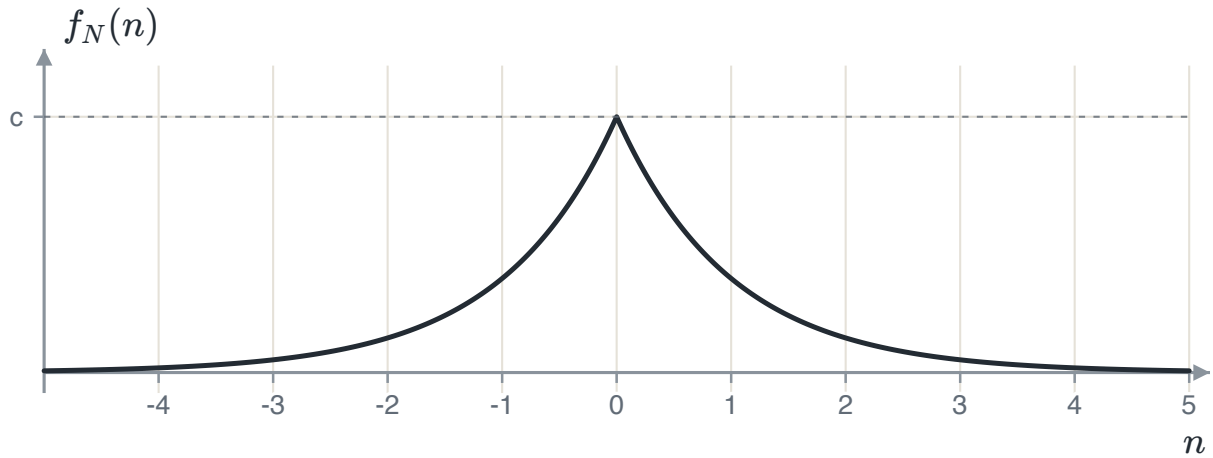
D2-18 · BINARY DECISION IN NOISE THAT IS NOT GAUSSIAN

In a binary digital communication system, the received sample is denoted by Y . If “0” is transmitted, Y is exponentially distributed with mean 3. If “1” is transmitted, Y is exponentially distributed with mean 12. The input bits are equiprobable. An exponential PDF with mean μ is $f_Y(y) = \frac{1}{\mu} e^{-y/\mu}$ for $y \geq 0$.

- a) [9 pts] Calculate the average probability of bit error for the decision rule $Y \underset{0}{\overset{1}{\geq}} 6$, that is, decide “1” if $Y > 6$ and “0” if $Y < 6$.
- b) [8 pts] Determine the optimal decision threshold (λ).
- c) [8 pts] Calculate the average probability of bit error (P_b) for the λ value obtained in part (b).

D2-19 · BINARY DECISION IN NOISE THAT IS NOT GAUSSIAN

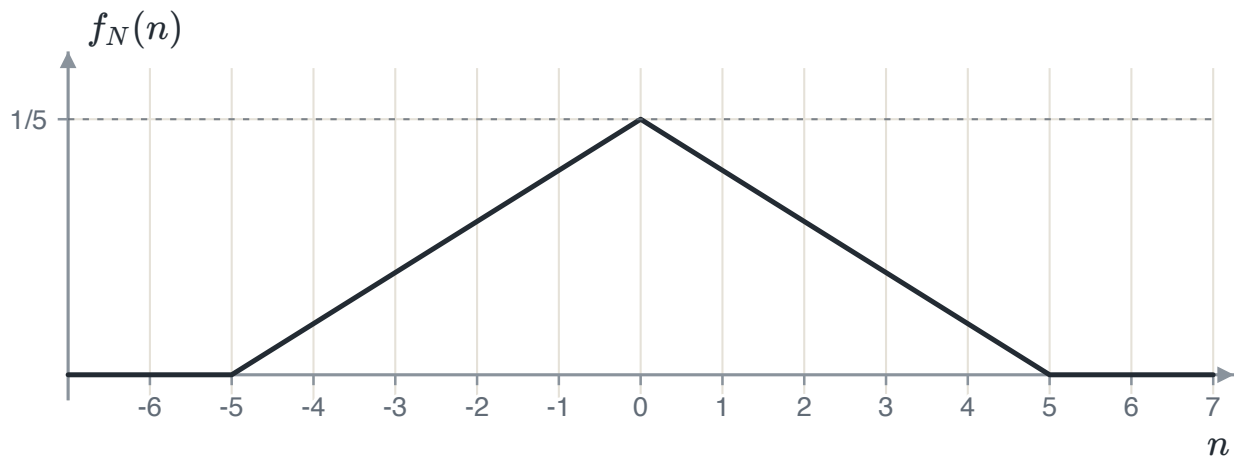
In a binary digital communication system, the received sample is $Y = -3 + N$ if “0” is sent and $Y = 3 + N$ if “1” is sent. The PDF of the noise sample N is $f_N(n) = c e^{-|n|}$ for all n , where c is a constant. The bit “0” is sent with probability 0.8 and “1” with probability 0.2.



- a) [9 pts] Find c . Then calculate P_b for the decision rule $Y \underset{0}{\overset{1}{\geq}} 0$.
- b) [8 pts] Determine the optimal decision threshold (λ) for these priors.
- c) [8 pts] Calculate P_b for the λ value obtained in part (b).

D2-20 · BINARY DECISION IN NOISE THAT IS NOT GAUSSIAN

In a binary digital communication system, the received sample is $Y = N$ if “0” is sent and $Y = 4 + N$ if “1” is sent. The noise N has the triangular PDF $f_N(n) = \frac{1}{5} \left(1 - \frac{|n|}{5}\right)$ for $|n| \leq 5$, and 0 otherwise. The bit “0” is sent with probability 0.6 and “1” with probability 0.4.



- a) [9 pts] Calculate P_b for the decision rule $Y \underset{0}{\overset{1}{\geq}} 2$, the midpoint between the two signal values.
- b) [8 pts] Determine the optimal decision threshold (λ) for these priors.
- c) [8 pts] Calculate P_b for the λ value obtained in part (b).

D2-21 · BINARY DECISION IN NOISE THAT IS NOT GAUSSIAN

In a binary digital communication system, the received sample is denoted by Y . If “0” is sent, Y is exponentially distributed with mean 1. If “1” is sent, Y is exponentially distributed with mean 3. The bit “0” is sent with probability 0.4 and “1” with probability 0.6.

- a) [9 pts] Calculate P_b for the decision rule $Y \underset{0}{\overset{1}{\geq}} 2$.
- b) [8 pts] Determine the optimal decision threshold (λ) for these priors.
- c) [8 pts] Calculate P_b for the λ value obtained in part (b).

D2-22 · BINARY PAM WITH PRIORS, UNEQUAL VARIANCES OR FOLDED OUTPUTS

In a binary pulse amplitude modulation (PAM) communication system, “0” and “1” bits are transmitted. At the output of the matched filter, we observe $Y = |N_0|$ when “0” is transmitted and $Y = |N_1|$ when “1” is transmitted. Here N_0 and N_1 are independent Gaussian random variables with $N_0 \sim \mathcal{N}(\mu = 0, \sigma^2 = 1)$ and $N_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 4)$. Assume that “0” and “1” are transmitted with equal probabilities.

- a) [8 pts] Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- b) [9 pts] Find the optimal decision threshold (λ).
- c) [8 pts] Calculate the average probability of bit error (P_b). Give the result as a numerical value, using a table of the Q function.

D2-23 · BINARY PAM WITH PRIORS, UNEQUAL VARIANCES OR FOLDED OUTPUTS

Assume that in a binary pulse amplitude modulation (PAM) communication system, “0” and “1” bits occur with probabilities 0.6 and 0.4. At the output of the matched filter, the output signal has a Gaussian distribution with mean 3 and variance 4 if “1” is transmitted. It has a Gaussian distribution with mean 0 and variance 1 if “0” is transmitted. According to this information,

- a) [8 pts] Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- b) [9 pts] Find the optimal decision rule. Show that it needs two thresholds, and give both.
- c) [8 pts] Calculate the average probability of bit error (P_b) as a numerical value, using a table of the Q function.

D2-24 · BINARY PAM WITH PRIORS, UNEQUAL VARIANCES OR FOLDED OUTPUTS

Assume that in a binary pulse amplitude modulation (PAM) communication system, “0” and “1” bits occur with probabilities 0.7 and 0.3. Consider a signal detector with the input

$$Y = \begin{cases} 5 + N, & \text{if 1 is sent} \\ N, & \text{if 0 is sent} \end{cases}$$

where N is a Gaussian random variable with mean 0 and variance 4. According to this information,

- a) [8 pts] Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- b) [8 pts] Find the optimal decision threshold (λ).
- c) [9 pts] Calculate the average probability of bit error (P_b) as a numerical value, using a table of the Q function.

D2-25 · BINARY PAM WITH PRIORS, UNEQUAL VARIANCES OR FOLDED OUTPUTS

In a binary pulse amplitude modulation (PAM) communication system, the matched-filter output is $Y = -2 + N$ when “0” is sent and $Y = 2 + N$ when “1” is sent. The noise N is Gaussian with mean 0 and variance 1. The bit “1” occurs three times as often as the bit “0”. According to this information,

- a) [6 pts] Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- b) [7 pts] Find the optimal decision threshold (λ).
- c) [7 pts] Calculate P_b as a numerical value, using a table of the Q function.
- d) [5 pts] A receiver ignores the priors and uses $\lambda = 0$. Calculate its P_b and compare.

D2-26 · BINARY PAM WITH PRIORS, UNEQUAL VARIANCES OR FOLDED OUTPUTS

In a binary pulse amplitude modulation (PAM) communication system, the matched-filter output is $Y = |N_0|$ when “0” is transmitted and $Y = |N_1|$ when “1” is transmitted. N_0 and N_1 are independent Gaussian random variables with $N_0 \sim \mathcal{N}(\mu = 0, \sigma^2 = 1)$ and $N_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 9)$. The bit “0” is sent with probability 0.6 and “1” with probability 0.4.

- a) [8 pts] Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$.
- b) [9 pts] Find the optimal decision threshold (λ) for these priors.
- c) [8 pts] Calculate P_b as a numerical value, using a table of the Q function.

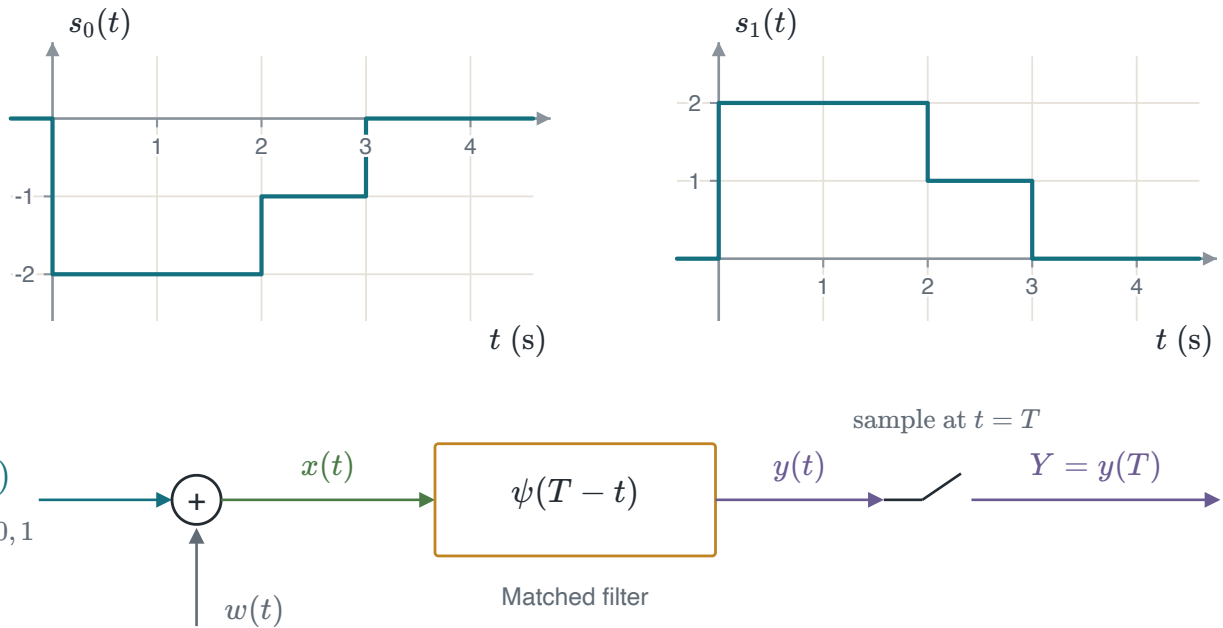
D2-27 · WORKING BACK FROM A TARGET ERROR OR THRESHOLD

In a binary pulse amplitude modulation (PAM) communication system, the matched-filter output is $Y = N$ when “0” is sent and $Y = 3 + N$ when “1” is sent. The noise N is Gaussian with mean 0 and variance 1. A designer wants the optimal threshold to sit at $\lambda = 2$. According to this information,

- a) [6 pts] Write $f_Y(y | 1)$ and $f_Y(y | 0)$. Find the optimal threshold and P_b for equal priors.
- b) [7 pts] Find the probability p_0 of “0” that makes $\lambda = 2$ the optimal threshold.
- c) [7 pts] Calculate P_b for this prior with $\lambda = 2$.
- d) [5 pts] With this prior, the receiver keeps the threshold of part (a). Calculate its P_b and compare.

D2-28 · WORKING BACK FROM A TARGET ERROR OR THRESHOLD

Two equiprobable messages, $s_0(t)$ for “0” and $s_1(t)$ for “1”, are transmitted by the waveforms below, with $T = 4$ s. They pass through the matched-filter type demodulator shown, with the unit-energy basis signal $\psi(t)$. The channel adds zero-mean white Gaussian noise of two-sided power spectral density $N_0/2$, which is not yet known. The link must reach $P_b \leq 10^{-3}$. According to the information given above,



- [5 pts] Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [8 pts] Write the optimal threshold and P_b as a function of $N_0/2$.
- [7 pts] Find the largest $N_0/2$ that meets $P_b \leq 10^{-3}$. Use $Q(3.09) = 1.00 \times 10^{-3}$.
- [5 pts] The message “0” is now sent as $s_0(t) = 0$, with the same $s_1(t)$. Find the largest $N_0/2$ again.

D2-29 · DETECTION WITH A BANDWIDTH OR WITH INTERFERENCE

A binary antipodal baseband link sends $R_b = 12$ kb/s with raised-cosine pulses of roll-off factor $\alpha = 0.25$. The matched-filter sample is $Y = \sqrt{E_b} + N$ for “1” and $Y = -\sqrt{E_b} + N$ for “0”, and the bits are equiprobable. The received power is $P = 3.6 \mu\text{W}$. N is Gaussian with mean 0 and variance $N_0/2$, where $N_0 = 5 \times 10^{-11}$ W/Hz. According to this information,

- [6 pts] Find the Nyquist bandwidth W , the frequency f_1 where the roll-off starts, and the transmission bandwidth B_T .
- [7 pts] Find E_b . Write $f_Y(y | 1)$ and $f_Y(y | 0)$, and give the optimal threshold.
- [6 pts] Calculate P_b with a table of the Q function.
- [6 pts] The channel offers $B_T = 9$ kHz. The rate is raised to fill it, with the same α and the same received power. Find the new R_b and P_b .

D2-30 · DETECTION WITH A BANDWIDTH OR WITH INTERFERENCE

In a binary PAM system, the pulse of each bit spreads into the next bit interval. The matched-filter sample of the current bit is $Y = a + c + N$. Here $a = 1.8$ for “1” and $a = -1.8$ for “0”. The term c comes from the previous bit: $c = 0.6$ if that bit was “1” and $c = -0.6$ if it was “0”. All bits are independent and equiprobable, and N is Gaussian with mean 0 and variance 0.36. According to this information,

- a) [8 pts] Determine the conditional PDFs, $f_Y(y | 1)$ and $f_Y(y | 0)$, averaged over the previous bit.
- b) [7 pts] Find the optimal decision threshold (λ).
- c) [10 pts] Calculate P_b with a table of the Q function. Compare it with P_b when the interference is removed ($c = 0$).

MODULE

3

Geometric Representation of Signals

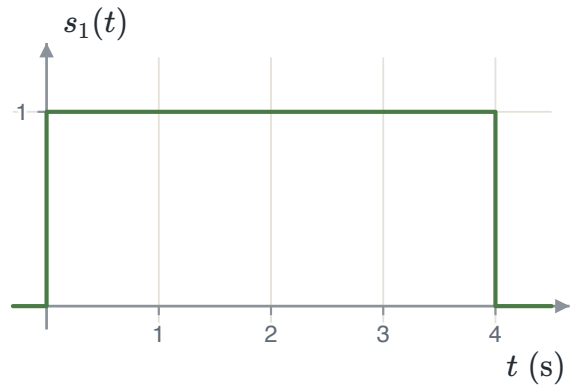
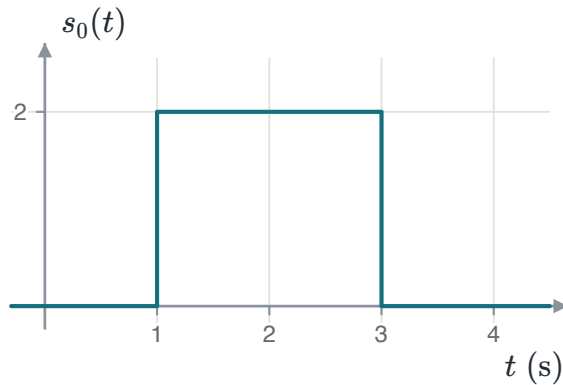
30 questions on geometric representation of signals. Write your reasoning in the space under each one.

D3-01 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

In an additive white Gaussian noise channel, two equiprobable messages are transmitted by the waveforms $s_0(t)$ and $s_1(t)$ below, with $T = 4$ s. Before a receiver is designed, the pair is written in geometric form. The correlation coefficient of the pair is

$$\rho_{01} = \frac{1}{\sqrt{E_0 E_1}} \int_0^T s_0(t) s_1(t) dt.$$

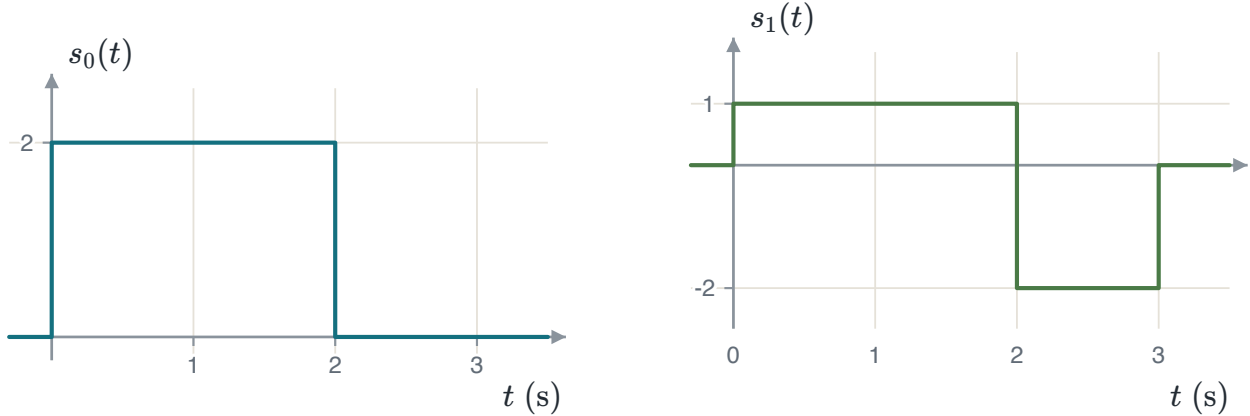
According to the information given above,



- [8 pts] Find an orthonormal basis for $\{s_0(t), s_1(t)\}$ by the Gram-Schmidt procedure, starting with $s_0(t)$. Plot the basis functions.
- [7 pts] Find the signal vectors $\mathbf{s}_0, \mathbf{s}_1$ and the energies E_0, E_1 .
- [5 pts] Calculate the distance $d_{01} = \|\mathbf{s}_0 - \mathbf{s}_1\|$ and the correlation coefficient ρ_{01} .
- [5 pts] Draw the signal constellation. How many dimensions does the signal set need?

D3-02 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

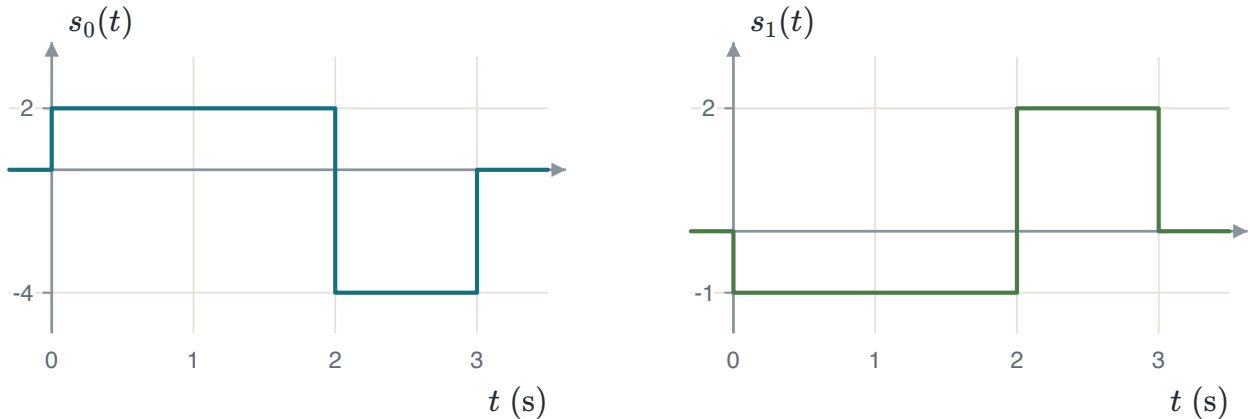
Two equiprobable messages are transmitted by the waveforms $s_0(t)$ and $s_1(t)$ shown below, with $T = 3$ s. The correlation coefficient of the pair is $\rho_{01} = \langle s_0, s_1 \rangle / \sqrt{E_0 E_1}$, where $\langle s_0, s_1 \rangle = \int_0^T s_0(t) s_1(t) dt$. According to the information given above,



- [8 pts] Use the Gram–Schmidt procedure, starting with $s_0(t)$, to find an orthonormal basis. Plot the basis functions.
- [6 pts] Find the signal vectors and the energies E_0 and E_1 .
- [6 pts] Calculate d_{01} and ρ_{01} .
- [5 pts] Draw the signal constellation and calculate the average symbol energy $E_{s,av}$.

D3-03 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

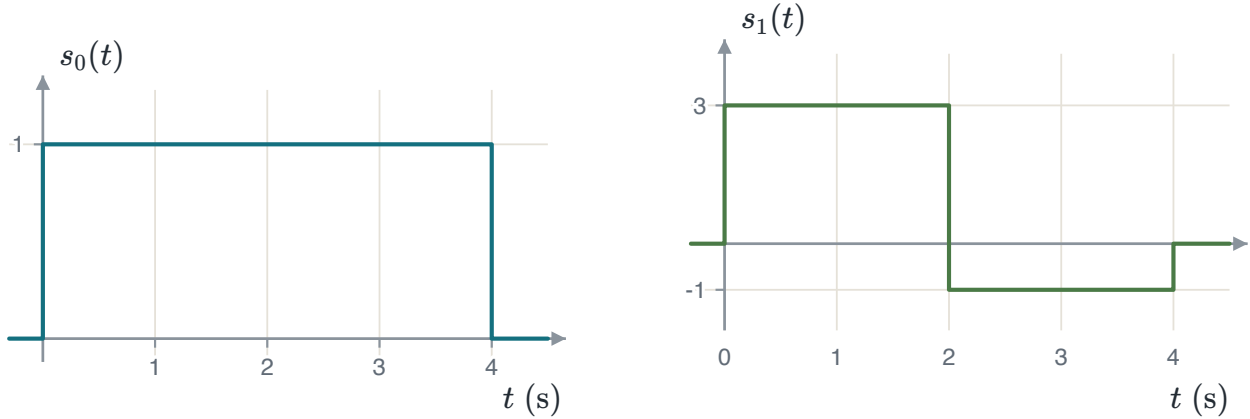
In an additive white Gaussian noise channel, two equiprobable messages are transmitted by the waveforms below, with $T = 3$ s. According to the information given above,



- [9 pts] Apply the Gram–Schmidt procedure, starting with $s_0(t)$. Show that it stops after one basis function, and plot that function.
- [6 pts] Find the coordinates of the two signals and their energies.
- [5 pts] Calculate the distance d_{01} and the correlation coefficient $\rho_{01} = \langle s_0, s_1 \rangle / \sqrt{E_0 E_1}$.
- [5 pts] Draw the signal constellation and give $E_{s,av}$.

D3-04 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

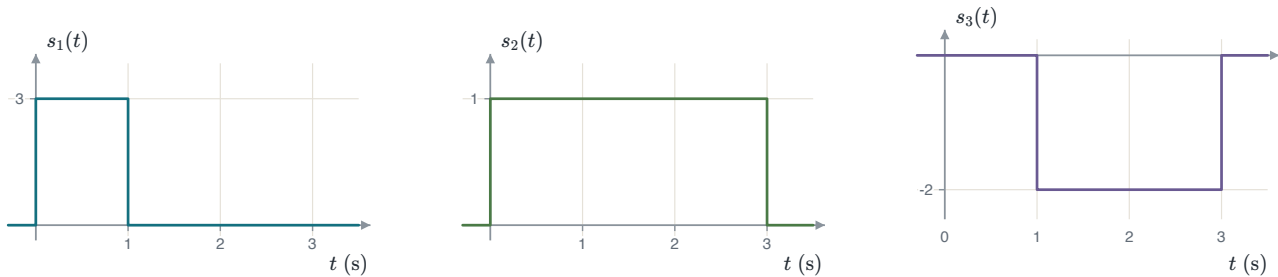
Two equiprobable messages are transmitted by the waveforms $s_0(t)$ and $s_1(t)$ shown below, with $T = 4$ s. According to the information given above,



- [8 pts] Find an orthonormal basis for the two signals by the Gram-Schmidt procedure, starting with $s_0(t)$. Plot the basis functions.
- [7 pts] Find s_0 and s_1 , and compute each energy both from the vector and from the waveform.
- [5 pts] Calculate d_{01} and $\rho_{01} = \langle s_0, s_1 \rangle / \sqrt{E_0 E_1}$.
- [5 pts] Draw the signal constellation and give $E_{s,av}$.

D3-05 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

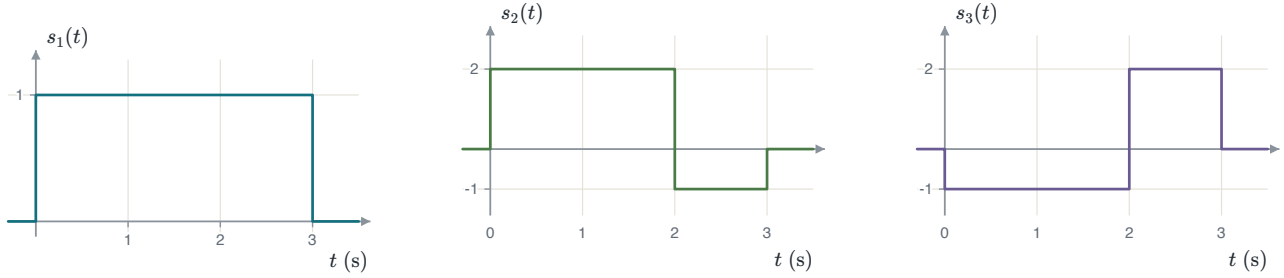
Three equiprobable messages are transmitted by the waveforms $s_1(t)$, $s_2(t)$ and $s_3(t)$ shown below, with $T = 3$ s. According to the information given above,



- [9 pts] Find an orthonormal basis by the Gram-Schmidt procedure, taking the signals in the order s_1, s_2, s_3 . Plot the basis functions.
- [6 pts] Find the three signal vectors and their energies.
- [6 pts] Calculate the three distances between the signals and give $d_{\min}$.
- [4 pts] Draw the signal constellation and state the number of dimensions.

D3-06 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

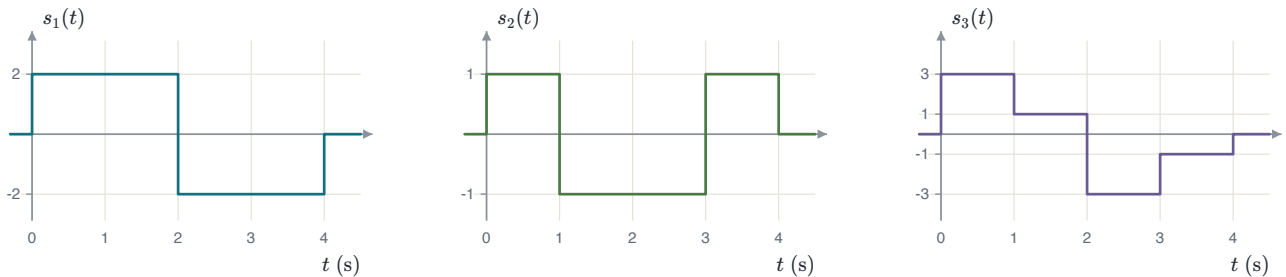
Three equiprobable symbols are represented by the waveforms shown below on $0 \leq t \leq T$, with $T = 3$ s. According to the information given above,



- [9 pts] Use the Gram–Schmidt procedure in the order s_1, s_2, s_3 to find an orthonormal basis. Plot the basis functions.
- [6 pts] Find the signal vectors.
- [5 pts] Calculate $E_{s,av}$ and the minimum distance $d_{\min}$.
- [5 pts] Draw the signal constellation, and write $s_3(t)$ as a combination of $s_1(t)$ and $s_2(t)$.

D3-07 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

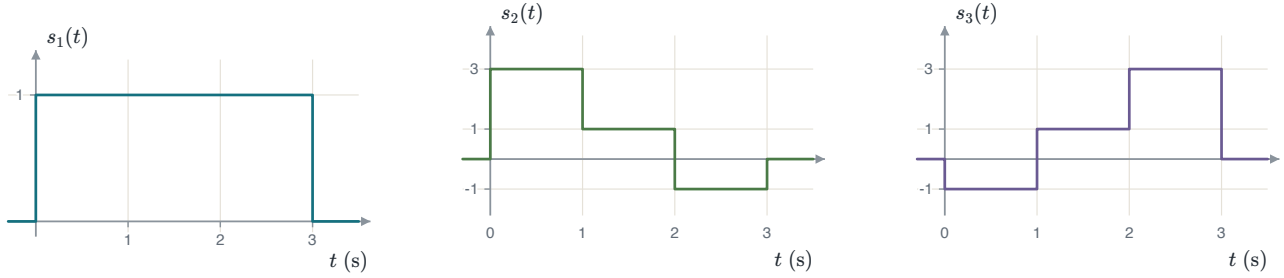
Three equiprobable messages are transmitted by the waveforms shown below, with $T = 4$ s. According to the information given above,



- [6 pts] Show that $s_1(t)$ and $s_2(t)$ are orthogonal, and give their correlation coefficient $\rho_{12} = \langle s_1, s_2 \rangle / \sqrt{E_1 E_2}$.
- [9 pts] Find an orthonormal basis for all three signals by the Gram–Schmidt procedure in the order s_1, s_2, s_3 , and the three signal vectors.
- [5 pts] Calculate the three distances and $d_{\min}$. Which pair is closest?
- [5 pts] Draw the signal constellation and calculate $E_{s,av}$.

D3-08 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

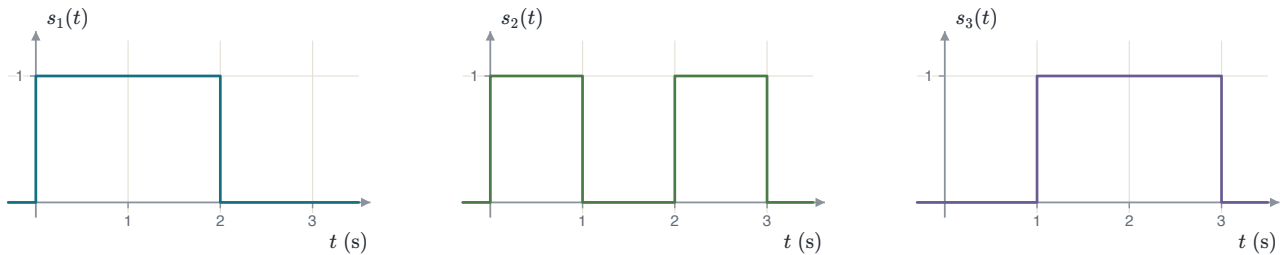
Three equiprobable symbols are transmitted with the staircase waveforms shown below on $0 \leq t \leq T$, with $T = 3$ s. According to the information given above,



- [9 pts] Find an orthonormal basis by the Gram–Schmidt procedure in the order s_1, s_2, s_3 . Plot the basis functions.
- [6 pts] Find the signal vectors and the symbol energies.
- [6 pts] Calculate all distances, $d_{\min}$, and the number of nearest neighbours of each symbol.
- [4 pts] Draw the constellation. The three points lie on one straight line. Explain why the set still needs two dimensions.

D3-09 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

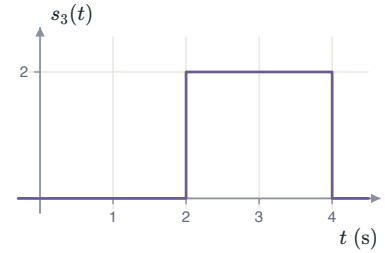
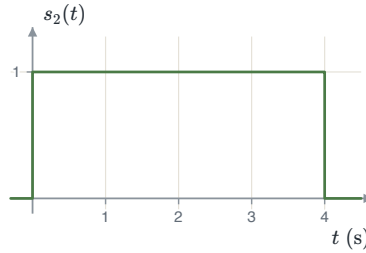
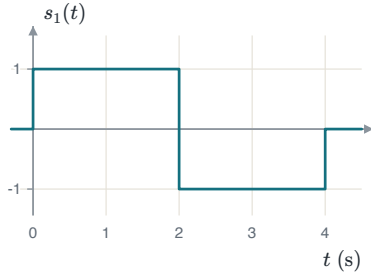
Three equiprobable messages are transmitted by the waveforms shown below, with $T = 3$ s. Each waveform is 1 on two of the three unit intervals and 0 on the third. According to the information given above,



- [10 pts] Apply the Gram–Schmidt procedure in the order s_1, s_2, s_3 and find an orthonormal basis. Plot the basis functions.
- [6 pts] Find the three signal vectors.
- [5 pts] Calculate the energies and the three distances.
- [4 pts] How many dimensions does the set need, and what shape do the three points form?

D3-10 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

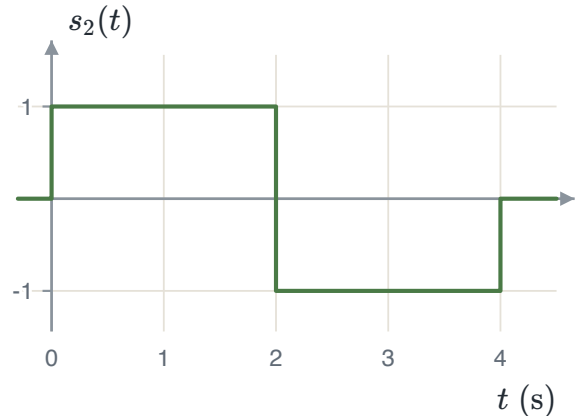
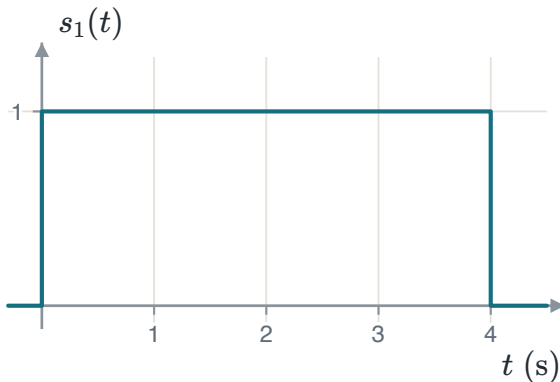
Three equiprobable messages are transmitted by the waveforms shown below, with $T = 4$ s. According to the information given above,



- [8 pts] Find an orthonormal basis by the Gram-Schmidt procedure in the order s_1, s_2, s_3 . Plot the basis functions.
- [6 pts] Find the signal vectors and the energies.
- [6 pts] Calculate ρ_{12} and ρ_{23} , where $\rho_{ij} = \langle s_i, s_j \rangle / \sqrt{E_i E_j}$, and all three distances.
- [5 pts] Draw the signal constellation, give $d_{\min}$, and write $s_3(t)$ in terms of $s_1(t)$ and $s_2(t)$.

D3-11 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

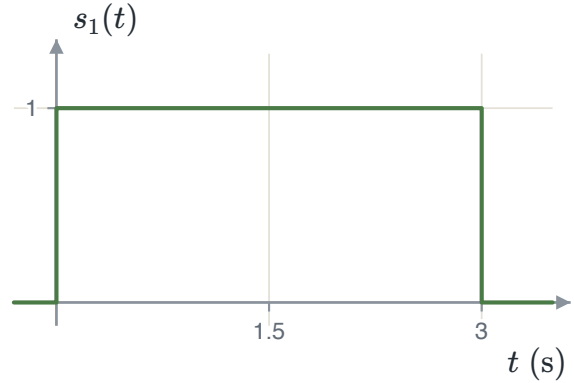
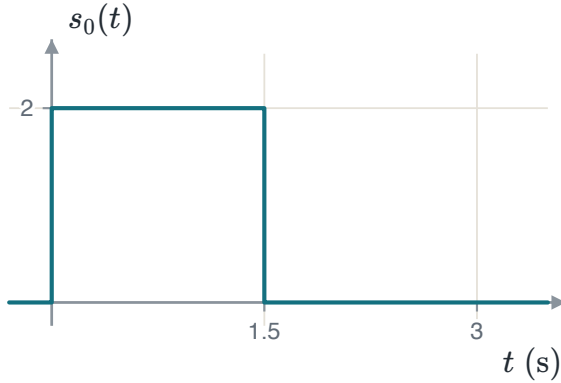
Four equiprobable symbols are transmitted by the waveforms $s_1(t)$ and $s_2(t)$ shown below and by their negatives, $s_3(t) = -s_1(t)$ and $s_4(t) = -s_2(t)$, with $T = 4$ s. According to the information given above,



- [8 pts] Find an orthonormal basis for the four signals by the Gram-Schmidt procedure, in the order s_1, s_2, s_3, s_4 .
- [6 pts] Find the four signal vectors.
- [6 pts] Calculate $E_{s,av}$, the minimum distance $d_{\min}$, and the number of nearest neighbours of each symbol.
- [5 pts] Draw the signal constellation and give the ratio $d_{\min}^2 / E_{s,av}$.

D3-12 · DRAWN WAVEFORMS AND GRAM-SCHMIDT

Two equiprobable messages are transmitted by the waveforms $s_0(t)$ and $s_1(t)$ shown below, with $T = 3$ s. Note that $s_0(t)$ changes value at $t = 1.5$ s. According to the information given above,



- [8 pts] Find an orthonormal basis by the Gram-Schmidt procedure, starting with $s_0(t)$. Plot the basis functions.
- [7 pts] Find the signal vectors and the energies.
- [5 pts] Calculate d_{01} and $\rho_{01} = \langle s_0, s_1 \rangle / \sqrt{E_0 E_1}$.
- [5 pts] Draw the signal constellation and calculate $d_{01}^2 / E_{s,av}$.

D3-13 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$s_k(t) = \sqrt{18} \cos \left(4000\pi t + \frac{\pi(2k-1)}{6} \right), \quad k \in \{1, \dots, 6\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- [8 pts] Find an orthonormal basis for the signal set and show that it is orthonormal.
- [7 pts] Find the signal vectors and draw the signal constellation.
- [5 pts] Calculate the symbol energies and the average symbol energy $E_{s,av}$.
- [5 pts] Determine the minimum distance $d_{\min}$ and the number of nearest neighbours of each symbol.

D3-14 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$\begin{aligned} s_1(t) &= 0, \\ s_2(t) &= 3\sqrt{2} \cos(6000\pi t), \\ s_3(t) &= 3\sqrt{2} \cos\left(6000\pi t + \frac{\pi}{2}\right), \\ s_4(t) &= 6 \cos\left(6000\pi t + \frac{\pi}{4}\right), \end{aligned} \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- [8 pts] Find an orthonormal basis for the signal set and show that it is orthonormal.
- [7 pts] Find the four signal vectors and draw the signal constellation.
- [5 pts] Calculate the symbol energies and $E_{s,av}$.
- [5 pts] Determine $d_{\min}$, the number of nearest neighbours of each symbol, and $d_{\min}^2/E_{s,av}$.

D3-15 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$s_k(t) = (2k - 5)\sqrt{2} \cos(5000\pi t), \quad k \in \{1, 2, 3, 4\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- [8 pts] Show that one basis function is enough for this set. Give it and show that it has unit energy.
- [7 pts] Find the signal coordinates and draw the signal constellation.
- [5 pts] Calculate the symbol energies and $E_{s,av}$.
- [5 pts] Determine $d_{\min}$, the number of nearest neighbours of each symbol, and their average.

D3-16 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$s_k(t) = 2 \cos\left(2000\pi t + \frac{k\pi}{4}\right), \quad k \in \{0, 1, \dots, 7\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- [8 pts] Find an orthonormal basis for the signal set and show that it is orthonormal.
- [7 pts] Find the signal vectors and draw the signal constellation.
- [5 pts] Calculate $E_{s,av}$ and the minimum distance $d_{\min}$.
- [5 pts] Give the number of nearest neighbours of each symbol and the ratio $d_{\min}^2/E_{s,av}$.

D3-17 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$s_k(t) = \sqrt{10} \cos\left(6000\pi t + \frac{(2k-1)\pi}{4}\right), \quad k \in \{1, 2, 3, 4\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- a) [8 pts] Find an orthonormal basis for the signal set and show that it is orthonormal.
- b) [7 pts] Find the signal vectors and draw the signal constellation.
- c) [5 pts] Calculate $E_{s,av}$.
- d) [5 pts] Determine $d_{\min}$, the number of nearest neighbours, and the largest distance between two symbols.

D3-18 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$s_{m,n}(t) = \sqrt{2} \left[(2m-5) \cos(4000\pi t) - (2n-5) \sin(4000\pi t) \right], \quad m, n \in \{1, 2, 3, 4\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- a) [8 pts] Find an orthonormal basis for the signal set and show that it is orthonormal.
- b) [6 pts] Find the signal vector of $s_{m,n}$ and draw the signal constellation.
- c) [6 pts] Calculate the symbol energies that occur and $E_{s,av}$.
- d) [5 pts] Determine $d_{\min}$ and the number of nearest neighbours of a corner, an edge and an inner symbol. Give the average.

D3-19 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$s_k(t) = \begin{cases} 2\sqrt{2} \cos\left(2000\pi t + \frac{(k-1)\pi}{2}\right), & k \in \{1, 2, 3, 4\}, \\ 4 \cos\left(2000\pi t + \frac{(2k-9)\pi}{4}\right), & k \in \{5, 6, 7, 8\}, \end{cases} \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- a) [8 pts] Find an orthonormal basis for the signal set and show that it is orthonormal.
- b) [7 pts] Find the signal vectors and draw the signal constellation.
- c) [5 pts] Calculate the symbol energies and the average symbol energy $E_{s,av}$.
- d) [5 pts] Determine the minimum distance $d_{\min}$ and the number of nearest neighbours of each symbol.

D3-20 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$s_1(t) = 0, \quad s_k(t) = 2\sqrt{2} \cos\left(2000\pi t + \frac{2\pi(k-2)}{3}\right), \quad k \in \{2, 3, 4\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- a) [8 pts] Find an orthonormal basis for the signal set and show that it is orthonormal.
- b) [7 pts] Find the four signal vectors and draw the signal constellation.
- c) [5 pts] Calculate the symbol energies and $E_{s,av}$.
- d) [5 pts] Determine $d_{\min}$, the number of nearest neighbours of each symbol, and $d_{\min}^2/E_{s,av}$.

D3-21 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$\begin{aligned} s_1(t) &= 3\sqrt{2} \cos(2000\pi t), & s_3(t) &= -s_1(t), \\ s_2(t) &= 3\sqrt{2} \cos(3000\pi t), & s_4(t) &= -s_2(t), \end{aligned} \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- a) [8 pts] Show that $\psi_1(t) = \sqrt{2} \cos(2000\pi t)$ and $\psi_2(t) = \sqrt{2} \cos(3000\pi t)$ form an orthonormal basis for the set.
- b) [7 pts] Find the signal vectors and draw the signal constellation.
- c) [5 pts] Calculate the symbol energies and $E_{s,av}$.
- d) [5 pts] Determine $d_{\min}$ and the number of nearest neighbours of each symbol.

D3-22 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$s_k(t) = (3k - 6)\sqrt{2} \cos(2500\pi t), \quad k \in \{0, 1, 2, 3, 4\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- a) [8 pts] Find a basis for the signal set, show that it has unit energy, and state the number of dimensions.
- b) [7 pts] Find the signal coordinates and draw the signal constellation.
- c) [5 pts] Calculate the symbol energies and $E_{s,av}$.
- d) [5 pts] Determine $d_{\min}$, the number of nearest neighbours of each symbol, and $d_{\min}^2/E_{s,av}$.

D3-23 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

$$s_{m,n}(t) = \sqrt{2} [(2m - 5) \cos(2000\pi t) - (2n - 3) \sin(2000\pi t)], \quad m \in \{1, 2, 3, 4\}, n \in \{1, 2\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- a) [8 pts] Find an orthonormal basis for the signal set and show that it is orthonormal.
- b) [6 pts] Find the signal vectors and draw the signal constellation.
- c) [5 pts] Calculate the symbol energies and $E_{s,av}$.
- d) [6 pts] Determine $d_{\min}$ and the number of nearest neighbours of each symbol. Give the average.

D3-24 · A CARRIER WAVEFORM FAMILY AND ITS CONSTELLATION

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

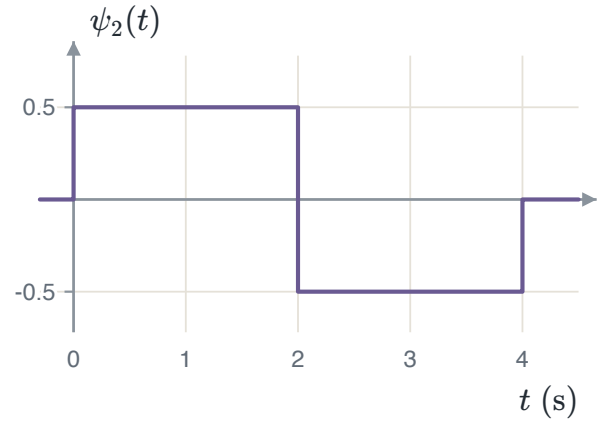
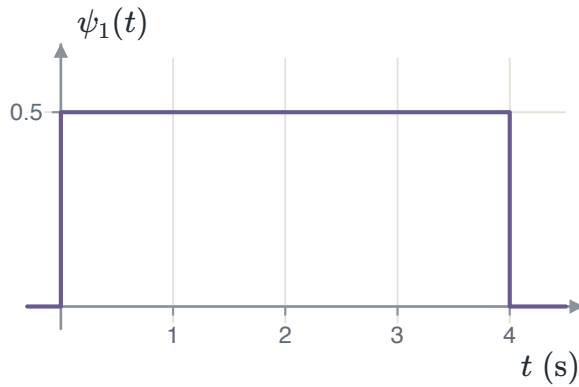
$$s_k(t) = k\sqrt{2} \cos\left(3000\pi t + \frac{k\pi}{2}\right), \quad k \in \{1, 2, 3, 4\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel. According to the information given above,

- a) [8 pts] Find an orthonormal basis for the signal set and show that it is orthonormal.
- b) [7 pts] Find the four signal vectors and draw the signal constellation.
- c) [4 pts] Calculate the symbol energies and $E_{s,av}$.
- d) [6 pts] Calculate all six distances, $d_{\min}$, and the number of nearest neighbours of each symbol.

D3-25 · FROM A CONSTELLATION BACK TO THE WAVEFORMS

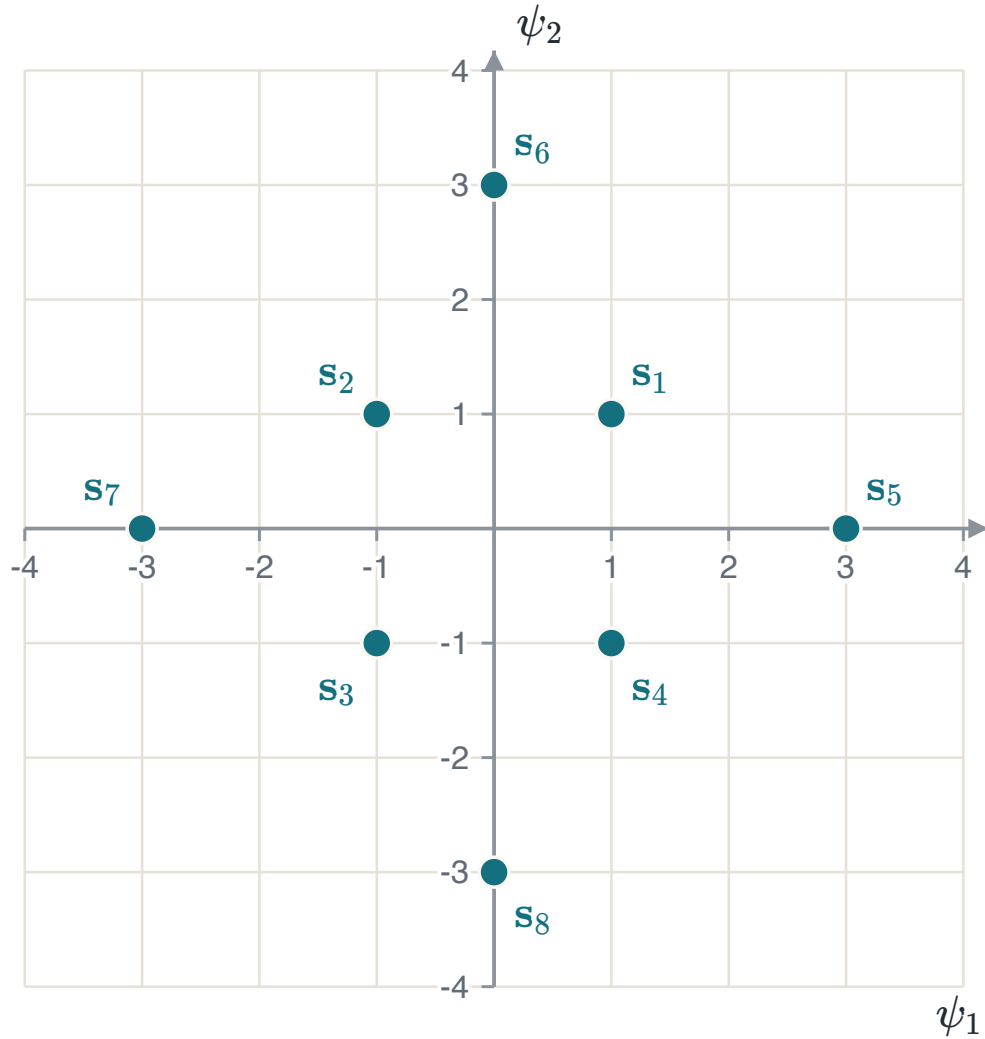
A signal set is built on the two basis functions $\psi_1(t)$ and $\psi_2(t)$ shown below, with $T = 4$ s. The four equiprobable symbols have the signal vectors $\mathbf{s}_1 = (2, 2)$, $\mathbf{s}_2 = (2, -2)$, $\mathbf{s}_3 = (-2, -2)$ and $\mathbf{s}_4 = (-2, 2)$. According to the information given above,



- [6 pts] Show that $\psi_1(t)$ and $\psi_2(t)$ are orthonormal.
- [8 pts] Write the four waveforms $s_1(t), \dots, s_4(t)$ and sketch them.
- [6 pts] Calculate the energy of each waveform from its sketch, and $E_{s,av}$.
- [5 pts] Determine $d_{\min}$ and the number of nearest neighbours. Which pairs of waveforms are orthogonal?

D3-26 · FROM A CONSTELLATION BACK TO THE WAVEFORMS

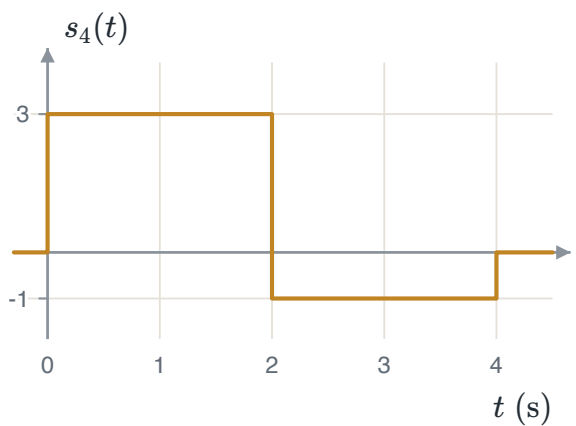
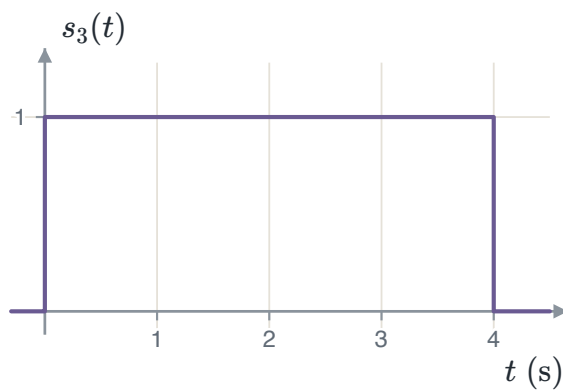
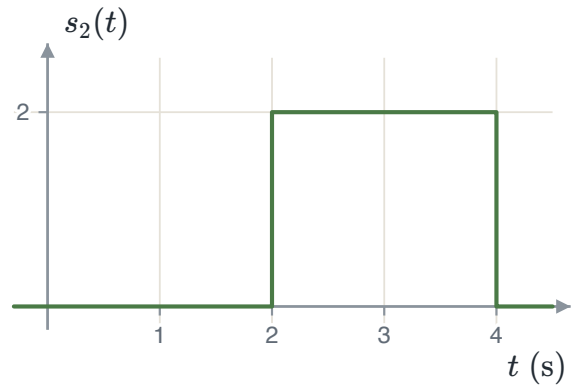
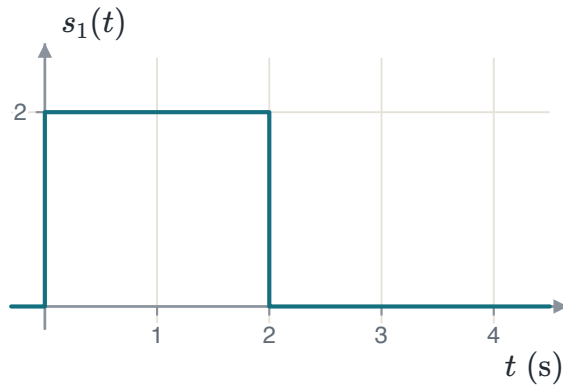
An M -ary modulation scheme uses the basis $\psi_1(t) = \sqrt{2} \cos(4000\pi t)$ and $\psi_2(t) = -\sqrt{2} \sin(4000\pi t)$ on $0 \leq t \leq 1$, which is orthonormal. Its eight equiprobable symbols form the constellation below. According to the information given above,



- [10 pts] Write each waveform in the form $s_k(t) = A_k \cos(4000\pi t + \theta_k)$.
- [5 pts] Calculate the symbol energies from the waveforms, and $E_{s,av}$.
- [5 pts] Determine $d_{\min}$ and the number of nearest neighbours of each symbol.
- [5 pts] An eight-phase set with the same $E_{s,av}$ is proposed. Which of the two sets has the larger $d_{\min}$?

D3-27 · HOW MANY DIMENSIONS A SET NEEDS

Four equiprobable messages are transmitted by the waveforms shown below, with $T = 4$ s. According to the information given above,



- [8 pts] Apply the Gram–Schmidt procedure in the order s_1, s_2, s_3, s_4 . How many basis functions does the set need?
- [6 pts] Find the four signal vectors and the energies.
- [6 pts] Determine $d_{\min}$ and the number of nearest neighbours of each symbol.
- [5 pts] Draw the constellation, give $E_{s,av}$, and write $s_3(t)$ and $s_4(t)$ in terms of $s_1(t)$ and $s_2(t)$.

D3-28 · HOW MANY DIMENSIONS A SET NEEDS

Consider an M -ary modulation scheme where the equiprobable symbols have the following waveforms:

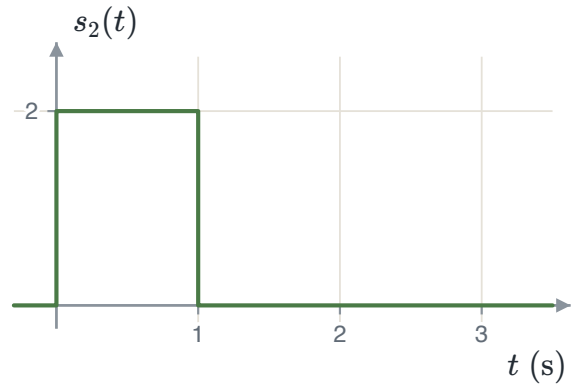
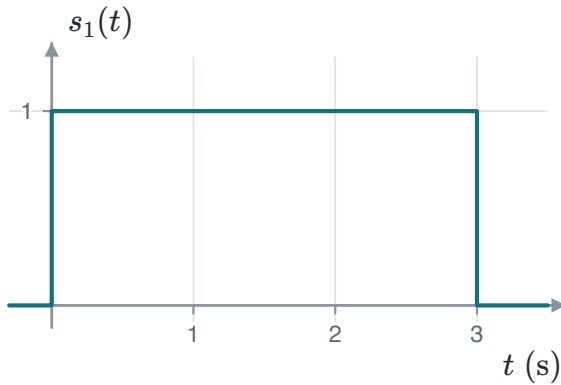
$$s_k(t) = 2(2k - 5) \cos\left(2000\pi t + \frac{\pi}{3}\right), \quad k \in \{1, 2, 3, 4\}, \quad 0 \leq t \leq 1.$$

The functions $\psi_1(t) = \sqrt{2} \cos(2000\pi t)$ and $\psi_2(t) = -\sqrt{2} \sin(2000\pi t)$ are orthonormal on this interval. According to the information given above,

- [8 pts] Find the signal vectors against ψ_1 and ψ_2 , and draw the constellation.
- [7 pts] Show that the set needs only one dimension. Give a single unit-energy basis function $\phi(t)$ and the coordinates against it.
- [5 pts] Calculate the symbol energies and $E_{s,av}$.
- [5 pts] Determine $d_{\min}$ and the number of nearest neighbours, in both descriptions.

D3-29 · COMPARING TWO DESCRIPTIONS OR TWO DESIGNS

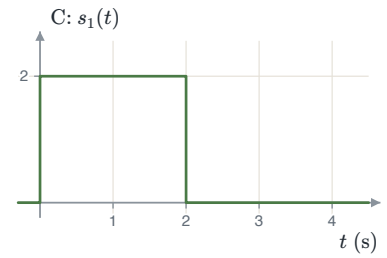
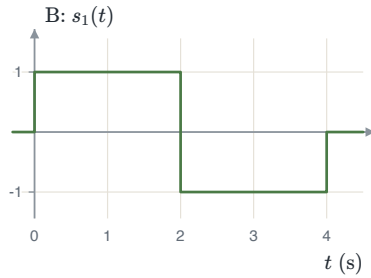
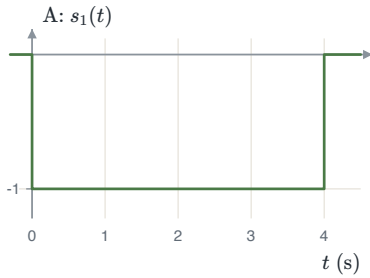
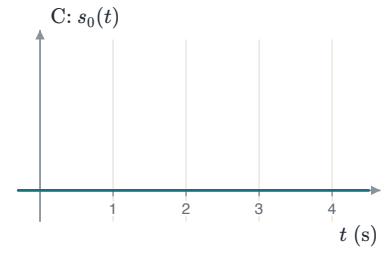
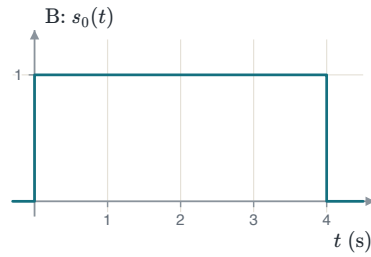
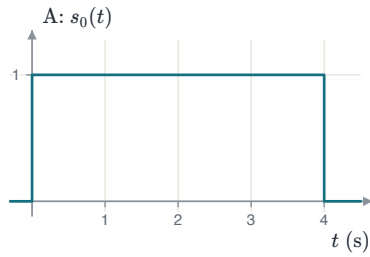
Two equiprobable messages are transmitted by the waveforms $s_1(t)$ and $s_2(t)$ shown below, with $T = 3$ s. According to the information given above,



- [8 pts] Apply the Gram–Schmidt procedure starting with $s_1(t)$. Give the basis $\{\psi_1, \psi_2\}$ and the two signal vectors.
- [8 pts] Apply it again starting with $s_2(t)$. Give the basis $\{\phi_1, \phi_2\}$ and the two signal vectors.
- [5 pts] In each basis, calculate E_1 , E_2 , d_{12} and $\rho_{12} = \langle \mathbf{s}_1, \mathbf{s}_2 \rangle / \sqrt{E_1 E_2}$.
- [4 pts] Draw both constellations. State what changed between them and what did not.

D3-30 · COMPARING TWO DESCRIPTIONS OR TWO DESIGNS

Three binary designs are proposed for two equiprobable messages on $[0, T]$, with $T = 4$ s. Each design uses the pair of waveforms in one column below. According to the information given above,



- [6 pts] Find an orthonormal basis for each design and give its number of dimensions.
- [6 pts] Find the signal vectors and $E_{s,av}$ of each design.
- [6 pts] Calculate the distance d_{01} of each design.
- [7 pts] Compare the designs by $d_{01}^2/E_{s,av}$. By how many decibels does the best design exceed the others?

MODULE

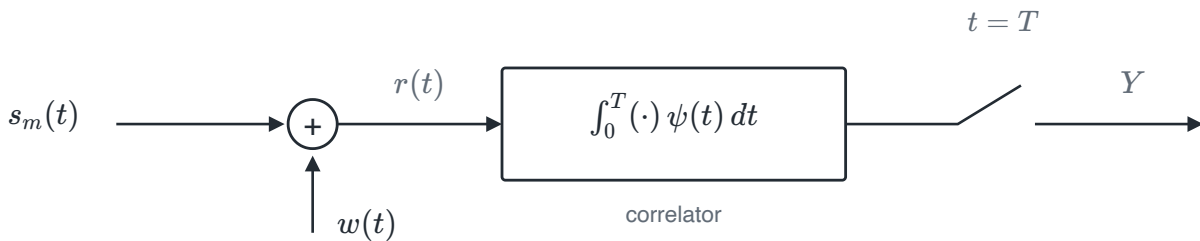
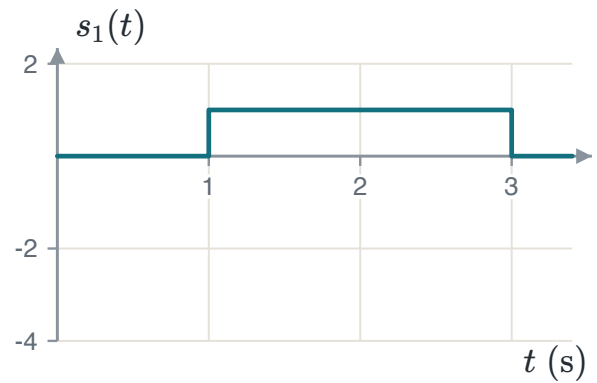
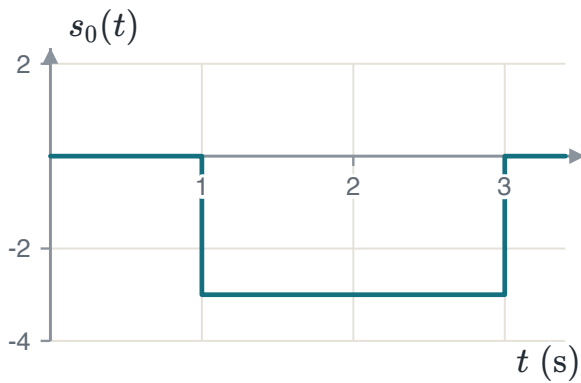
4

The Optimal Receiver in AWGN

30 questions on the optimal receiver in awgn. Write your reasoning in the space under each one.

D4-01 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

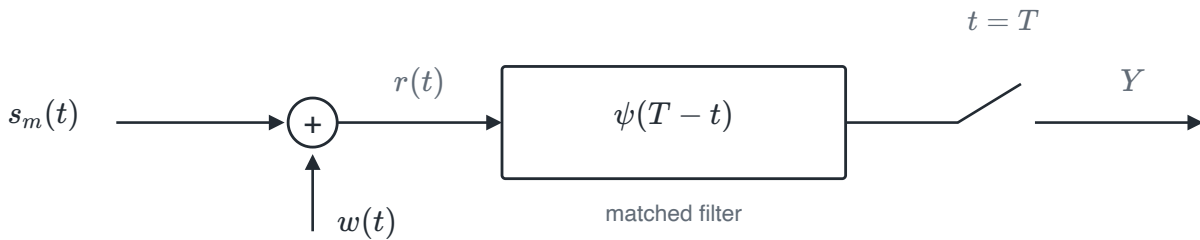
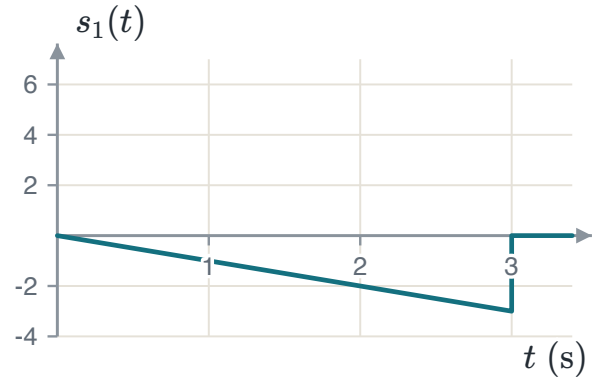
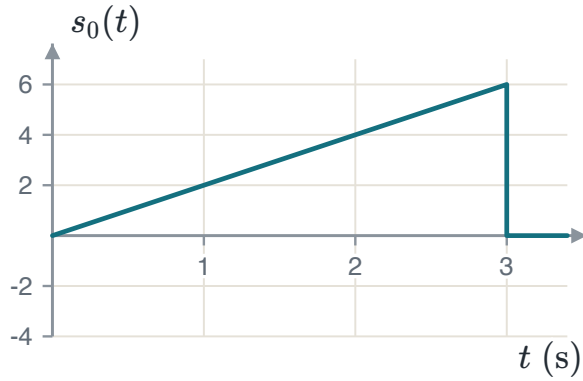
Two equiprobable messages, "0" and "1", are sent over an additive white Gaussian noise channel. The two-sided noise power spectral density is $N_0/2 = 2$ W/Hz. Message m is carried by the waveform $s_m(t)$ drawn below, with $T = 3$ s. These signals are passed through the following correlator-type demodulator. Here $\psi(t)$ is the unit-energy basis function of the two signals, and $w(t)$ is the white Gaussian noise. According to the information given above,



- [7 pts] Find $\psi(t)$ and the coordinates s_0 and s_1 of the two signals on it.
- [8 pts] Determine the conditional probability density functions $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts] Find the optimal decision threshold λ and calculate the average probability of bit error P_b .

D4-02 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

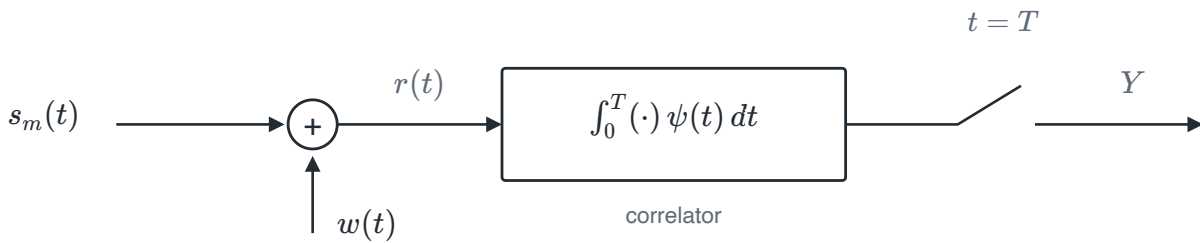
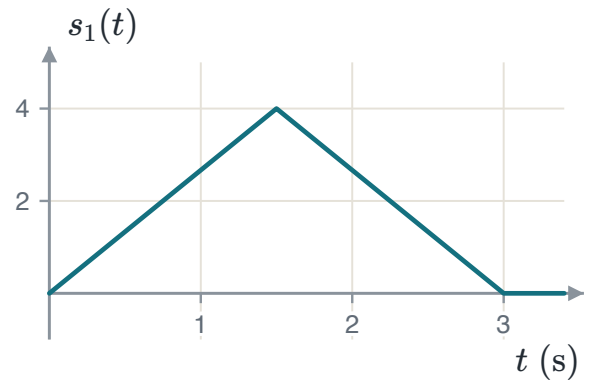
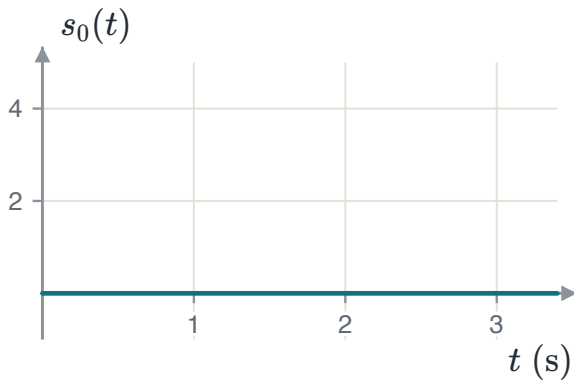
In an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 9$ W/Hz, two equiprobable messages, "0" and "1", are transmitted. Their waveforms $s_0(t)$ and $s_1(t)$ are drawn below. These signals are passed through the following matched-filter type demodulator, sampled at $t = T = 3$ s. Here $\psi(t)$ is the unit-energy basis signal that represents both waveforms and is also the impulse response of the matched filter. According to the information given above,



- [5 pts] Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [10 pts] Determine the conditional PDFs $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts] Find the optimal decision threshold λ and calculate the average probability of bit error P_b . Use a Q -function table.

D4-03 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

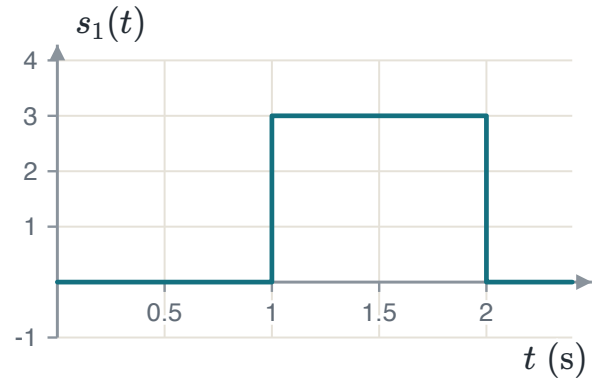
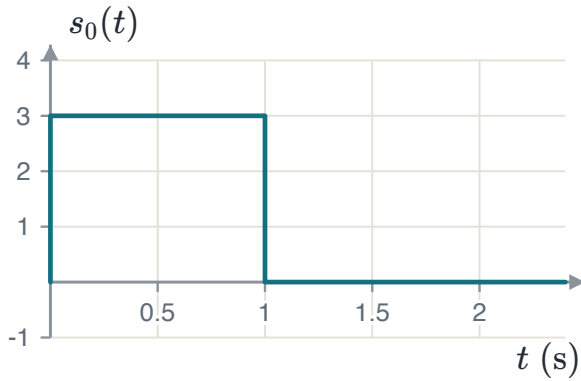
Two equiprobable messages are sent over an additive white Gaussian noise channel with two-sided noise power spectral density $N_0/2 = 0.64$ W/Hz. Message "0" is sent by $s_0(t) = 0$. Message "1" is sent by the triangular pulse $s_1(t)$ drawn below, with $T = 3$ s. The received signal passes through the following correlator-type demodulator. According to the information given above,



- [6 pts] Find the basis function $\psi(t)$, the coordinates s_0 and s_1 , and the energies E_0 and E_1 .
- [6 pts] Determine the conditional PDFs $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [7 pts] Find the optimal decision threshold λ and the average probability of bit error P_b .
- [6 pts] The receiver is rebuilt to compare $\int_0^T r(t)s_i(t) dt - E_i/2$ for $i = 0, 1$ and pick the larger. Show that it makes the same decisions as part (c).

D4-04 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

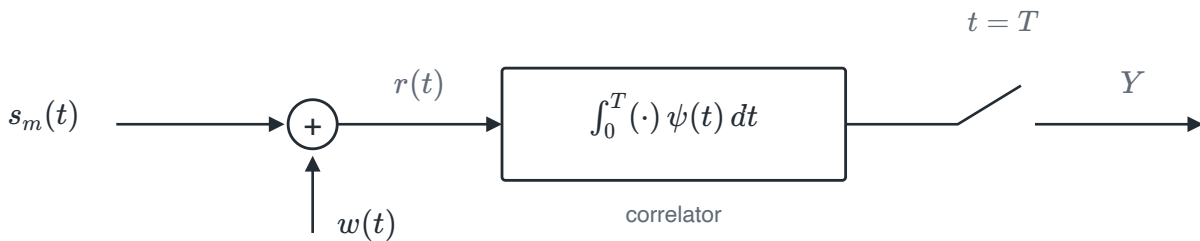
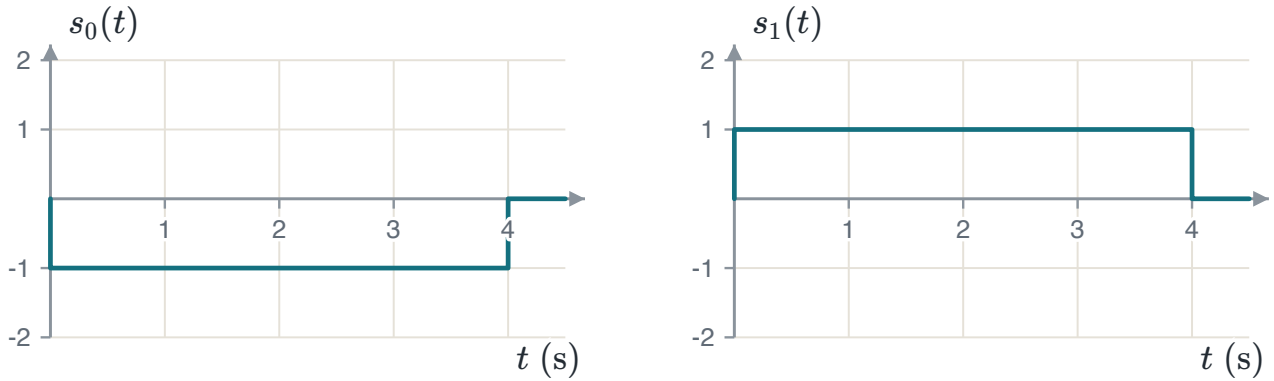
Two equiprobable messages, "0" and "1", are sent over an additive white Gaussian noise channel with $N_0/2 = 0.5$ W/Hz. The waveforms are $s_0(t) = 3$ for $0 \leq t < 1$ and $s_1(t) = 3$ for $1 \leq t < 2$, each zero elsewhere in $[0, 2]$. The two waveforms are not multiples of one pulse. The receiver therefore uses two correlators, with basis functions $\psi_1(t)$ and $\psi_2(t)$, sampled at $T = 2$ s. Their outputs form $\mathbf{Y} = (Y_1, Y_2)$. According to the information given above,



- [6 pts] Find an orthonormal basis $\psi_1(t)$, $\psi_2(t)$ and the signal vectors $\mathbf{s}_0$ and $\mathbf{s}_1$.
- [6 pts] Write the conditional joint PDFs $f_{\mathbf{Y}}(y_1, y_2 | 0)$ and $f_{\mathbf{Y}}(y_1, y_2 | 1)$.
- [6 pts] Find the ML decision rule and draw the two decision regions.
- [7 pts] Calculate the average probability of bit error P_b .

D4-05 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

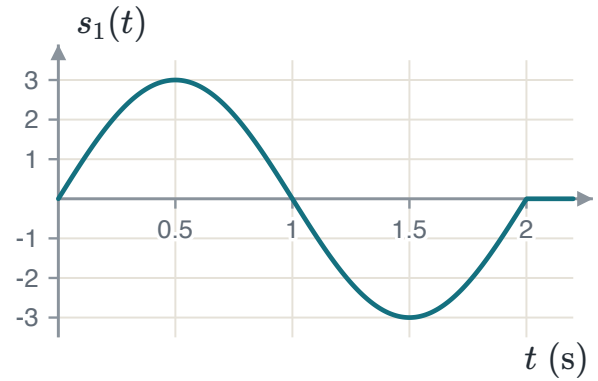
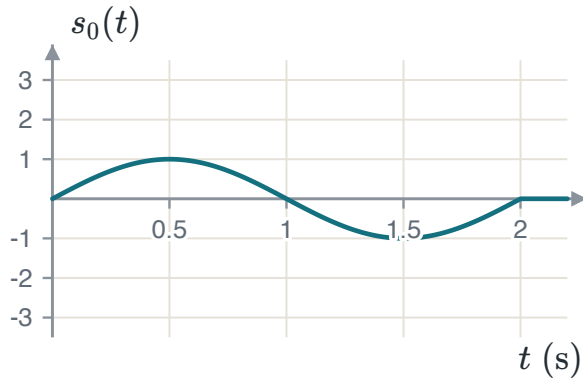
Two messages are sent over an additive white Gaussian noise channel with $N_0/2 = 1$ W/Hz. Message "0" is sent with probability 0.3 and message "1" with probability 0.7. The waveforms $s_0(t)$ and $s_1(t)$ are drawn below, with $T = 4$ s. The received signal passes through the following correlator-type demodulator. According to the information given above,



- [5 pts] Find $\psi(t)$ and the coordinates s_0 and s_1 .
- [6 pts] Determine the conditional PDFs $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [7 pts] Find the optimal (MAP) decision threshold λ .
- [7 pts] Calculate P_b for this threshold, and compare it with the receiver that uses the ML threshold.

D4-06 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

Two equiprobable messages, "0" and "1", are sent over an additive white Gaussian noise channel with $N_0/2 = 0.36$ W/Hz. The waveforms are $s_0(t) = \sin(\pi t)$ and $s_1(t) = 3 \sin(\pi t)$ for $0 \leq t \leq 2$ s, drawn below. The received signal passes through a correlator-type demodulator with $T = 2$ s. According to the information given above,



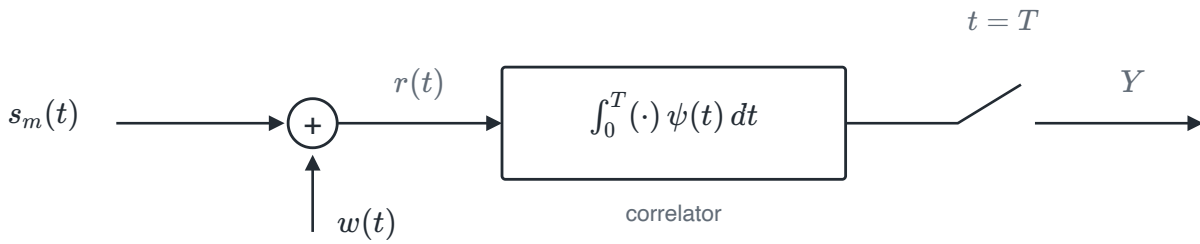
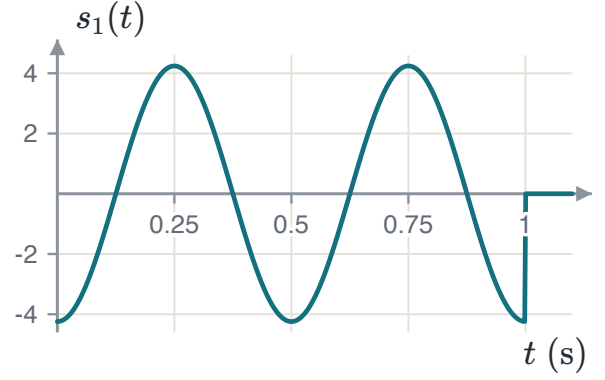
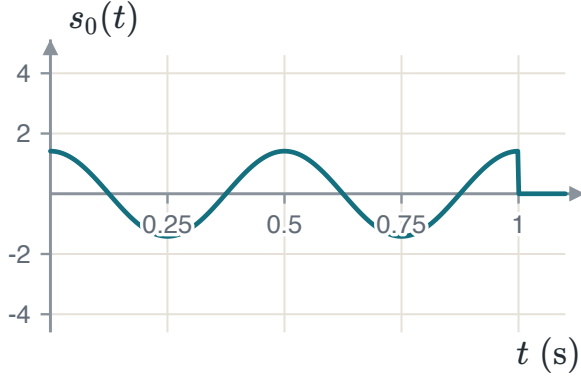
- [6 pts] Find $\psi(t)$, the coordinates s_0 and s_1 , and the energies E_0 and E_1 .
- [6 pts] Determine the conditional PDFs $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [7 pts] Find the optimal decision threshold λ and calculate P_b .
- [6 pts] A designer builds a receiver that picks the larger of $\int_0^T r(t)s_i(t) dt$ and leaves out $-E_i/2$. Find its P_b .

D4-07 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

Two equiprobable messages are sent over an additive white Gaussian noise channel with $N_0/2 = 0.49$ W/Hz. They are transmitted by

$$s_0(t) = \sqrt{2} \cos(4\pi t), \quad s_1(t) = -3\sqrt{2} \cos(4\pi t), \quad 0 \leq t \leq 1 \text{ s.}$$

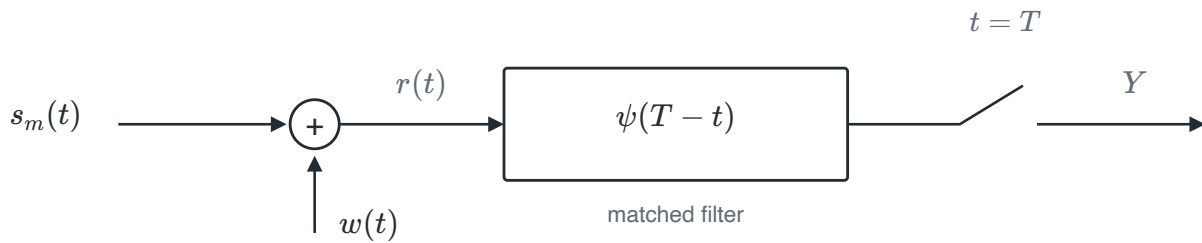
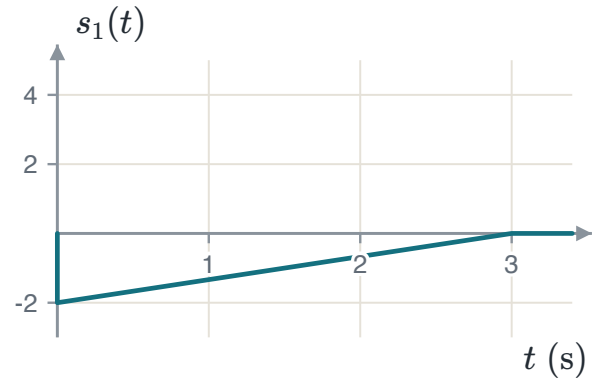
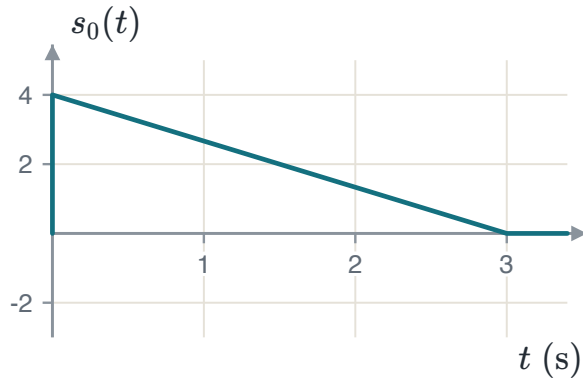
These signals are passed through a correlator-type demodulator with $T = 1$ s, drawn below. According to the information given above,



- [7 pts] Find the unit-energy basis $\psi(t)$ and the coordinates s_0 and s_1 .
- [8 pts] Determine the conditional PDFs $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts] Find the optimal decision threshold λ and calculate the average probability of bit error P_b .

D4-08 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

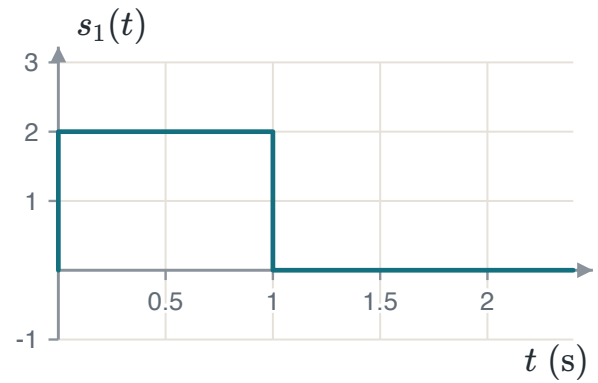
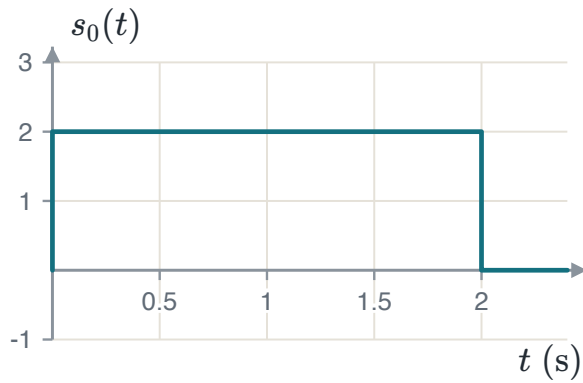
In an additive white Gaussian noise channel with $N_0/2 = 4$ W/Hz, "0" and "1" bits occur with probabilities 0.6 and 0.4. They are transmitted by the waveforms $s_0(t)$ and $s_1(t)$ drawn below. These signals are passed through the following matched-filter type demodulator, sampled at $t = T = 3$ s. According to the information given above,



- [5 pts] Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [8 pts] Determine the conditional PDFs $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [12 pts] Find the optimal decision threshold λ and calculate the average probability of bit error P_b .

D4-09 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

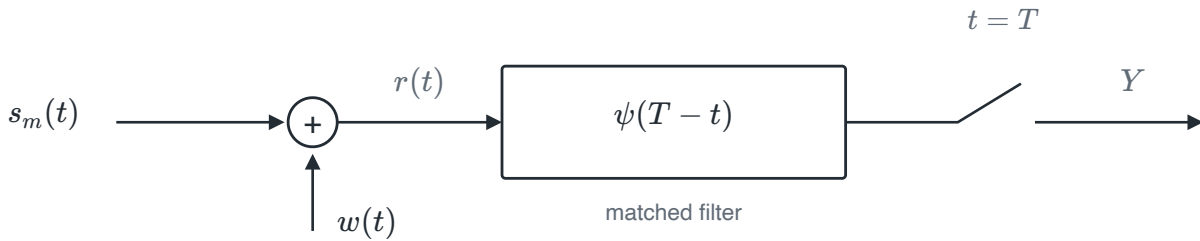
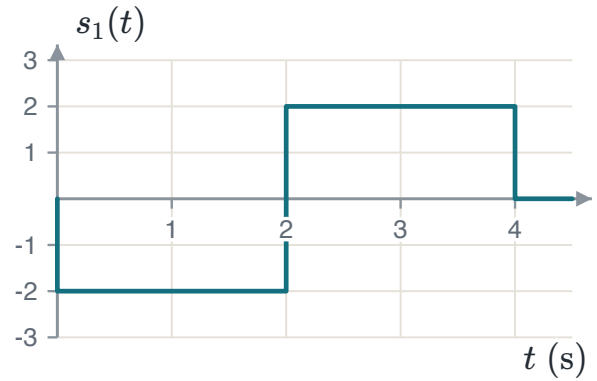
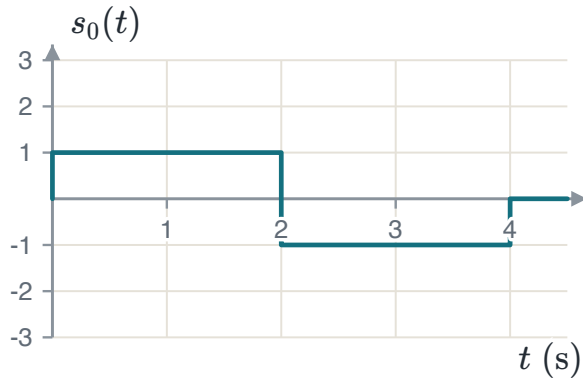
Two equiprobable messages are sent over an additive white Gaussian noise channel with $N_0/2 = 0.2$ W/Hz. The waveforms are $s_0(t) = 2$ for $0 \leq t < 2$, and $s_1(t) = 2$ for $0 \leq t < 1$ and $s_1(t) = 0$ for $1 \leq t < 2$. They are drawn below, with $T = 2$ s. The receiver uses two correlators, with the basis functions of part (a). According to the information given above,



- [7 pts] Apply Gram–Schmidt, starting from $s_1(t)$, to find $\psi_1(t)$, $\psi_2(t)$ and the vectors $\mathbf{s}_0$ and $\mathbf{s}_1$.
- [6 pts] Find the ML decision rule and draw the decision regions.
- [6 pts] Calculate the average probability of bit error P_b .
- [6 pts] Show that one correlator with $s_0(t) - s_1(t)$ makes the same decisions, and give its threshold.

D4-10 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

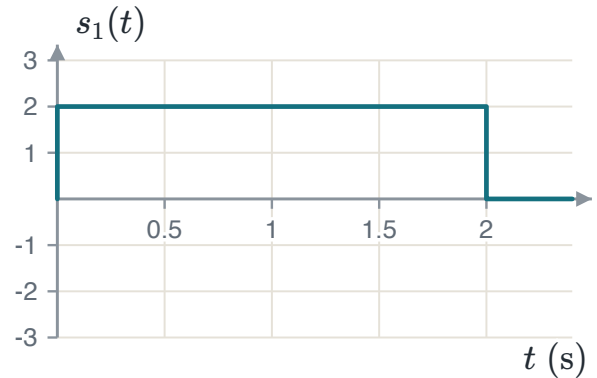
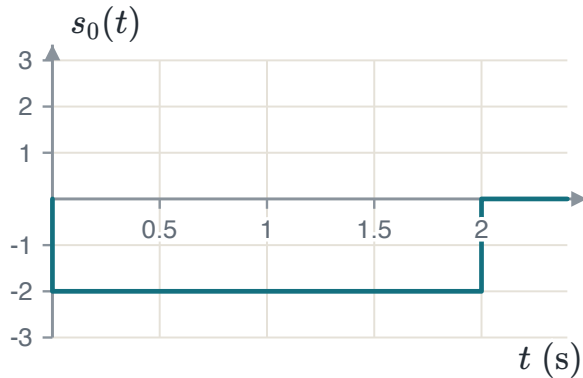
In an additive white Gaussian noise channel with noise power spectral density $N_0/2 = 9$ W/Hz, two equiprobable messages are sent by the waveforms drawn below. These signals are passed through the following matched-filter type demodulator, sampled at $t = T = 4$ s. According to the information given above,



- [5 pts] Plot the impulse response of the matched filter, $h(t) = \psi(T - t)$.
- [10 pts] Determine the conditional PDFs $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [10 pts] Find the optimal decision threshold λ and calculate the average probability of bit error P_b .

D4-11 · TWO WAVEFORMS, THE OPTIMAL RECEIVER AND THE BIT ERROR PROBABILITY

A binary link uses $s_1(t) = 2$ and $s_0(t) = -2$ for $0 \leq t < 2$ s, with equal priors. The noise is white and Gaussian with an unknown two-sided power spectral density $N_0/2$. The receiver is the optimal correlator receiver with $T = 2$ s. The link must reach $P_b \leq 1.0 \times 10^{-3}$. A Q table gives $Q(3.09) = 1.001 \times 10^{-3}$ and $Q(3.10) = 0.968 \times 10^{-3}$. According to the information given above,



- [8 pts] Find the coordinates s_0 and s_1 , and write P_b as a function of $N_0/2$.
- [9 pts] Find the largest $N_0/2$ that meets the requirement.
- [8 pts] The link is changed to on-off signalling, $s_0(t) = 0$ and $s_1(t) = A$ on $[0, 2)$, with the same average energy per bit. Find A , the largest $N_0/2$ now, and the loss in decibels.

D4-12 · BINARY PAM WITH UNEQUAL PRIORS: THE MAP THRESHOLD

Assume that in a binary pulse amplitude modulation (PAM) communication system, "0" and "1" bits occur with probabilities 0.4 and 0.6. At the output of the matched filter the sample is

$$Y = \begin{cases} 2 + N, & \text{if "1" is sent} \\ -2 + N, & \text{if "0" is sent} \end{cases}.$$

Here N is a Gaussian random variable with mean 0 and variance 2.25. According to this information,

- [8 pts] Determine the conditional PDFs $f_Y(y | 1)$ and $f_Y(y | 0)$.
- [8 pts] Find the optimal decision threshold λ .
- [9 pts] Calculate the average probability of bit error P_b as a number. Use a Q -function table.

D4-13 · BINARY PAM WITH UNEQUAL PRIORS: THE MAP THRESHOLD

Assume that in a binary PAM communication system, "0" and "1" bits occur with probabilities 0.6 and 0.4. Consider a signal detector with the input

$$Y = \begin{cases} 5 + N, & \text{if "1" is sent} \\ N, & \text{if "0" is sent} \end{cases}.$$

Here N is a Gaussian random variable with mean 0 and variance 4. According to this information,

- a) [8 pts] Find the optimal decision threshold λ .
- b) [10 pts] Calculate the average probability of bit error P_b as a number. Use a Q -function table.
- c) [7 pts] A simpler detector uses the midpoint 2.5 as its threshold. Calculate its P_b and compare.

D4-14 · BINARY PAM WITH UNEQUAL PRIORS: THE MAP THRESHOLD

In a binary PAM system the matched-filter output is $Y = 1 + N$ when "1" is sent and $Y = -2 + N$ when "0" is sent. N is Gaussian with mean 0 and variance 1. The bits "0" and "1" occur with probabilities 0.75 and 0.25. According to this information,

- a) [6 pts] Determine the conditional PDFs $f_Y(y | 1)$ and $f_Y(y | 0)$.
- b) [6 pts] Find the ML threshold and the P_b of a detector that uses it.
- c) [7 pts] Find the optimal (MAP) decision threshold λ .
- d) [6 pts] Calculate P_b of the MAP detector as a number, and compare.

D4-15 · BINARY PAM WITH UNEQUAL PRIORS: THE MAP THRESHOLD

Bits are sent by antipodal signalling with energy $E_b = 2.25$ J per bit. The two signal points are $s_0 = -\sqrt{E_b}$ and $s_1 = +\sqrt{E_b}$ on one basis function. The noise is white and Gaussian with $N_0 = 0.5$ W/Hz. The bit "0" occurs with probability 0.8 and "1" with probability 0.2. According to this information,

- a) [6 pts] Find the distance d between the two points and the P_b of the ML detector.
- b) [7 pts] Find the MAP boundary: its distance μ from s_0 , and the threshold λ .
- c) [7 pts] Calculate the conditional error probabilities $P(e | 0)$ and $P(e | 1)$ of the MAP detector.
- d) [5 pts] Calculate P_b of the MAP detector and compare it with part (a).

D4-16 · BINARY PAM WITH UNEQUAL PRIORS: THE MAP THRESHOLD

A three-level PAM system sends $Y = s + N$ with $s \in \{-2, 0, 2\}$. The priors are $P(-2) = P(2) = 0.25$ and $P(0) = 0.5$. N is Gaussian with mean 0 and variance 0.5. According to this information,

- a) [6 pts] Write the three conditional PDFs $f_Y(y | s)$.
- b) [7 pts] Find the two optimal (MAP) thresholds $\lambda_1 < \lambda_2$.
- c) [7 pts] Calculate the average symbol error probability P_e as a number.
- d) [5 pts] Calculate P_e of the ML detector, with thresholds ± 1 , and compare.

D4-17 · BINARY PAM WITH UNEQUAL PRIORS: THE MAP THRESHOLD

In a binary PAM system the bit "1" is twice as likely as the bit "0". The matched-filter output is $Y = 1 + N$ when "1" is sent and $Y = -1 + N$ when "0" is sent. N is Gaussian with mean 0 and variance 0.25. According to this information,

- a) [8 pts] Find the two priors and the optimal decision threshold λ .
- b) [9 pts] Calculate the average probability of bit error P_b as a number.
- c) [8 pts] The noise variance rises to 1. Find the new threshold and P_b , and say how far λ moved.

D4-18 · BINARY PAM WITH UNEQUAL PRIORS: THE MAP THRESHOLD

In a binary PAM system the matched-filter output is $Y = 2 + N$ for "1" and $Y = -2 + N$ for "0". N is Gaussian with mean 0 and variance 4. The bit "0" occurs with probability 0.9 and "1" with probability 0.1. According to this information,

- a) [6 pts] For the sample $y = 0.5$, compute $P_0 f_Y(y | 0)$ and $P_1 f_Y(y | 1)$. Give the MAP and the ML decisions.
- b) [6 pts] Find the MAP threshold λ .
- c) [7 pts] Calculate P_b of the MAP detector as a number.
- d) [6 pts] Show that $P(e | 1)$ exceeds one half, and compare P_b with the ML detector.

D4-19 · BINARY PAM WITH UNEQUAL PRIORS: THE MAP THRESHOLD

In a binary PAM system the matched-filter output is $Y = 1 + N$ for "1" and $Y = -1 + N$ for "0". N is Gaussian with mean 0 and variance 0.5. The priors are not known. The MAP detector of this system uses the threshold $\lambda = -0.25$. According to this information,

- a) [9 pts] Find the priors P_0 and P_1 .
- b) [8 pts] Calculate P_b of this MAP detector as a number.
- c) [8 pts] Calculate P_b of the ML detector with the same priors, and compare.

D4-20 · BINARY PAM WITH UNEQUAL PRIORS: THE MAP THRESHOLD

Two messages use $s_0(t) = 2\sqrt{2} \cos(2\pi t)$ and $s_1(t) = 2\sqrt{2} \sin(2\pi t)$ for $0 \leq t \leq 1$ s. The channel adds white Gaussian noise with $N_0/2 = 0.5$ W/Hz. The priors are $P(s_0) = 0.75$ and $P(s_1) = 0.25$. The receiver has two correlators with $\psi_1(t) = \sqrt{2} \cos(2\pi t)$ and $\psi_2(t) = \sqrt{2} \sin(2\pi t)$. According to the information given above,

- [6 pts] Find the signal vectors $\mathbf{s}_0$ and $\mathbf{s}_1$, and write the MAP metric of each.
- [7 pts] Find the MAP decision boundary and draw the two decision regions.
- [6 pts] The observation is $\mathbf{r} = (0.9, 1.1)$. Give the MAP and the ML decisions.
- [6 pts] Calculate P_b of the MAP detector and compare it with the ML detector.

D4-21 · M-ARY CONSTELLATIONS: DECISION REGIONS AND THE NEAREST-NEIGHBOUR APPROXIMATION

Consider an M -ary modulation scheme where the equally probable symbols have the waveforms

$$s_k(t) = \sqrt{6} \cos\left(2000\pi t + \frac{(2k-1)\pi}{6}\right), \quad k \in \{1, \dots, 6\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. According to the information given above,

- [8 pts] Choose an orthonormal basis, find the coordinates of the six points, and give $E_{s,\text{avg}}$.
- [7 pts] Draw the signal constellation and the optimal decision regions.
- [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$.

D4-22 · M-ARY CONSTELLATIONS: DECISION REGIONS AND THE NEAREST-NEIGHBOUR APPROXIMATION

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$\begin{aligned} s_1(t) &= 0 \\ s_2(t) &= 2\sqrt{2} \cos(2000\pi t) \\ s_3(t) &= 4 \cos\left(2000\pi t + \frac{\pi}{4}\right) \\ s_4(t) &= 2\sqrt{2} \cos\left(2000\pi t + \frac{\pi}{2}\right) \end{aligned}$$

all for $0 \leq t \leq 1$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. According to the information given above,

- [7 pts] Find the four signal points and $E_{s,\text{avg}}$.
- [6 pts] Draw the signal constellation and the optimal decision regions.
- [7 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$.
- [5 pts] Moving the constellation so that its centre is at the origin keeps P_e . How much energy does it save, in decibels?

D4-23 · M-ARY CONSTELLATIONS: DECISION REGIONS AND THE NEAREST-NEIGHBOUR APPROXIMATION

Consider an M -ary modulation scheme where the equally probable symbols have the waveforms

$$s_k(t) = (2k - 5)\sqrt{2} \cos(4000\pi t), \quad k \in \{1, 2, 3, 4\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard additive white Gaussian noise (AWGN) channel with $\mathcal{N}(0, N_0/2)$. According to the information given above,

- a) [8 pts] Find the basis, the four signal points and $E_{s,\text{avg}}$.
- b) [7 pts] Draw the signal constellation and the optimal decision regions.
- c) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$.

D4-24 · M-ARY CONSTELLATIONS: DECISION REGIONS AND THE NEAREST-NEIGHBOUR APPROXIMATION

Consider an M -ary modulation scheme where the equally probable symbols have the waveforms

$$s_k(t) = 2 \cos\left(1000\pi t + \frac{(2k-1)\pi}{8}\right), \quad k \in \{1, \dots, 8\}, \quad 0 \leq t \leq 2.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. According to the information given above,

- a) [7 pts] Choose an orthonormal basis and find the signal points and $E_{s,\text{avg}}$.
- b) [6 pts] Draw the signal constellation and the optimal decision regions.
- c) [7 pts] Determine the nearest-neighbour approximation of P_e as a function of $E_{s,\text{avg}}/N_0$.
- d) [5 pts] Evaluate it at $E_{s,\text{avg}}/N_0 = 20$.

D4-25 · M-ARY CONSTELLATIONS: DECISION REGIONS AND THE NEAREST-NEIGHBOUR APPROXIMATION

Consider an M -ary modulation scheme where the equally probable symbols have the waveforms

$$s_{a,b}(t) = a\sqrt{2} \cos(2000\pi t) - b\sqrt{2} \sin(2000\pi t), \quad 0 \leq t \leq 1,$$

with $a \in \{-3, -1, 1, 3\}$ and $b \in \{-1, 1\}$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. According to the information given above,

- a) [8 pts] Find the signal points and $E_{s,\text{avg}}$.
- b) [7 pts] Draw the signal constellation and the optimal decision regions.
- c) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$.

D4-26 · M-ARY CONSTELLATIONS: DECISION REGIONS AND THE NEAREST-NEIGHBOUR APPROXIMATION

Consider an M -ary modulation scheme where the equally probable symbols have the waveforms $s_1(t) = 0$ and

$$s_k(t) = 2\sqrt{2} \cos\left(2000\pi t + \frac{(k-2)\pi}{2}\right), \quad k \in \{2, 3, 4, 5\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. According to the information given above,

- [8 pts] Find the five signal points and $E_{s,\text{avg}}$.
- [8 pts] Draw the signal constellation and the optimal decision regions.
- [9 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$.

D4-27 · M-ARY CONSTELLATIONS: DECISION REGIONS AND THE NEAREST-NEIGHBOUR APPROXIMATION

Consider an M -ary modulation scheme where the equally probable symbols have the waveforms

$$s_k(t) = 2 \cos\left(2000\pi t + \frac{2\pi k}{3} + \frac{\pi}{2}\right), \quad k \in \{0, 1, 2\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. According to the information given above,

- [7 pts] Find the three signal points and $E_{s,\text{avg}}$.
- [6 pts] Draw the signal constellation and the optimal decision regions.
- [7 pts] Determine the nearest-neighbour approximation of P_e as a function of $E_{s,\text{avg}}/N_0$.
- [5 pts] Evaluate it at $E_{s,\text{avg}}/N_0 = 6$.

D4-28 · M-ARY CONSTELLATIONS: DECISION REGIONS AND THE NEAREST-NEIGHBOUR APPROXIMATION

Consider an M -ary modulation scheme where the equally probable symbols have the waveforms $s_1(t) = 0$ and

$$s_k(t) = 2\sqrt{2} \cos\left(2000\pi t + \frac{(k-2)\pi}{3}\right), \quad k \in \{2, \dots, 7\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. According to the information given above,

- [8 pts] Find the seven signal points and $E_{s,\text{avg}}$.
- [8 pts] Draw the signal constellation and the optimal decision regions.
- [9 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$.

D4-29 · UNION BOUND AND EQUAL-ENERGY COMPARISON

Two designs carry three bits per symbol with the same average symbol energy $E_{s,\text{avg}}$ over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. Design A is 8-PSK: eight equally likely points on a circle, 45° apart. Design B is the rectangle of points $c(a, b)$ with $a \in \{\pm 1, \pm 3\}$, $b \in \{\pm 1\}$ and a scale $c > 0$. According to the information given above,

- a) [7 pts] Find $d_{\min}^2$ of each design in terms of $E_{s,\text{avg}}$.
- b) [6 pts] Find $N_{\min}$ of each design.
- c) [7 pts] Evaluate the nearest-neighbour approximation of P_e for both at $E_{s,\text{avg}}/N_0 = 20$.
- d) [5 pts] Say which design is better, and give its gain in $d_{\min}^2$ in decibels.

D4-30 · UNION BOUND AND EQUAL-ENERGY COMPARISON

Three equally likely messages use $s_1(t) = 0$, $s_2(t) = 2$ for $0 \leq t < 1$ (zero after), and $s_3(t) = 3$ for $1 \leq t < 2$ (zero before). $T = 2$ s. The channel adds white Gaussian noise with $N_0/2 = 0.25$ W/Hz. The receiver is the optimal ML receiver. According to the information given above,

- a) [6 pts] Find the basis, the three signal points, and draw the decision regions.
- b) [6 pts] Find all pairwise distances, $d_{\min}$ and $N_{\min}$.
- c) [7 pts] Evaluate the union bound on the symbol error probability P_e .
- d) [6 pts] Evaluate the nearest-neighbour approximation and the minimum-distance bound, and put the three numbers in order.

MODULE

5

Digital Modulation Methods

30 questions on digital modulation methods. Write your reasoning in the space under each one.

D5-01 · EQUAL-ENERGY PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = \sqrt{18} \cos\left(3000\pi t + \frac{\pi(2k+1)}{6}\right), \quad k \in \{1, \dots, 6\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] Evaluate the approximation of part (b) at $E_{s,\text{avg}}/N_0 = 18$.

D5-02 · EQUAL-ENERGY PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = \sqrt{24} \cos\left(2000\pi t + \frac{\pi(4k-3)}{6}\right), \quad k \in \{1, 2, 3\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$.
- c) [5 pts] How many bits does one symbol carry? Rewrite the result of part (b) as a function of E_b/N_0 .

D5-03 · EQUAL-ENERGY PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = 2 \cos\left(4000\pi t + \frac{(2k-1)\pi}{8}\right), \quad k \in \{1, \dots, 8\}, \quad 0 \leq t \leq 2.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [8 pts] Find $E_{s,\text{avg}}$, the number of bits per symbol and the average energy per bit E_b .
- b) [8 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- c) [9 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, and then as a function of E_b/N_0 .

D5-04 · AMPLITUDE SETS ON ONE CARRIER

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = 2(2k-5) \cos(5000\pi t), \quad k \in \{1, 2, 3, 4\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] Evaluate the approximation of part (b) at $E_{s,\text{avg}}/N_0 = 22.5$.

D5-05 · AMPLITUDE SETS ON ONE CARRIER

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = 3\sqrt{2}(k-1) \cos(3000\pi t), \quad k \in \{1, 2, 3\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] The set is shifted so that its points are symmetric about the origin, with the same $d_{\min}$. Find the saving in $E_{s,\text{avg}}$ in decibels.

D5-06 · AMPLITUDE SETS ON ONE CARRIER

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = 2(k-3) \cos(4000\pi t), \quad k \in \{1, \dots, 5\}, \quad 0 \leq t \leq 0.5.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [8 pts] Find $E_{s,\text{avg}}$ and the number of bits carried by one symbol.
- b) [8 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- c) [9 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$. Evaluate it at $E_{s,\text{avg}}/N_0 = 36$.

D5-07 · AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_1(t) = 0,$$

$$s_k(t) = 2\sqrt{2} \cos\left(2000\pi t + \frac{\pi}{2} + \frac{2\pi(k-2)}{3}\right), \quad k \in \{2, 3, 4\},$$

all on $0 \leq t \leq 1$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] Evaluate the approximation of part (b) at $E_{s,\text{avg}}/N_0 = 6$.

D5-08 · AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_{mn}(t) = \sqrt{2}[(2m-5) \cos(3000\pi t) - (2n-3) \sin(3000\pi t)],$$

with $m \in \{1, 2, 3, 4\}$, $n \in \{1, 2\}$ and $0 \leq t \leq 1$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] Evaluate the approximation of part (b) at $E_{s,\text{avg}}/N_0 = 27$.

D5-09 · AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_{mn}(t) = (2m - 5) \cos(4000\pi t) - (2n - 5) \sin(4000\pi t),$$

with $m, n \in \{1, 2, 3, 4\}$ and $0 \leq t \leq 2$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [8 pts] Find $E_{s,\text{avg}}$, the number of bits per symbol and the average energy per bit E_b .
- b) [8 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- c) [9 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$ and of E_b/N_0 . Evaluate it at $E_b/N_0 = 11.25$.

D5-10 · AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

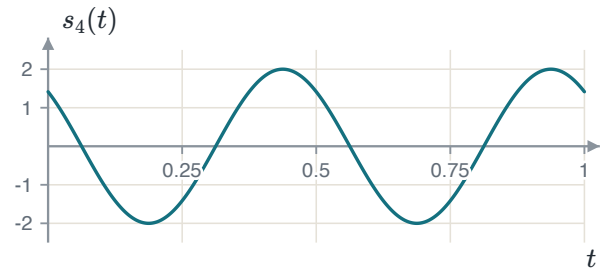
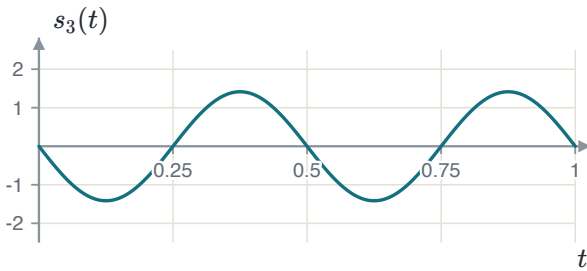
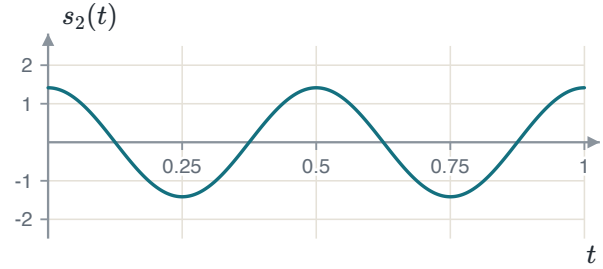
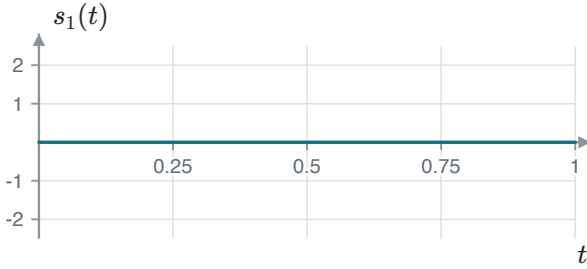
$$s_k(t) = A_k \cos\left(2000\pi t + \frac{(k-1)\pi}{4}\right), \quad k \in \{1, \dots, 8\}, \quad 0 \leq t \leq 1,$$

where $A_k = \sqrt{2}$ for odd k and $A_k = 4$ for even k . These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] Which symbols have no neighbour at $d_{\min}$? Evaluate the approximation at $E_{s,\text{avg}}/N_0 = 40.5$.

D5-11 • AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the four equally probable symbols have the waveforms drawn below, each on $0 \leq t \leq 1$. Each waveform is zero or a sinusoid at 2 Hz. The peak value is $\sqrt{2}$ for s_2 and s_3 and 2 for s_4 . These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.



- [7 pts] Write each waveform in the form $A \cos(4\pi t + \theta)$ and find its signal-space coordinates.
- [8 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- [6 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$.
- [4 pts] The four points are moved together so that their centre is at the origin. Find the saving in $E_{s,\text{avg}}$ in decibels.

D5-12 • EQUAL-ENERGY PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = 2\sqrt{3} \cos\left(2000\pi t + \frac{\pi}{6} + \frac{(k-1)\pi}{2}\right), \quad k \in \{1, 2, 3, 4\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- [8 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- [7 pts] With Gray labelling, one symbol error costs about one bit error. Find E_b , then compare the bit error probability with binary PSK at $E_b/N_0 = 4.5$.

D5-13 · AMPLITUDE SETS ON ONE CARRIER

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = \sqrt{2} (2k - 9) \cos(2000\pi t), \quad k \in \{1, \dots, 8\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- [8 pts] Find $E_{s,\text{avg}}$, the number of bits per symbol and the average energy per bit E_b .
- [8 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- [9 pts] Determine the nearest-neighbour approximation as a function of $E_{s,\text{avg}}/N_0$ and of E_b/N_0 . Find the E_b/N_0 in decibels that makes the Q argument 3.00, and the resulting P_e .

D5-14 · EQUAL-ENERGY PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = 5\sqrt{2} \cos\left(3000\pi t + \frac{2\pi k}{5} - \frac{\pi}{10}\right), \quad k \in \{1, \dots, 5\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- [5 pts] Evaluate the approximation of part (b) at $E_{s,\text{avg}}/N_0 = 13$.

D5-15 · AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$\begin{aligned} s_1(t) &= 0, \\ s_k(t) &= 2 \cos\left(2000\pi t + \frac{(k-2)\pi}{3}\right), \quad k \in \{2, \dots, 7\}, \end{aligned}$$

all on $0 \leq t \leq 1$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- [5 pts] At $E_{s,\text{avg}}/N_0 = 12$, compare the conditional error probability of s_1 with that of an outer symbol.

D5-16 · AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = \begin{cases} \sqrt{2} \cos(2000\pi t + (k-1)\frac{\pi}{2}), & k \in \{1, 2, 3, 4\}, \\ 2\sqrt{2} \cos(2000\pi t + (k-5)\frac{\pi}{2}), & k \in \{5, 6, 7, 8\}, \end{cases}$$

all on $0 \leq t \leq 1$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] Compare this set with 8-PSK at the same $E_{s,\text{avg}}$, in decibels.

D5-17 · EQUAL-ENERGY PHASE SETS

Consider a binary modulation scheme where the equally probable symbols have the following waveforms:

$$s_1(t) = 3 \cos(5000\pi t), \quad s_2(t) = 3 \cos\left(5000\pi t + \frac{2\pi}{3}\right), \quad 0 \leq t \leq 2.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [8 pts] Draw the signal constellation and the optimal decision boundary.
- b) [9 pts] Determine the symbol error probability as a function of $E_{s,\text{avg}}/N_0$. Explain why the nearest-neighbour approximation is exact here.
- c) [8 pts] Which phase difference minimises the error probability at the same energy? Find the loss of the given set in decibels, and both error probabilities at $E_{s,\text{avg}}/N_0 = 6$.

D5-18 · SETS BUILT FROM TWO FREQUENCIES

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_{1,2}(t) = \pm 2 \cos(2000\pi t), \quad s_{3,4}(t) = \pm 2 \cos(3000\pi t), \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [8 pts] Show that the two carriers are orthogonal on $0 \leq t \leq 1$, and choose an orthonormal basis.
- b) [8 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- c) [9 pts] Determine the nearest-neighbour approximation as a function of $E_{s,\text{avg}}/N_0$. Evaluate it at 9 and compare with a four-point PSK set of the same energy.

D5-19 · SETS BUILT FROM TWO FREQUENCIES

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_1(t) = 0, \quad s_2(t) = 3 \cos(4000\pi t), \quad s_3(t) = 3 \cos(5000\pi t), \quad 0 \leq t \leq 2.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] Evaluate the approximation of part (b) at $E_{s,\text{avg}}/N_0 = 12$.

D5-20 · AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_{mn}(t) = \sqrt{2} [2(m-2) \cos(2000\pi t) - (2n-3) \sin(2000\pi t)],$$

with $m \in \{1, 2, 3\}$, $n \in \{1, 2\}$ and $0 \leq t \leq 1$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] How many bits does one symbol carry? Evaluate the approximation at $E_{s,\text{avg}}/N_0 = 16.5$.

D5-21 · AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = 2 \cos\left(2000\pi t + \frac{(k-1)\pi}{2}\right), \quad k \in \{1, 2, 3\},$$

$$s_4(t) = 4 \cos\left(2000\pi t + \frac{3\pi}{2}\right),$$

all on $0 \leq t \leq 1$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] Compare with a regular four-point PSK set of the same $E_{s,\text{avg}}$, in decibels.

D5-22 · AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

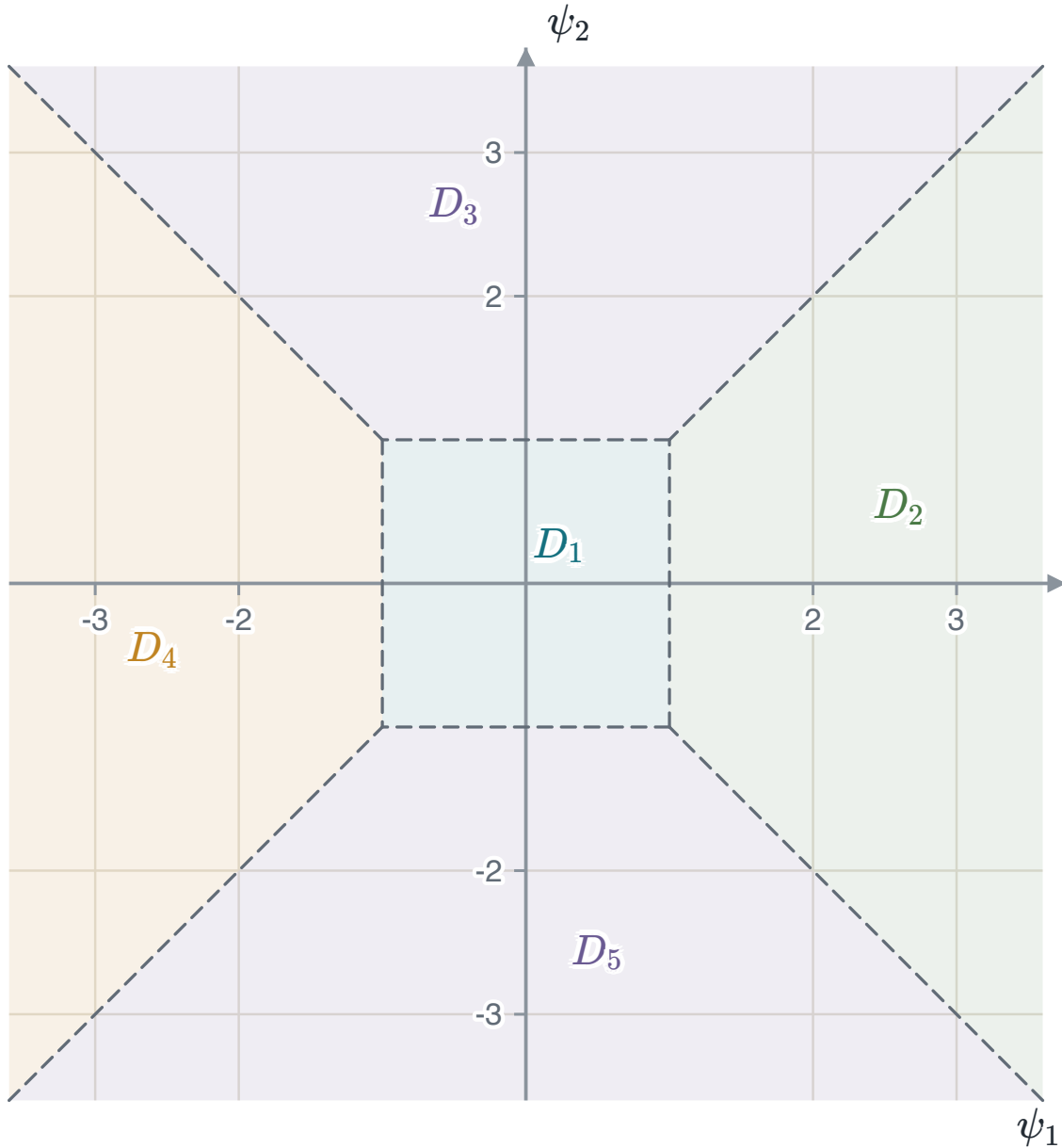
$$s_{1,2}(t) = \pm\sqrt{2} \cos(3000\pi t), \quad s_{3,4}(t) = 2\sqrt{2} \cos\left(3000\pi t \pm \frac{\pi}{2}\right), \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [10 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [5 pts] The next distance is only slightly larger than $d_{\min}$. Add its terms to the approximation and evaluate both at $E_{s,\text{avg}}/N_0 = 11.25$.

D5-23 · REVERSED QUESTIONS AND COMPARISONS

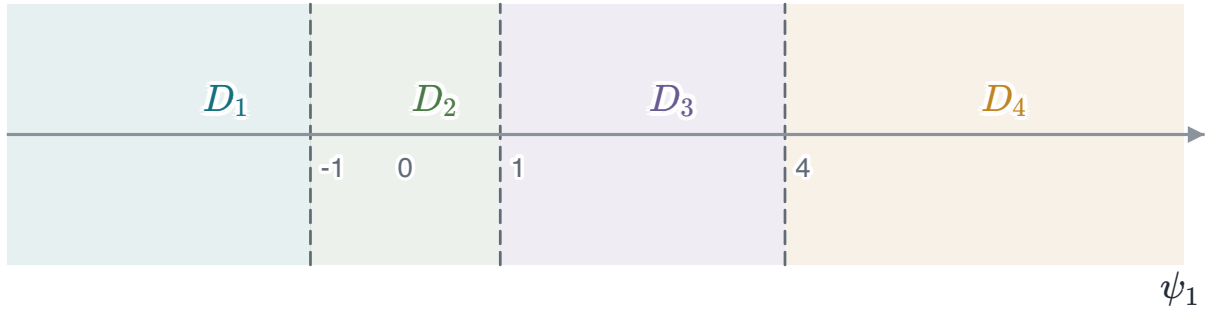
Five equally probable waveforms are transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. The figure shows their optimal decision regions $D_1, \dots, D_5$ on the basis $\psi_1(t) = \sqrt{2} \cos(2000\pi t)$, $\psi_2(t) = -\sqrt{2} \sin(2000\pi t)$, $0 \leq t \leq 1$. The central region is the square $|\psi_1| < 1$, $|\psi_2| < 1$, and the diagonals split the rest. One waveform is the zero signal. The other four have equal energy, and $E_{s,\text{avg}} = 3.2$.



- [8 pts] Locate the five signal points from the regions, and confirm the value of $E_{s,\text{avg}}$.
- [7 pts] Write the five waveforms $s_1(t), \dots, s_5(t)$, with s_k the symbol decided in D_k .
- [10 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$. Evaluate it at $E_{s,\text{avg}}/N_0 = 14.4$.

D5-24 · REVERSED QUESTIONS AND COMPARISONS

Four equally probable waveforms $s_k(t) = c_k \sqrt{2} \cos(2000\pi t)$, $0 \leq t \leq 1$, with $c_1 < c_2 < c_3 < c_4$, are detected optimally after a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. The figure shows the decision thresholds on the axis $\psi_1(t) = \sqrt{2} \cos(2000\pi t)$. The smallest coefficient is $c_1 = -2$.



- [8 pts] Find c_2 , c_3 and c_4 from the thresholds.
- [7 pts] Find $E_{s,\text{avg}}$ and write the four waveforms.
- [6 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$.
- [4 pts] Give the four-point set with the same $d_{\min}$ and the least $E_{s,\text{avg}}$. Find its saving in decibels.

D5-25 · AMPLITUDE SETS ON ONE CARRIER

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$s_k(t) = c_k \sqrt{2} \cos(2000\pi t), \quad c_k \in \{-2, 0, 1, 3\}, \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- [8 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- [7 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- [5 pts] Evaluate the approximation of part (b) at $E_{s,\text{avg}}/N_0 = 28$.
- [5 pts] Find the exact symbol error probability at the same value from the thresholds, and compare.

D5-26 · REVERSED QUESTIONS AND COMPARISONS

Two sets of sixteen equally probable waveforms are proposed, both on $0 \leq t \leq 1$:

$$\text{A: } s_k(t) = 2\sqrt{5} \cos\left(2000\pi t + \frac{k\pi}{8}\right), \quad k \in \{1, \dots, 16\},$$

$$\text{B: } s_{mn}(t) = \sqrt{2}[(2m - 5) \cos(2000\pi t) - (2n - 5) \sin(2000\pi t)], \quad m, n \in \{1, 2, 3, 4\}.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- [8 pts] Show that the two sets have the same $E_{s,\text{avg}}$, and draw both constellations with their optimal decision regions.
- [9 pts] Determine the nearest-neighbour approximation of the average symbol error probability of each set as a function of $E_{s,\text{avg}}/N_0$.
- [8 pts] Which set is better at equal $E_{s,\text{avg}}$, and by how many decibels? Evaluate both approximations at $E_{s,\text{avg}}/N_0 = 45$.

D5-27 · SETS BUILT FROM TWO FREQUENCIES

Consider a binary modulation scheme where the equally probable symbols have the following waveforms:

$$s_1(t) = 2 \cos(2000\pi t), \quad s_2(t) = 2 \cos(2\pi(1000.75)t), \quad 0 \leq t \leq 1.$$

These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- [8 pts] Find the energies and the correlation coefficient $\rho = \langle s_1, s_2 \rangle / E$. Neglect terms at the double frequency.
- [8 pts] Use Gram–Schmidt to find the signal points, and draw the constellation with its optimal decision boundary.
- [9 pts] Find the error probability as a function of $E_{s,\text{avg}}/N_0$. Compare it with the orthogonal choice $f_2 = 1000.5$ Hz at $E_{s,\text{avg}}/N_0 = 7.5$.

D5-28 · REVERSED QUESTIONS AND COMPARISONS

Eight equally probable waveforms are formed from two rings on $0 \leq t \leq 1$:

$$s_k(t) = \begin{cases} \sqrt{2} \cos(2000\pi t + (k-1)\frac{\pi}{2}), & k \in \{1, 2, 3, 4\}, \\ R\sqrt{2} \cos(2000\pi t + (k-5)\frac{\pi}{2}), & k \in \{5, 6, 7, 8\}, \end{cases}$$

with $R > 1$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- [7 pts] Find $d_{\min}$ as a function of R .
- [8 pts] Find the R that maximises $d_{\min}^2/E_{s,\text{avg}}$, and that maximum.
- [6 pts] For this R , draw the constellation and regions, and determine the nearest-neighbour approximation as a function of $E_{s,\text{avg}}/N_0$.
- [4 pts] Compare the result with 8-PSK.

D5-29 · REVERSED QUESTIONS AND COMPARISONS

A set of M equally probable waveforms $s(t) = a\sqrt{2}\cos(2000\pi t) - b\sqrt{2}\sin(2000\pi t)$, $0 \leq t \leq 1$, is transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$. The points (a, b) form a square grid centred on the origin. The nearest-neighbour approximation of the set is

$$P_e \approx 3Q\left(\sqrt{\frac{E_{s,\text{avg}}}{5N_0}}\right),$$

and $E_{s,\text{avg}} = 2.5$.

- a) [8 pts] Find M and $d_{\min}$.
- b) [7 pts] Give the values that a and b take, and confirm $N_{\min} = 3$.
- c) [6 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- d) [4 pts] Find the bits per symbol and E_b , and write the approximation as a function of E_b/N_0 .

D5-30 · AMPLITUDE-AND-PHASE SETS

Consider an M -ary modulation scheme where the equally probable symbols have the following waveforms:

$$\begin{aligned} s_0(t) &= 0, \\ s_k(t) &= 3\cos\left(3000\pi t + \frac{k\pi}{4}\right), \quad k \in \{1, \dots, 8\}, \end{aligned}$$

all on $0 \leq t \leq 2$. These signals are planned to be transmitted over a standard AWGN channel with $\mathcal{N}(0, N_0/2)$.

- a) [8 pts] Draw the signal constellation and the optimal decision regions for this signal constellation.
- b) [9 pts] Determine the nearest-neighbour approximation of the average symbol error probability as a function of $E_{s,\text{avg}}/N_0$, where $E_{s,\text{avg}}$ is the average symbol energy.
- c) [8 pts] Compare this set with 8-PSK at the same $E_{s,\text{avg}}$, both in decibels and in bits per symbol.

MODULE

6

An Introduction to Information Theory

30 questions on an introduction to information theory. Write your reasoning in the space under each one.

D6-01 · A FUNCTION OF ONE UNIFORM SOURCE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between -4 and 4 . Let $Y \triangleq X^3 \pmod{5}$ be another DMS which is a function of X . Here $a \pmod{m}$ denotes the remainder of a on division by m , taken in $\{0, 1, \dots, m-1\}$.

- [10 pts] Calculate the entropy of the source, Y .
- [10 pts] Design a *binary* Huffman code for the source, Y .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-02 · A FUNCTION OF ONE UNIFORM SOURCE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between 0 and 9 . Let $Y \triangleq X^2 \pmod{10}$ be another DMS which is a function of X . Here $a \pmod{m}$ denotes the remainder of a on division by m , taken in $\{0, 1, \dots, m-1\}$. In words, Y is the last decimal digit of X^2 .

- [10 pts] Calculate the entropy of the source, Y .
- [10 pts] Design a *binary* Huffman code for the source, Y .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-03 · A FUNCTION OF ONE UNIFORM SOURCE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between 1 and 12 . Let $Y \triangleq \lfloor 12/X \rfloor$ be another DMS which is a function of X . Here $\lfloor u \rfloor$ is the largest integer not greater than u .

- [10 pts] Calculate the entropy of the source, Y .
- [10 pts] Design a *binary* Huffman code for the source, Y .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-04 · A FUNCTION OF ONE UNIFORM SOURCE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between 0 and 15. Each value of X is written as a four-bit binary word. Let Y be the number of ones in that word. Y is another DMS which is a function of X .

- a) [10 pts] Calculate the entropy of the source, Y .
- b) [10 pts] Design a *binary* Huffman code for the source, Y .
- c) [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-05 · A FUNCTION OF ONE UNIFORM SOURCE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between -5 and 5 . Let $Y \triangleq \lceil |X|/2 \rceil$ be another DMS which is a function of X . Here $\lceil u \rceil$ is the smallest integer not less than u .

- a) [10 pts] Calculate the entropy of the source, Y .
- b) [10 pts] Design a *binary* Huffman code for the source, Y .
- c) [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-06 · A FUNCTION OF ONE UNIFORM SOURCE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between -2 and 9 . Let $Y \triangleq |X| \pmod{7}$ be another DMS which is a function of X . Here $a \pmod{m}$ denotes the remainder of a on division by m , taken in $\{0, 1, \dots, m-1\}$.

- a) [10 pts] Calculate the entropy of the source, Y .
- b) [10 pts] Design a *binary* Huffman code for the source, Y .
- c) [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-07 · A FUNCTION OF ONE UNIFORM SOURCE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between -6 and 6 . Let $Y \triangleq X^2 \pmod{11}$ be another DMS which is a function of X . Here $a \pmod{m}$ denotes the remainder of a on division by m , taken in $\{0, 1, \dots, m-1\}$.

- a) [10 pts] Calculate the entropy of the source, Y .
- b) [10 pts] Design a *binary* Huffman code for the source, Y .
- c) [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-08 · A FUNCTION OF ONE UNIFORM SOURCE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between -7 and 7 . Let $Y \triangleq \max(X, 0)$ be another DMS which is a function of X .

- [10 pts] Calculate the entropy of the source, Y .
- [10 pts] Design a *binary* Huffman code for the source, Y .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

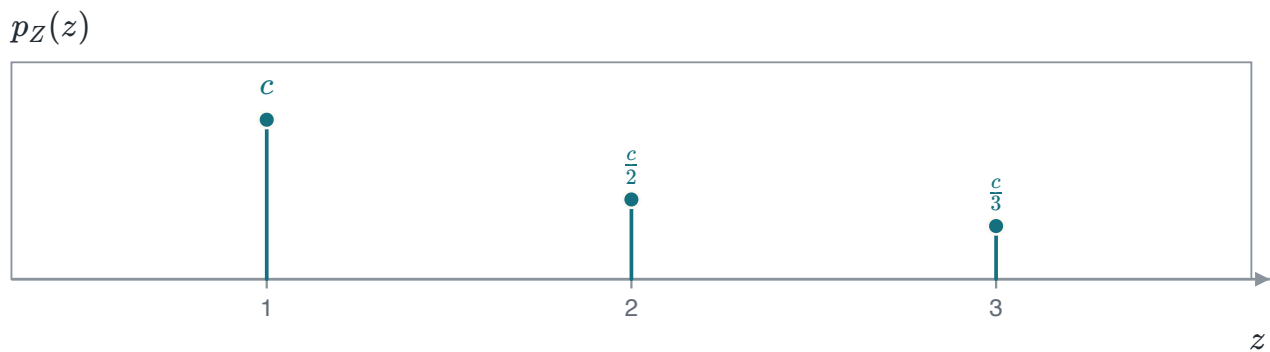
D6-09 · A FUNCTION OF TWO SOURCES, ONE WITH AN UNKNOWN CONSTANT

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable on the alphabet $\{1, 2, 3\}$. Let Z be a random variable (independent of X) with the probability mass function $p_Z(z) = c \left(\frac{1}{3}\right)^z$ for the integers $-1 \leq z \leq 1$. It is zero otherwise, and c is a constant. Finally, let $Y \triangleq X \times Z$ be another DMS which is a function of both X and Z .

- [10 pts] Calculate the entropy of the source, Y .
- [10 pts] Design a *binary* Huffman code for the source, Y .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-10 · A FUNCTION OF TWO SOURCES, ONE WITH AN UNKNOWN CONSTANT

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable on the alphabet $\{-1, 0, 1\}$. Let Z be a random variable (independent of X) with the probability mass function $p_Z(z) = c/z$ for $z \in \{1, 2, 3\}$. It is zero otherwise, and c is a constant. The pmf is shown below. Finally, let $Y \triangleq X \times Z$ be another DMS which is a function of both X and Z .



- [5 pts] Find the constant c .
- [8 pts] Calculate the entropy of the source, Y .
- [8 pts] Design a *binary* Huffman code for the source, Y .
- [4 pts] Calculate the coding efficiency of the Huffman code designed in part (c).

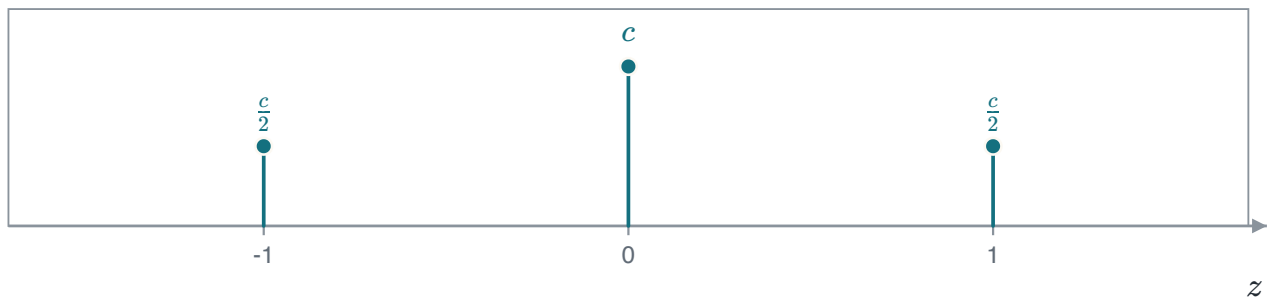
D6-11 · A FUNCTION OF TWO SOURCES, ONE WITH AN UNKNOWN CONSTANT

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable on the alphabet $\{0, 1, 2, 3\}$. Let Z be a random variable (independent of X) with the probability mass function $p_Z(z) = c \left(\frac{1}{2}\right)^z$ for the integers $0 \leq z \leq 2$. It is zero otherwise, and c is a constant. Finally, let $Y \triangleq \max(X, Z)$ be another DMS which is a function of both X and Z .

- [10 pts] Calculate the entropy of the source, Y .
- [10 pts] Design a *binary* Huffman code for the source, Y .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-12 · A FUNCTION OF TWO SOURCES, ONE WITH AN UNKNOWN CONSTANT

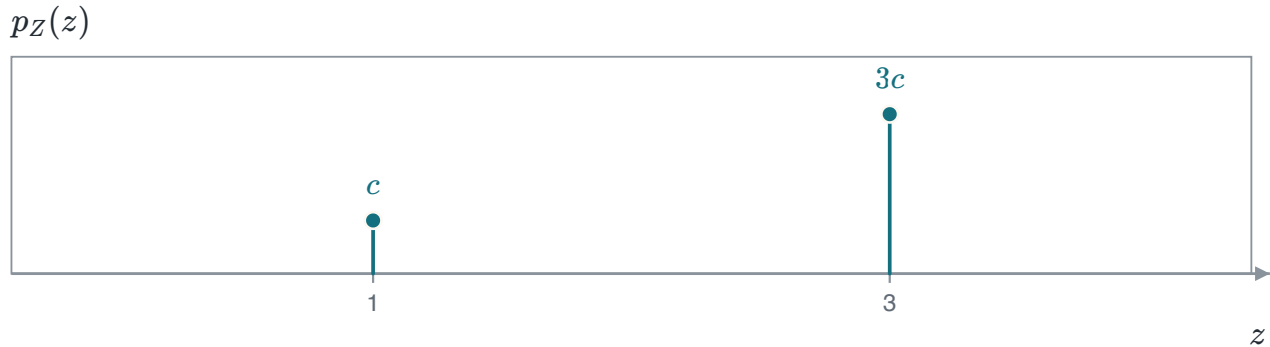
Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable on the alphabet $\{0, 1, 2, 3\}$. Let Z be a random variable (independent of X) with the probability mass function $p_Z(z) = c 2^{-|z|}$ for the integers $-1 \leq z \leq 1$. It is zero otherwise, and c is a constant. The pmf is shown below. Finally, let $Y \triangleq X + Z$ be another DMS which is a function of both X and Z .

 $p_Z(z)$


- [10 pts] Calculate the entropy of the source, Y .
- [10 pts] Design a *binary* Huffman code for the source, Y .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-13 · A FUNCTION OF TWO SOURCES, ONE WITH AN UNKNOWN CONSTANT

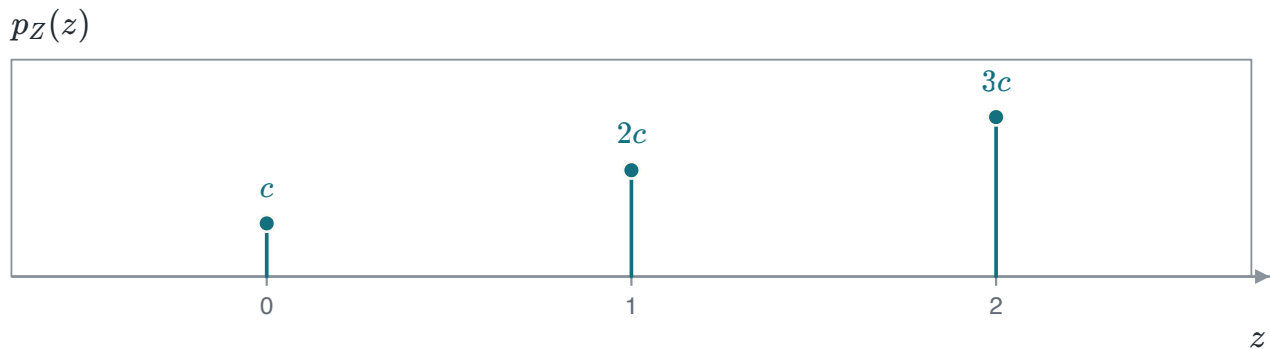
Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable on the integers $0, 1, \dots, 5$. Let Z be a random variable (independent of X) with the probability mass function $p_Z(z) = cz$ for $z \in \{1, 3\}$. It is zero otherwise, and c is a constant. The pmf is shown below. Finally, let $Y \triangleq \lfloor X/Z \rfloor$ be another DMS which is a function of both X and Z . Here $\lfloor u \rfloor$ is the largest integer not greater than u .



- [10 pts] Calculate the entropy of the source, Y .
- [10 pts] Design a *binary* Huffman code for the source, Y .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-14 · A FUNCTION OF TWO SOURCES, ONE WITH AN UNKNOWN CONSTANT

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable on the alphabet $\{-2, -1, 1, 2\}$. Let Z be a random variable (independent of X) with the probability mass function $p_Z(z) = c(z+1)$ for $z \in \{0, 1, 2\}$. It is zero otherwise, and c is a constant. The pmf is shown below. Finally, let $Y \triangleq X \times Z$ be another DMS which is a function of both X and Z .



- [5 pts] Find the constant c .
- [8 pts] Calculate the entropy of the source, Y .
- [8 pts] Design a *binary* Huffman code for the source, Y .
- [4 pts] Calculate the coding efficiency of the Huffman code designed in part (c).

D6-15 · A FUNCTION OF TWO SOURCES, ONE WITH AN UNKNOWN CONSTANT

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable on the alphabet $\{0, 1, 2, 3\}$. Let Z be a random variable (independent of X) with the probability mass function $p_Z(z) = c \left(\frac{1}{2}\right)^z$ for the integers $0 \leq z \leq 3$. It is zero otherwise, and c is a constant. Finally, let $Y \triangleq \min(X, Z)$ be another DMS which is a function of both X and Z .

- a) [10 pts] Calculate the entropy of the source, Y .
- b) [10 pts] Design a *binary* Huffman code for the source, Y .
- c) [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-16 · TWO INDEPENDENT SOURCES DESCRIBED BY RATIOS

Consider two *independent* discrete memoryless sources, X and Y . The source X is described by the alphabet $\mathcal{X} = \{0, 1, 2\}$, where $P(X = 0) = 2P(X = 1) = 4P(X = 2)$. Similarly, the source Y is described by the alphabet $\mathcal{Y} = \{0, 1, 2, 3\}$, where each symbol is generated with equal probability. Let $Z \triangleq X + Y$ be another discrete memoryless source.

- a) [10 pts] Calculate the entropy of the source, Z .
- b) [10 pts] Design a *binary* Huffman code for the source, Z .
- c) [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-17 · TWO INDEPENDENT SOURCES DESCRIBED BY RATIOS

Consider two *independent* discrete memoryless sources, X and Y . The source X is described by the alphabet $\mathcal{X} = \{1, 2, 3\}$, where each symbol is generated with equal probability. Similarly, the source Y is described by the alphabet $\mathcal{Y} = \{1, 2, 3\}$, where $P(Y = 1) = 2P(Y = 2) = 2P(Y = 3)$. Let $Z \triangleq X \times Y$ be another discrete memoryless source.

- a) [10 pts] Calculate the entropy of the source, Z .
- b) [10 pts] Design a *binary* Huffman code for the source, Z .
- c) [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-18 · TWO INDEPENDENT SOURCES DESCRIBED BY RATIOS

Consider two *independent* discrete memoryless sources, X and Y . The source X is described by the alphabet $\mathcal{X} = \{0, 1, 2, 3\}$, where each symbol is generated with equal probability. Similarly, the source Y is described by the alphabet $\mathcal{Y} = \{0, 1, \dots, 5\}$, where each symbol is generated with equal probability. Let $Z \triangleq |X - Y|$ be another discrete memoryless source.

- [10 pts] Calculate the entropy of the source, Z .
- [10 pts] Design a *binary* Huffman code for the source, Z .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-19 · TWO INDEPENDENT SOURCES DESCRIBED BY RATIOS

Consider two *independent* discrete memoryless sources, X and Y . The source X is described by the alphabet $\mathcal{X} = \{1, 2, 3, 4, 5\}$, where each symbol is generated with equal probability. Similarly, the source Y is described by the same alphabet, where each symbol is generated with equal probability. Let $Z \triangleq \min(X, Y)$ be another discrete memoryless source.

- [10 pts] Calculate the entropy of the source, Z .
- [10 pts] Design a *binary* Huffman code for the source, Z .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-20 · TWO INDEPENDENT SOURCES DESCRIBED BY RATIOS

Consider two *independent* discrete memoryless sources, X and Y . The source X is described by the alphabet $\mathcal{X} = \{0, 1, 2\}$, where $P(X = 1) = 2P(X = 0) = 2P(X = 2)$. Similarly, the source Y is described by the alphabet $\mathcal{Y} = \{0, 1, 2\}$, where $P(Y = 0) = 2P(Y = 1) = 4P(Y = 2)$. Let $Z \triangleq X - Y$ be another discrete memoryless source.

- [10 pts] Calculate the entropy of the source, Z .
- [10 pts] Design a *binary* Huffman code for the source, Z .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-21 · TWO INDEPENDENT SOURCES DESCRIBED BY RATIOS

Consider two *independent* discrete memoryless sources, X and Y . The source X is described by the alphabet $\mathcal{X} = \{1, 2, 3, 4\}$, where $P(X = k) = k P(X = 1)$. Similarly, the source Y is described by the alphabet $\mathcal{Y} = \{0, 2, 4\}$, where each symbol is generated with equal probability. Let $Z \triangleq X + Y$ be another discrete memoryless source.

- [10 pts] Calculate the entropy of the source, Z .
- [10 pts] Design a *binary* Huffman code for the source, Z .
- [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-22 · TWO INDEPENDENT SOURCES DESCRIBED BY RATIOS

Consider two *independent* discrete memoryless sources, X and Y . The source X is described by the alphabet $\mathcal{X} = \{0, 1, 2, 3\}$, where $P(X = 0) = P(X = 3) = \frac{1}{2}P(X = 1) = \frac{1}{2}P(X = 2)$. Similarly, the source Y is described by the alphabet $\mathcal{Y} = \{0, 1, 2\}$, where each symbol is generated with equal probability. Let $Z \triangleq X + Y$ be another discrete memoryless source.

- a) [10 pts] Calculate the entropy of the source, Z .
- b) [10 pts] Design a *binary* Huffman code for the source, Z .
- c) [5 pts] Calculate the coding efficiency of the Huffman code designed in part (b).

D6-23 · JUDGING A CODE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between 1 and 16. Let $Y \triangleq \lfloor \log_2 X \rfloor$ be another DMS which is a function of X . Here $\lfloor u \rfloor$ is the largest integer not greater than u .

- a) [8 pts] Calculate the entropy of the source, Y .
- b) [8 pts] Design a *binary* Huffman code for the source, Y .
- c) [5 pts] A fixed-length binary code is proposed for Y instead. Give its codeword length and its coding efficiency.
- d) [4 pts] Calculate the coding efficiency of the Huffman code of part (b), and explain the value.

D6-24 · JUDGING A CODE

Consider two *independent* discrete memoryless sources, X and Y . The source X is described by the alphabet $\mathcal{X} = \{0, 1, 2, 3\}$, where each symbol is generated with equal probability. Similarly, the source Y is described by the same alphabet, where each symbol is generated with equal probability. Let $Z \triangleq |X - Y|$ be another discrete memoryless source.

- a) [7 pts] Calculate the entropy of the source, Z .
- b) [8 pts] Design a *binary* Huffman code for Z , placing each merged probability as high as possible in the list.
- c) [6 pts] Design a second binary Huffman code, placing each merged probability as low as possible. Show that both codes have the same average length.
- d) [4 pts] Calculate the coding efficiency and the variance of the codeword lengths of both codes. Which code suits a transmitter with a finite buffer?

D6-25 · CODING THE SECOND-ORDER EXTENSION

Let X be a DMS which is modeled as a uniform random variable on $\{0, 1, 2, 3\}$. Let Z be independent of X and uniform on $\{0, 1\}$. Let $Y \triangleq (X \times Z) \pmod{2}$ be another DMS, the parity of the product.

- a) [6 pts] Calculate the probabilities and the entropy of the source, Y .
- b) [5 pts] Design a *binary* Huffman code for Y and calculate its coding efficiency.
- c) [9 pts] Design a binary Huffman code for the second-order extension of Y , whose symbols are pairs Y_1Y_2 of successive outputs.
- d) [5 pts] Calculate the average number of bits per symbol of Y and the coding efficiency for the code of part (c). Compare with part (b).

D6-26 · THE ENTROPY OF A FUNCTION AGAINST ITS INPUTS

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between -2 and 5 . Let $Y \triangleq |X - 1|$ be another DMS which is a function of X .

- a) [6 pts] Calculate the entropies $H(X)$ and $H(Y)$.
- b) [7 pts] Calculate the mutual information $I(X; Y)$, the conditional entropy $H(X | Y)$ and the joint entropy $H(X, Y)$.
- c) [8 pts] Design a *binary* Huffman code for the source, Y .
- d) [4 pts] Calculate the coding efficiency of the Huffman code designed in part (c).

D6-27 · THE ENTROPY OF A FUNCTION AGAINST ITS INPUTS

Consider two *independent* discrete memoryless sources, X and Y . The source X is described by the alphabet $\mathcal{X} = \{0, 1, 2\}$, where $P(X = 0) = 2P(X = 1) = 2P(X = 2)$. Similarly, the source Y is described by the alphabet $\mathcal{Y} = \{0, 1, 2, 3\}$, where each symbol is generated with equal probability. Let $Z \triangleq X + Y$ be another discrete memoryless source.

- a) [8 pts] Calculate the entropy of the source, Z .
- b) [6 pts] Calculate the conditional entropy $H(Z | X)$ and the mutual information $I(X; Z)$.
- c) [7 pts] Design a *binary* Huffman code for the source, Z .
- d) [4 pts] Calculate the coding efficiency of the Huffman code designed in part (c).

D6-28 · JUDGING A CODE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between 1 and 9. Let $Y \triangleq X \pmod{4}$ be another DMS which is a function of X . Here $a \pmod{m}$ denotes the remainder of a on division by m , taken in $\{0, 1, \dots, m-1\}$. A designer proposes the code $Y = 1 \rightarrow 0, Y = 0 \rightarrow 1, Y = 2 \rightarrow 01, Y = 3 \rightarrow 10$.

- a) [7 pts] Calculate the entropy of the source, Y .
- b) [6 pts] Use the Kraft inequality to decide whether a prefix code with the lengths of the proposed code exists. Give a bit string that the proposed code cannot decode uniquely.
- c) [8 pts] Design a *binary* Huffman code for the source, Y .
- d) [4 pts] Calculate the coding efficiency of the Huffman code, and compare it with a fixed-length code for Y .

D6-29 · JUDGING A CODE

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable taking the integer values between 1 and 12. Let $Y \triangleq X \pmod{5}$ be another DMS which is a function of X . Here $a \pmod{m}$ denotes the remainder of a on division by m , taken in $\{0, 1, \dots, m-1\}$.

- a) [7 pts] Calculate the entropy of the source, Y .
- b) [6 pts] Give each symbol the length $l(y) = \left\lceil \log_2 \frac{1}{P(Y=y)} \right\rceil$. Check the Kraft inequality for these lengths and calculate their average length.
- c) [8 pts] Design a *binary* Huffman code for the source, Y .
- d) [4 pts] Calculate the coding efficiency of both codes. Which one is optimal?

D6-30 · A FUNCTION OF TWO SOURCES, ONE WITH AN UNKNOWN CONSTANT

Let X be a discrete memoryless source (DMS) which is modeled as a uniform random variable on the alphabet $\{0, 1, 2, 3\}$. Let Z be a random variable (independent of X) with the probability mass function $p_Z(z) = c a^z$ for $z \in \{0, 1, 2\}$. It is zero otherwise, and c and $0 < a < 1$ are constants. Finally, let $Y \triangleq X \times Z$ be another DMS which is a function of both X and Z . It is known that $P(Y = 0) = \frac{10}{13}$.

- a) [7 pts] Find the constants c and a .
- b) [7 pts] Calculate the entropy of the source, Y .
- c) [7 pts] Design a *binary* Huffman code for the source, Y .
- d) [4 pts] Calculate the coding efficiency of the Huffman code designed in part (c).

**Digital Communications · Student Workbook**

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